Instructions

DC-300™

Adjustable Speed Drives

1 thru 300 HP

TM - TRADEMARK OF GENERAL ELECTRIC COMPANY, U.S.A.

DC-300

Adjustable Speed Drives
1 - 300 HP

GEK-85765J

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These instructions do not purport to cover all details or variations in equipment, not to provide for every possible contingency to be met during installation, operation, and maintenance. Should further information be desired or should particular problems arise that are not covered sufficiently for the purchaser's purpose, the matter should be referred to GE Drive Systems & Turbine Controls.

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1. SAFETY PRECAUTIONS

WARNING, CAUTION, AND NOTE LABELS PLACED ON THE EQUIPMENT

The following format is used on the safety and informative labels placed on the equipment. Read all labels and follow the directions of them whenever working on the equipment.

WARNING: Denotes operating procedures and practices that may result in personal injury or loss of life if not correctly followed.

WARNING labels will be red in color with black or white lettering.

CAUTION: Denotes operating procedures and practices that, if not strictly observed, may result in damage to, or destruction of, the equipment.

CAUTION labels will be amber in color with black lettering.

NOTE: Notes call attention to information that is especially significant in understanding and operating the equipment.

NOTE labels will be white in color with black lettering.

WARNING, CAUTION, AND NOTE PARAGRAPHS WITHIN THIS INSTRUCTION

The following paragraphs list some general safety reminders and safety recommendations to be followed when operating or installing this equipment. These safety precautions will be repeated throughout this instruction book where applicable.

WARNING: To prevent personal injury or equipment damage caused by equipment malfunction, only adequately trained persons should modify any programmable machine.

WARNING - ELECTRICAL SHOCK HAZARD: This equipment contians a potential hazard of electrical shock or burn. Only personnel who are adequately trained and thoroughly familiar with the equipment and the instructions should install, operate, or maintain this equipment.

WARNING - ELECTRICAL SHOCK HAZARD: Isolation of test equipment from the equipment under test presents potential electrical hazards. If the test equipment cannot be grounded to the equipment under test, the test equipment's case must be shielded to prevent contact by personnel.

WARNING - ELECTRICAL SHOCK HAZARD: To minimize hazard of electrical shock or burn, approved grounding practices and procedures must be strictly followed.

WARNING - STRAIN HAZARD: Improper lifting practices can cause serious or fatal injury. Lift only with adequate equipment and trained personnel.

WARNING - ELECTRICAL SHOCK HAZARD: Circuit breakers, if supplied as part of the total system, may not disconnect all power to the equipment (see system elementary diagrams). Whether the AC voltage is grounded or not, high voltage to ground will be present at many points.

WARNING - ELECTRICAL SHOCK AND BURN HAZARD: When using instruments such as oscilloscopes to work on live equipment, the oscilloscope's chassis should be grounded and a differential amplifier input should be used. Care should be used in the selection of probes and leads and in the adjustment of the oscilloscope so that accurate readings may be made. See instrument manufacturers instruction book for proper operation and adjustments to the instrument.

WARNING - ELECTRICAL SHOCK HAZARD: Some controllers are furnished with partial enclosures open at the top and bottom. These are intended only for mounting in another enclosure or in a control room having access by qualified personnel only.

WARNING - FIRE AND EXPLOSION HAZARD: Fires or explosions might result from mounting drive controllers in hazardous areas such as locations where flammable or combustible vapors or dusts are present. Drive controllers should be installed away from hazardous areas, even if used with DC motors suitable for use in these locations.

WARNING - ELECTRICAL SHOCK HAZARD: All motor bases and equipment enclosure housings should be connected to the factory or facility earth grounding systems.

WARNING - MECHANICAL MOTION HAZARD: Motor drives and control systems cause mechanical motion. It is the responsibility of the user to insure that any such motion does not result in an unsafe condition. Factory provided interlocks and operating limits should not be bypassed or modified.

CAUTION: Do not remove connections or printed circuit cards from the controller while power is applied. This can damage the equipment.

CAUTION: Do not remove input power from the drive until it has fully executed a stop sequence as this can damage the drive system.

CAUTION: Installation wiring must be in accordance with the National Electrical Code and be consistent with all local codes. Secondaries of the 115 volt control transformers typically have one side fused and the other grounded or available for grounding by the user.

CAUTION: The elementary diagrams supplied with the drive controller identify some signal connections as "low level analog and/or digital signals" and indicate special wiring practices. These signals should be isolated from all other wiring (particularly any power wires) and run in a separate conduit or wire run to prevent possible misoperation of the controller.

CAUTION: Meggering can damage electronic components. Do not megger or Hi-Pot the drive system without consulting **GE DRIVE SYSTEMS, SALEM, VA.**

CAUTION: Do not connect any external circuits other than those shown on the elementary diagram. Connection of devices such as ammeters on the shunt or voltmeters on the tachometer may degrade the performance of the drive system.

CAUTION: Do not use power factor correction capacitors with this equipment without consulting **GE DRIVE SYSTEMS, SALEM, VA.** Damage may result from high voltages generated when capacitors are switched.

NOTE: *Always read the complete instructions prior to applying power or troubleshooting the equipment and follow the start-up procedures step by step.*

NOTE: *Read and heed all* **WARNING, CAUTION,** *and* **NOTE** *labels posted on the equipment.*

NOTES:

2. RECEIVING AND STORAGE

RECEIVING

The controller should be placed under adequate cover immediately upon receipt as packing cases are not suitable for out-door or unprotected storage. Each shipment should be carefully examined upon arrival and checked with the packing list. Any shortage or damage should be reported promptly to the carrier. If required, assistance may be requested from **GE DRIVE SYSTEMS, SALEM, VA.** When seeking assistance please use the **serial number, requisition number, model number** and **drive code** to identify the equipment **(Telephone 703-387-7595)**.

STORAGE

If a controller is not to be installed immediately, it should be stored in a clean, dry location at ambient temperatures from -20°C (-4°F) to 55°C (131°F). The surrounding air must be free of chemical and electrically conductive or corrosive contaminants.

Precautions should be taken to prevent condensation from forming within the equipment enclosure. If the storage enviroment exceeds a 15°C (27°F) drop in temperature at 50% humidity over a four hour period, a space heater should be installed inside each enclosure to prevent condensation. (A 100 watt lamp can sometimes serve as a substitute source of heat). Higher humidities with smaller temperature changes can also cause condensation.

NOTES:

3. INSTALLATION GUIDELINES

LOCATION

DC-300 controllers are suitable for most areas where industrial equipment is installed. They should be installed in well ventilated areas with ambient temperatures ranging from 0°C to 40°C (104°F) and relative humidities up to 90%. It should be recognized, however, that since the life expectancy of any electronic component decreases with increased ambient temperature, reduction of the ambient temperature will bring about extended component life. For example, longer component life should be expected if the ambient temperature is held between 20 C (68°F) and 30 C (87°F). Proper performance and normal operational life can be expected by maintaining a proper environment for the drive system.

WIRING LEVELS

1. Wiring should be run in separate conduits or wireways for signal, control, and power wiring levels.
   a. Signal -- Low level analog and digital signals.
      -- Speed and/or position signals.
      -- Power supplies and logic signals.
      (See system schematics and notes to identify signal level wires)
   b. Control -- AC or DC control circuits.
   c. Power -- Field leads or armature leads.
      -- Brakes, 115 VAC circuits.

2. Signal wiring and power wiring may cross at right angles with a **minimum** one-inch separation. Avoid parallel runs between signal level wires and power or control wires. If signal wires must be run in parallel to control or power wires, a **minimum** of a four-inch separation must be maintained between the wires.

GROUNDING

1. Drive common (**PCOMX**) should be grounded at only one point. If the reference is supplied by numerical control or process instrument with a grounded common, the drive common should not be grounded separately.

2. If an isolation transformer is used and must be grounded, it is recommended that a high resistance ground be used.

3. Shields for shielded and twisted, shielded wire must be grounded on one end only. Provision is made to tie shields to chassis ground at the drive I/O. It is preferred to have the shield grounded only at the drive end.

4. Proper case grounding that conforms to the applicable standards should utilize (minimum) wire sizes and crimp-type terminals per the following chart:

Circuit Rating (Max.)	Copper Ground Wire AWG	AMP Terminal Number	AMP Tool Number
20	14	34120	49900
60	10	33457	49900
100	8	33458	69062
200	6	33465	69062
300	4	31811	69062
500	2	320383	69062
800	1/0	36917	69120
1000	2/0	36923	69120
1200	3/0	36925	69120
1600	4/0	321878	69120

COMMONS

Connection of reference, meter, or any other external, commons can only be made at **3TB-PCOMX,** unless otherwise indicated by the system elementaries.

SUPPRESSION

Control system relays, solenoids, or brake coils can produce erratic drive behavior due to electrical noise transients. To eliminate this possibility, an RC suppressor should be added in parallel with the coils of these devices. A 220 ohm, 2 watt resistor in series with 0.5 mfd, 600 volt capacitor can be used.

TIGHTEN ALL CONNECTIONS

Check that all electrical connections are tight and secure to be sure that no loosening occurred during shipment and installation

- -- Incoming AC line connections
- -- Power connections to motor armature and motor field
- -- **1TB, 2TB, 3TB, 4TB,** and **MATB** ("A" Frame only*) if furnished

ADDITIONAL CONSIDERATIONS

Attention should be given to the National Electrical Code and any applicable local codes when installing any drives. Wire size and insulation type, conduit sizing, enclosures, etc., should be determined per these codes.

Environments which include excessive amounts of one or more of the following characteristics should be considered hostile to drive performance and life:

1. Dust, dirt, or other foreign matter.
2. Vibration or shock.
3. Moisture or vapors.
4. Rapid temperature excursions or high ambient temperatures.
5. Caustic fumes.
6. Power line fluctuations.
7. Electromagnetic interference or **"noise"** introduced by:
   a. Radio frequency signals, typically from portable transmitters used in the vicinity of the equipment or its wiring.
   b. Stray high voltage or high frequency signals such as might be provided by arc welders, or by the operation of unsuppressed relay, contactor, or brake coils as a part of, or in the vicinity of drive control circuits.

MOUNTING

- -- Totally enclosed controllers should be positioned to permit heat radiation from all surfaces.
- -- A wall-mounted enclosure may be placed side by side with another enclosure. Clearance at least to the width of the enclosure doors should be available in front so that the doors may be fully opened for easy access.
- -- Wall-mounted controllers may be mounted on any firm, reasonably flat, heat resistent, vertical surface.

* *"A" Frame no longer available as of 3/94.*

NOTES:

4. MAIN CONTROLLER ELEMENTS AND THEIR FUNCTIONS

GENERAL DESCRIPTION

The DC-300 adjustable speed DC drive is a digitally controlled programmable drive utilizing micro-processors to provide digital regulation. Drive parameters are stored in EEROM memory, and therefore, are accurate and drift free.

Four versions or frame sizes make up the 1 - 300 HP range of the DC-300 controllers covered in this instruction book. The **"A"** frame* is rated up to 25 HP, the **"B"** frame* up to 75 HP, the **"G"** frame up to 75 HP, and the **"C"** frame thru 300 HP.

NOTE: *In addition, there are **"D"** and **"E"** frame sizes that provide ratings thru 2500 HP regenerative (4000 HP non-regenerative) that are covered in other instruction books. Some applications may require use of a larger frame size for lower horsepower applications due to required derating of the drive.*

This instruction book covers the **A, B, G,** and **C** frame DC-300 drives, as these have hardware and software commonality. Attention is drawn to any variations between these frame sizes where necessary.

The DC-300 controller provides over 80 diagnostic messages that can be decoded via a LED diagnostic readout or an optional Programmer module with alpha-numeric readout. Diagnostic messages can also be read via an optional coded 8-wire output or over the optional RS422/232C communication link. Drive operating parameters stored in the EEROM may also be examined and changed via the RS422/232C link. In addition, the DC-300 controller provides diagnostics that can check the SCRs, provide a power-up self check down to the printed circuit board level, and offers optional self-tuning procedures that can optimize the drive operating parameters automatically, to provide appropriate response for system conditions such as reflected WK^2 or friction.

The DC-300 controller requires only **three** printed circuit cards (Control Card, Power Supply Card, and Power Connection Card) for operation and offers optional cards and terminal boards for additional functions.

* *"A" and "B" Frames no longer available as of 12/93 (replaced by "G" Frame).*

Optional cards include the Application Card, Microapplication Card, Process Interface Card, Encoder Process Interface Card, Programmer Card, and Terminal Board Cards (**3TB** and **4TB**).

The Programmer can be purchased in a **Local** or **Hand-Held** version that plugs into the Control Card for inspection and adjustment of parameters when tuning-up the drive, as well as diagnostic functions.

The Control Card provides an outer speed regulator loop with an inner current loop regulator (or optional voltage loop). In situations where speed accuracy is not a requirement, the speed regulator can be converted to a **CEMF** regulator. Individually timed linear acceleration and deceleration is adjustable over a range of zero (current limit acceleration) to 999.9 seconds. Optional extended linear time and "S-Curve" are also available.

Output metering signals for armature current, armature voltage, field current, speed, torque, and horsepower are also available from the DC-300 controller. In addition, all signals can be brought out as an analog signal via one of three optional digital-to-analog converters or the optional RS422/232C serial link. (These signals and others can be programed to activate optional programmable signal level detectors.)

The block or "flow" diagram (Figure 4-1) shows the operation of a basic DC-300 drive. Power is fed through the AC fuses (or an optional circuit breaker), current transformers (regenerative controllers only), and **MA** contactor, and enters the power conversion SCR modules where it is converted to adjustable DC voltage. DC current is fed through a shunt and a DC link fuse (regenerative controllers only) to the DC motor armature.

The speed of the motor is proportional to the DC voltage applied to its armature. Speed is measured by motor **CEMF** (armature voltage feedback with IR compensation). As an optional feature, a speed can be measured by DC, AC , or Pulse tachometer(s), in which case the motor field may also be controlled to vary the speed.

Analog voltage (or optional current), frequency, or digital inputs can be used as the references. Likewise, a wide variety of tach generators can be interfaced to the drive, including both AC or DC analog tachs or pulse tachs with frequency rates up to 40 KHz.

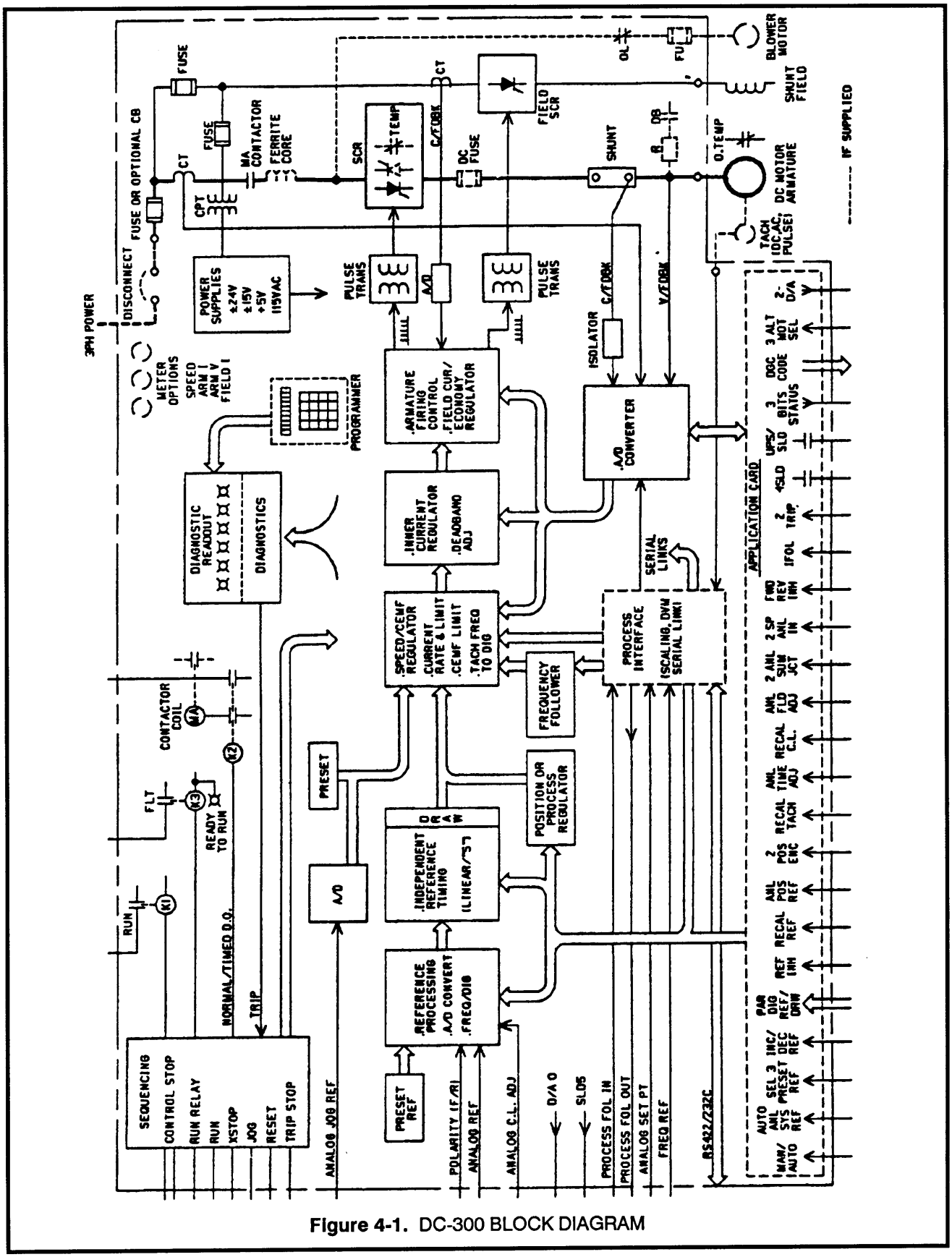


Figure 4-1. DC-300 BLOCK DIAGRAM

POWER CIRCUITS

(See Figures 4-2 thru 4-9)

Motor Field Control (MFC) may provided by an internal 10 Amp maximum field supply or by one of twooptional external field suppies rated to 24 Amps and 72 Amps.

NOTE: *A 24 Amp* ***"Motor Field Remote Supply"*** *is also available. It is used as an adjustable voltage exciter for highly inductive loads. See* **GEK-24979** *for information on this component.*

For the 10 Amp MFC, the Power Supply Card contains the field supply fuses (**CFU1-1, CFU2-1,** and **CFU3-1**), metal oxide varistors (**MOV**), field current feedback transformer (**T3**), and line reactor (**L4**). The field power module is mounted on the main controller heatsink. (See Figure 4-2.)

The 24 Amp external MFC provides a separate heatsink with field supply fuses (**CFU1-1, CFU2-1,** and **CFU3-1**), line reactor (**L4**), field power module, and MFC Suppression card containing the MOVs and field current feedback transformer (**T1**). (See Figure 4-3.)

NOTE: *There is a variation in fuse and transformer designations between the* ***10 Amp*** *and* ***24 Amp*** *motor field controls.*

NOTE: *Refer to* **GEK-85791** *for information on the* ***72 Amp*** *external MFC.*

The operation of the MFC may be best understood by reading the following while referring to the appropriate figure. Field power is supplied through fuses **CFU1, 2,** and **3**. MOVs protect from line spikes and transients.

The power to the motor field control is "two-thirds" wave rectified via the field power module. The SCR in the field power module controls field current as follows: When AC line **L1** is high with respect to **L2** and **L3,** the SCR is fired and current passes through the diode between **DX** and **DW** to the motor field. The circuit is completed through the SCR and returns through the line reactor to AC line **L3**. When **L2** becomes high, current flows through the diode between **DY** and **DW** to the field and back through the SCR, line reactor, and **L3**. When **L3** becomes high, the SCR is reversed biased and the energy stored in the field winding circulates current through the diode between **DZ** and **DW** (acting as a free wheeling diode) until **L1** becomes high again and the SCR is fired.

Field current is sensed by the current feedback transformer (**T1** or **T3** depending on the MFC rating) and fed into the field gain select circuit on the Power Supply Card (through **FCPL**) and to the Control Card.

The Power Supply Card provides input voltage to the control power transformer through **CFU4** and **5** via terminals **CPT H1** and **CPT H2. RT1, 2,** and **3** are the AC line sync signals to the Control Card.

Fuses **CFU6, CFU7,** and **CFU8** are located on the Power Supply Card. The internal DC power supplies are protected by **CFU6** and **CFU7**. The 115V AC internal supply is protected by **CFU8**. These circuits are fed from the control power transformer through plug **CPTPL**.

The SCR Power Conversion Bridge diagrams for regenerative and non-regenerative drives, **A, B, G,** and **C** frames, are shown in Figures 4-4 thru 4-9. Three-phase power is delivered to the conversion bridge through AC line fuses (**FU1-3**). Two line current transformers (**CT1** and **CT3**) provide a current feedback signal for commutation failure indication (regenerative drives only) and the reactors provide protection against AC line transients.

RC snubber networks located on the Power Connection Card provide protection for the SCRs and are connected across a forward/reverse SCR pair on regenerative drives, and in a delta configuration across the incoming 3-phase line for non-regenerative drives. The SCR firing pulses are generated through pulse transformers located on the Power Connection Card in the firing order 1-6-2-4-3-5 for both regenerative and non-regenerative drives, forward or reverse.

DC power is supplied to the armature through the shunt and **DA1** and returns through **DA2** (which is fused on regenerative drives only).

An RC network across **P1** and **DA2** aids latching of the SCRs. The voltage feedback signal is in parallel with the snubber, and current feedback is derived from the shunt. An optional thermal switch mounted on the heatsink is also available.

AC POWER REQUIREMENTS

The DC-300 drive is available in either 230 VAC +10%, -5% (1 to 100 HP) or 460 VAC +10%, -5% (2 to 250 HP), three-phase power configurations. Nominal line frequency is 50/60 Hz with tolerance of $\pm$2 Hz.

AC LINE FUSES (FU1-3)

AC line fuses provide short circuit protection for the SCRs, and the wiring inside the controller. A molded case disconnect or circuit breaker are also available as options. Additional overcurrent protection is provided by a timed overcurrent function.

FERRITE CORE ASSEMBLY (REACTORS)

(Regenerative Controllers Only)

The ferrite core assembly is provided on the AC power lines between the AC line fuses and the SCRs to protect the SCRs from possible misoperation due to transient currents and voltages during regeneration.

MAIN CONTACTOR (MA)

The MA contactor provides a controllable disconnect between the AC power line and the power conversion bridge. The contactor picks up when running mode is selected and no faults exist in the drive. The MA contactor drops out under the following conditions:

- -- When "**STOP**" is commanded on non-regenerative drives.
- -- When "**STOP**" is commanded and the motor slows to near zero speed on regenerative drives.
- -- When a fault condition occurs.

DC LINE FUSE (FU4)

(Regenerative Controllers Only)

A DC line fuse is provided in the DC output line to the motor armature and protects the SCRs and motor from overcurrent sourced from the motor.

CONTROL POWER TRANSFORMER (CPT)

This transformer is energized through its primary fuses (**CFU4** and **CFU5**) on the Power Supply Card. The secondary winding provides two isolated voltages:

- -- **115 VAC** to operate the coil of the main (MA) contactor, RUN (**K1**) Relay, and controller cooling fan (if required).
- -- **38 VAC** center tapped for the Power Supply Card to generate the DC control voltages necessary for controller operation.

SHUNT (SH)

The shunt provides the armature current feedback signal to the control circuit. A nominal 100mV output signal is generated at the current rating stamped on the shunt.

CURRENT TRANSFORMER ASM (CT)

The current transformer assembly is used to provide AC line current feedback signals (regenerative controllers only) that are used for circulating current fault detection.

CONNECTION DIAGRAMS

(See Figures 4-10 thru 4-15)

The connection diagrams for regenerative and non-regenerative controllers are shown in Figures 4-10 thru 4-15. These graphically describe the internal connections for all plugs and cables. These diagrams aid control hook up as well as troubleshooting when connection integrity of one of the cables is in question. The functions of the connectors are described in the "Electrical Connection" Section of this Instruction Book.

CONTROL CARD

This card provides the intelligence for the DC-300 drive controller. It contains three microprocessors and performs most of the regulating functions, SCR burst firing, diagnostic, and protective functions for the drive along with many customer and motor interfacing functions. The functions performed on the card are implemented in hardware and/or software.

The Control Card contains six Diagnostic LEDs to display the status of the DC-300 drive. The LEDs that are lit are interpreted as a binary number indicative of the fault. See the Troubleshooting Section for listing of Fault Codes. Note that during proper controller operation, the LEDs will blink sequentially right to left when the drive is stopped and sequentially in pairs when the drive is running.

MFC/POWER SUPPLY CARD

The Power Supply Card provides the necessary +5 VDC, +15 VDC, and +24 VDC power used throughout the controller. The armature current feedback isolation circuit as well as three relays, "RUN" (**K1**), "MAX" (**K2**), and "FAULT" (**K3**) are also on this card. Additionally, the motor field control is provided by this card.

The **G1** version card is used with motors requiring field current up to 10 Amps and contains the pulse transformer, current transformer, choke, MOVs, and fuses for the motor field circuitry as previously described. The **G2** version card is used in conjunction with external field control assemblies for field current control up to 24 Amps or 72 Amps.

NOTE: *A 24 Amp* ***"Motor Field Remote Supply"*** *is also available. It is used as an adjustable voltage exciter for highly inductive loads. See* **GEK-24979** *for information on this component.*

NOTE: *Refer to* **GEK-85791** *for information on the* **72 Amp** *external MFC.*

POWER CONNECTION CARD

This card contains SCR and DC output resistor-capacitor (**RC**) snubbers and the gate pulse transformers for the SCRs. The DC voltage feedback isolation resistors are also mounted on this card.

OPTIONAL APPLICATION CARD

This optional card contains a combination of analog interfaces between the Control Card and external equipment. These interfaces allow the core drive to be expanded to handle optional functions required in applications such as machine tool spindles, material handling cranes, etc. Two versions of the Application Card are available. The cards contain the following optional functions:

- Eight Analog Inputs
- Eight Digital Inputs
- Sixteen-Bit Parallel Word Input
- Five Programmable Signal-Level-Detector Relays (SLDs)
- Seven-Bit Digital Fault Status Output
- Three Programmable Status Bit Outputs
- Two Encoder Interfaces
- Two Analog Outputs
- Two General Purpose Amplifiers

The second type of Appplication Card available is the Microapplication Card. In addition to the above features, it also contains the DC-300 Local Area Network (**LAN**) feature for communication over a single pair of wires with up to 30 other drives, to a Series Six™ Programmable Controller. The Microapplication Card and **LAN** are described in the Local Area Network Users Guide, **GEK-85789**.

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OPTIONAL PROCESS INTERFACE CARD

This optional card contains a mixture of analog and digital interfaces between the Control Card and the equipment. The optional functions on this card are:

- Analog (AC or DC) tachometer interface.
- Analog speed meter driver (analog tachs only).
- Digital (reluctance or square wave) tachometer interface.
- Frequency (pulse train) reference interface.
- Process follower (current or voltage reference) interface.
- Digital voltmeter probe.
- RS232C and RS422 serial link interface.

OPTIONAL TERMINAL BOARD CARDS (3TB and 4TB)

These optional cards provide convenient termination points for control interface to the drive. Connections to the Control, Power Supply, and optional Process Interface Card are made thru **3TB**, which also contains an optional signal-level-detector (**SLD5**). Connections to the optional Application Card are made thru **4TB**. In addition, three 115 VAC relays are available on this card.

OPTIONAL PROGRAMMER MODULE

The Programmer is available as either a "**Local**" Programmer, mounted on the Control Card or as a "**Hand-held**" Programmer, which is plugged into **18PL** on either the Control Card or Diagnostic Readout Card. The Programmer contains a 10-digit fluorescent alphanumeric display and keypad for communicating with the drive.The Programmer has three modes:

- **Operate Mode** provides all necessary functions to run the drive and displays the drive status fault messages.
- **Parameter Mode** allows examination and changes, if necessary, of the system operating parameters.
- **Diagnostic Mode** allows the user to monitor drive values while in operation, self test the drive, and access diagnostic **RUN** and **STATIC** Mode. Diagnostic Mode also provides a number of tests, including optional Drive Self Tune features. Refer to "Use of Optional Programmer Module" Section for further information.

SERVICE AND PARTS INSTRUCTION

Each drive contains instructions placed inside the door to provide information on connection point and component locations essential in connecting or troubleshooting the drive. These instructions contain the following information:

- -- Card location
- -- Power module locations
- -- Fuse information and nomenclature
- -- Plug locations
- -- Incoming control connections at 3TB and 4TB
- -- Power connections

NOTES:

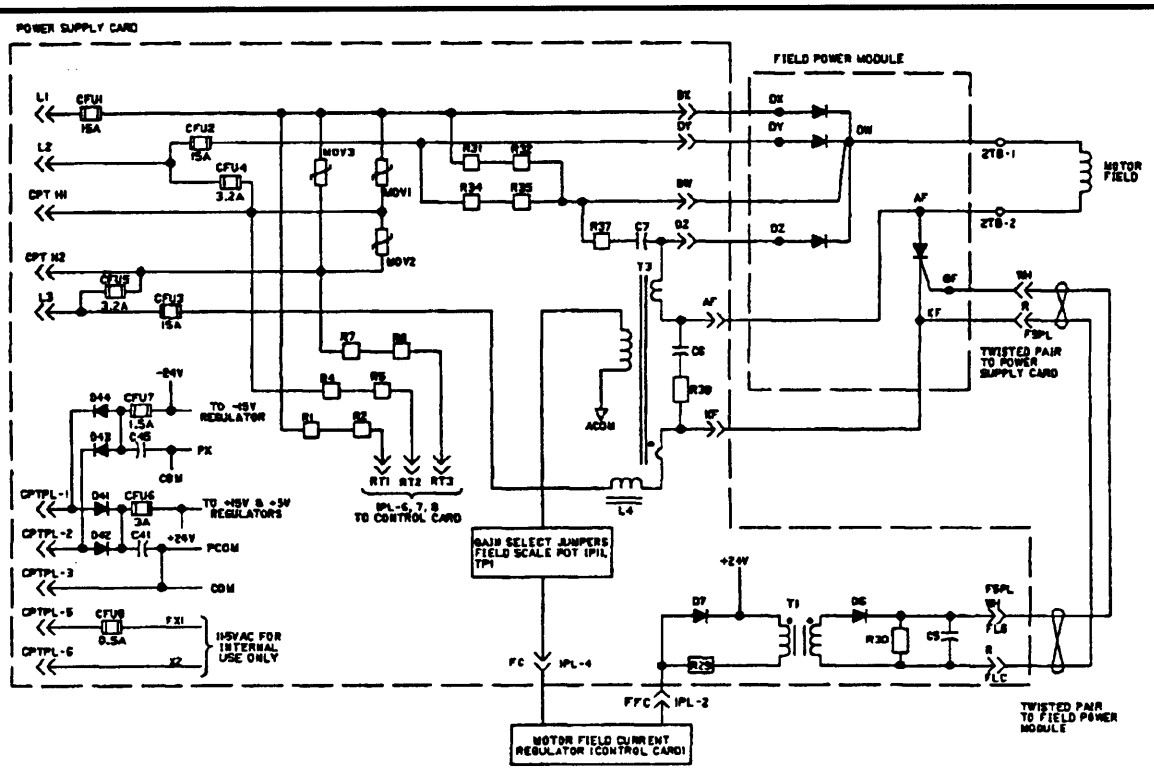
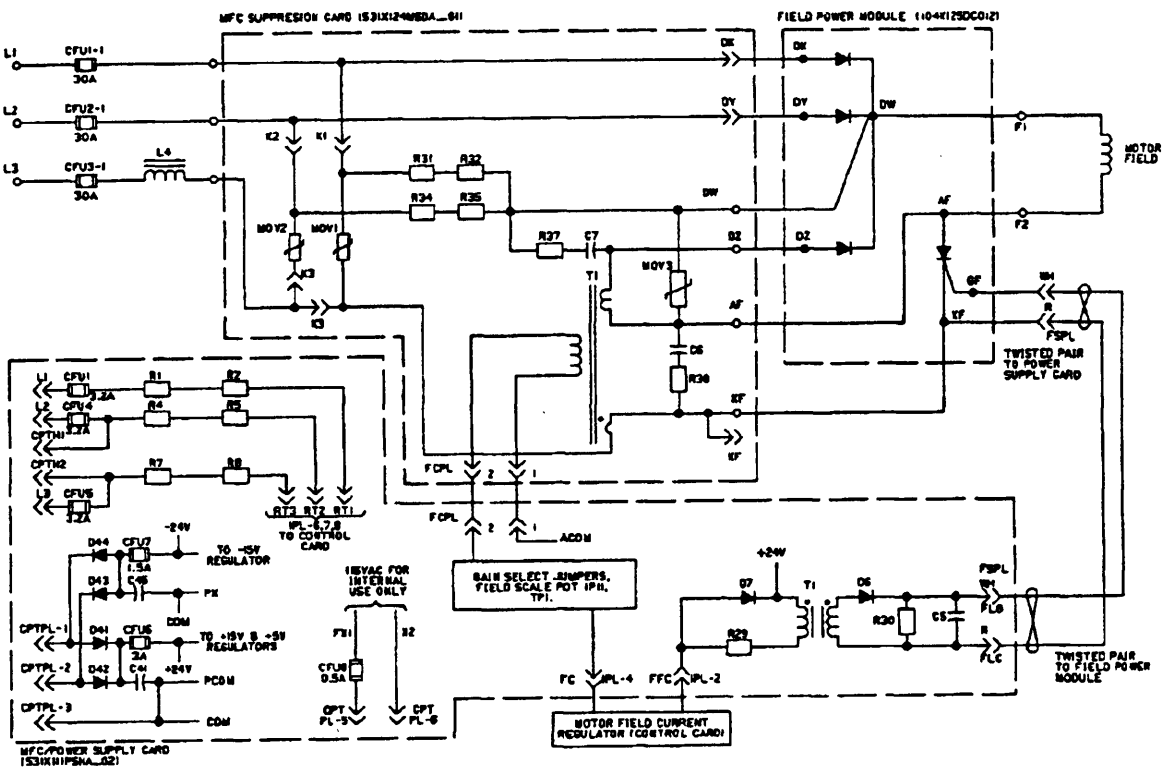


Figure 4-2. MOTOR FIELD CONTROL (10 AMP)



NOTE: This diagram shows the 531X124MFC Suppression Card that was used until 1/92. See Figure 4-3A for the DS200SSBAG1A card that was used from 1/92 until 6/94 and Figure 4-3B for the DS200SSBAG1B card that has been used since 6/94.

Figure 4-3. MOTOR FIELD CONTROL (24 AMP)

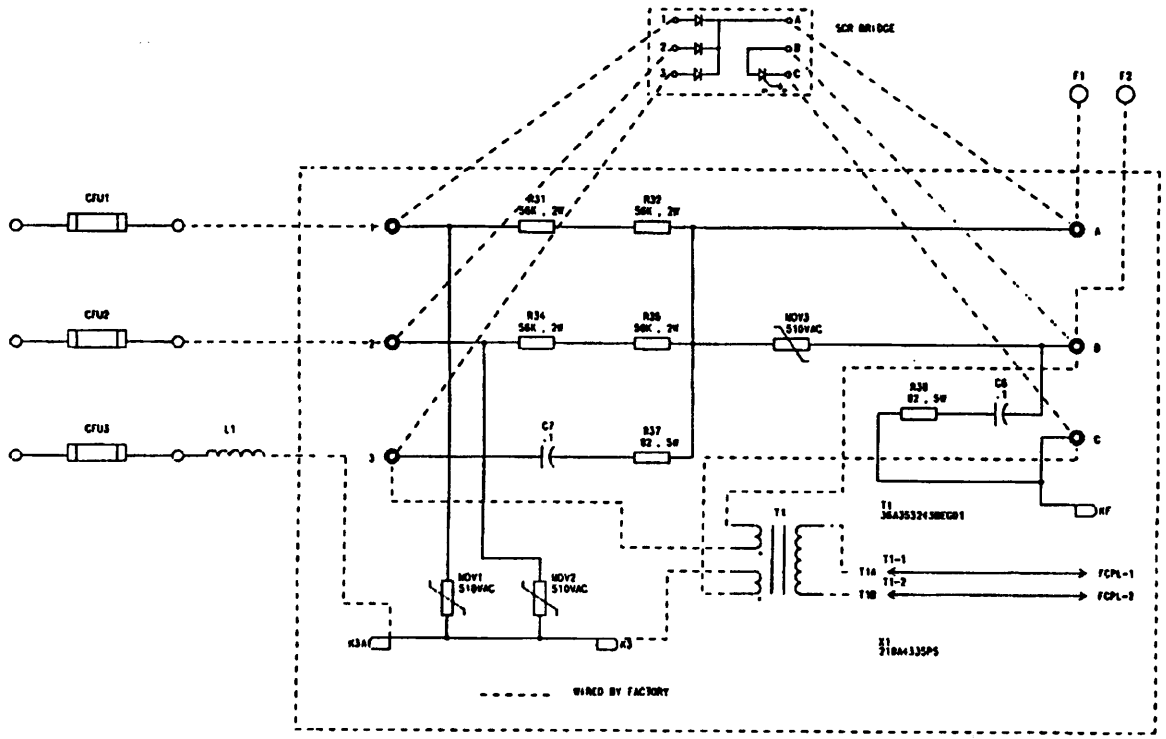


Figure 4-3A. DS200SSBAG1A SUPPRESSION CARD ELEMENTARY DIAGRAM

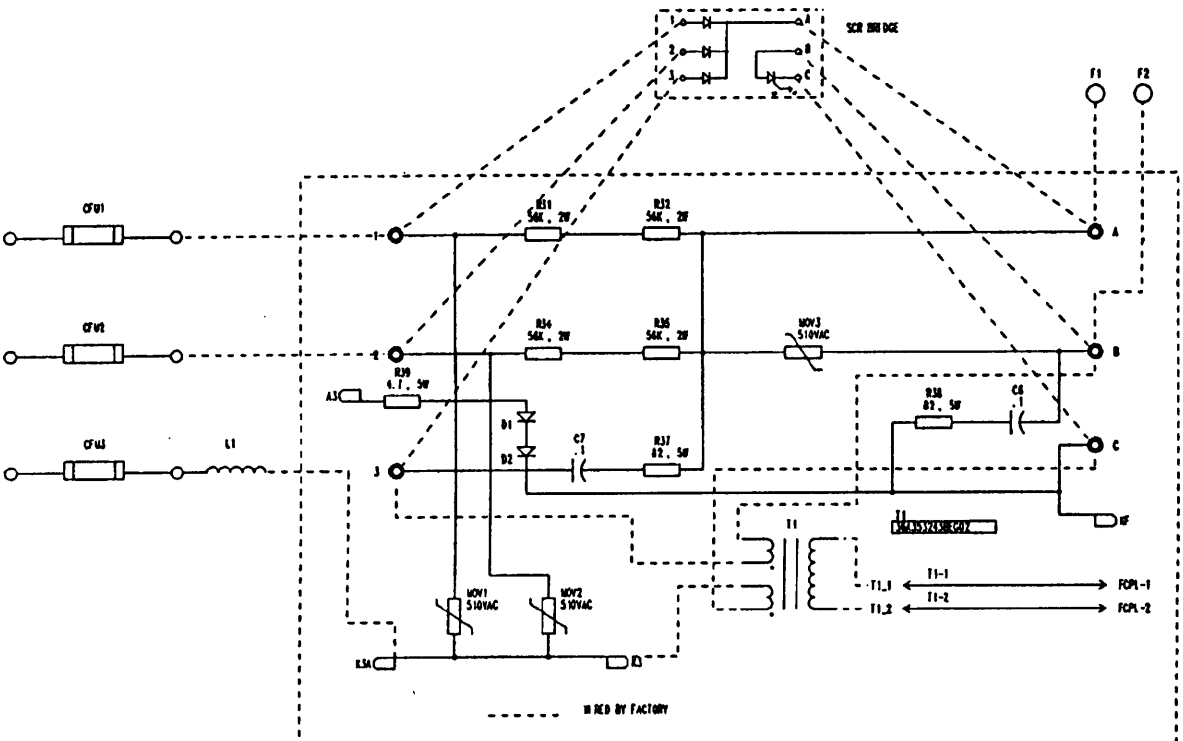


Figure 4-3B. DS200SSBAG1B SUPPRESSION CARD ELEMENTARY DIAGRAM

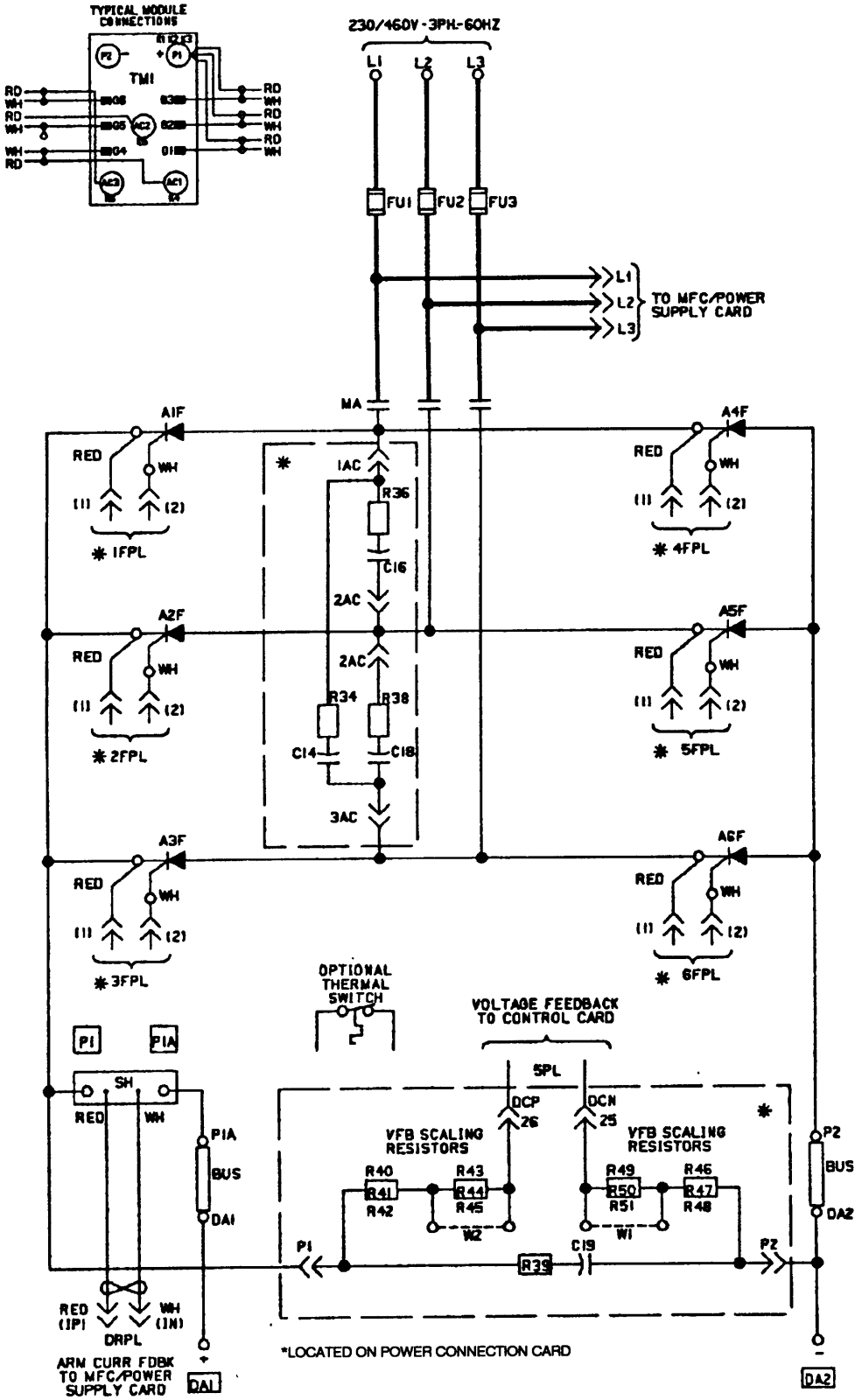


Figure 4-4. SCR POWER CONVERSION BRIDGE NON-REGENERATIVE (A FRAME*) (Pre 1/87 SCR Module)

** "A" and "B" Frames no longer available as of 12/93 (replaced by "G" Frame).*

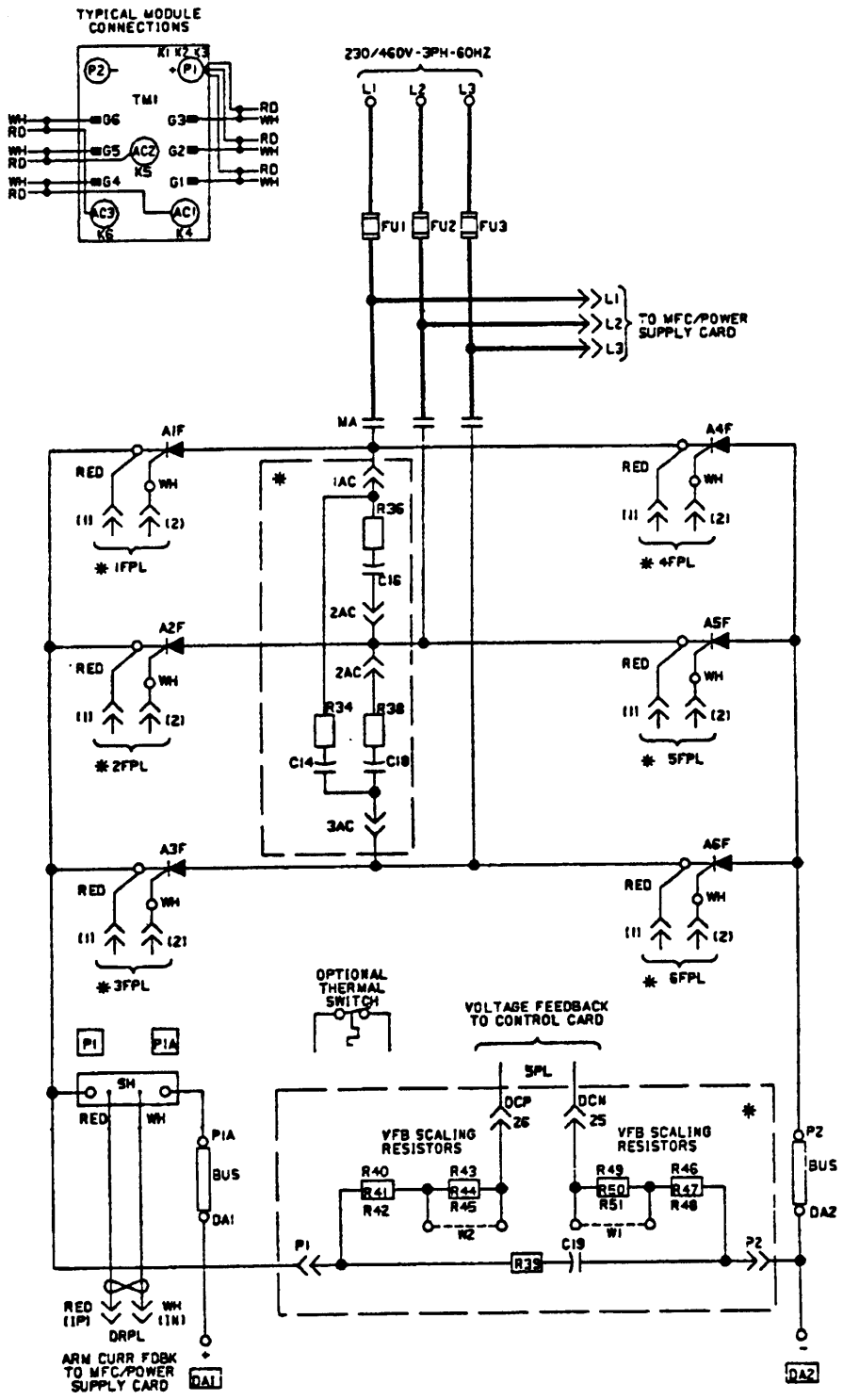


Figure 4-5. SCR POWER CONVERSION BRIDGE
NON-REGENERATIVE (A FRAME*) (1/87 and Later SCR Module)

** "A" and "B" Frames no longer available as of 12/93 (replaced by "G" Frame).*

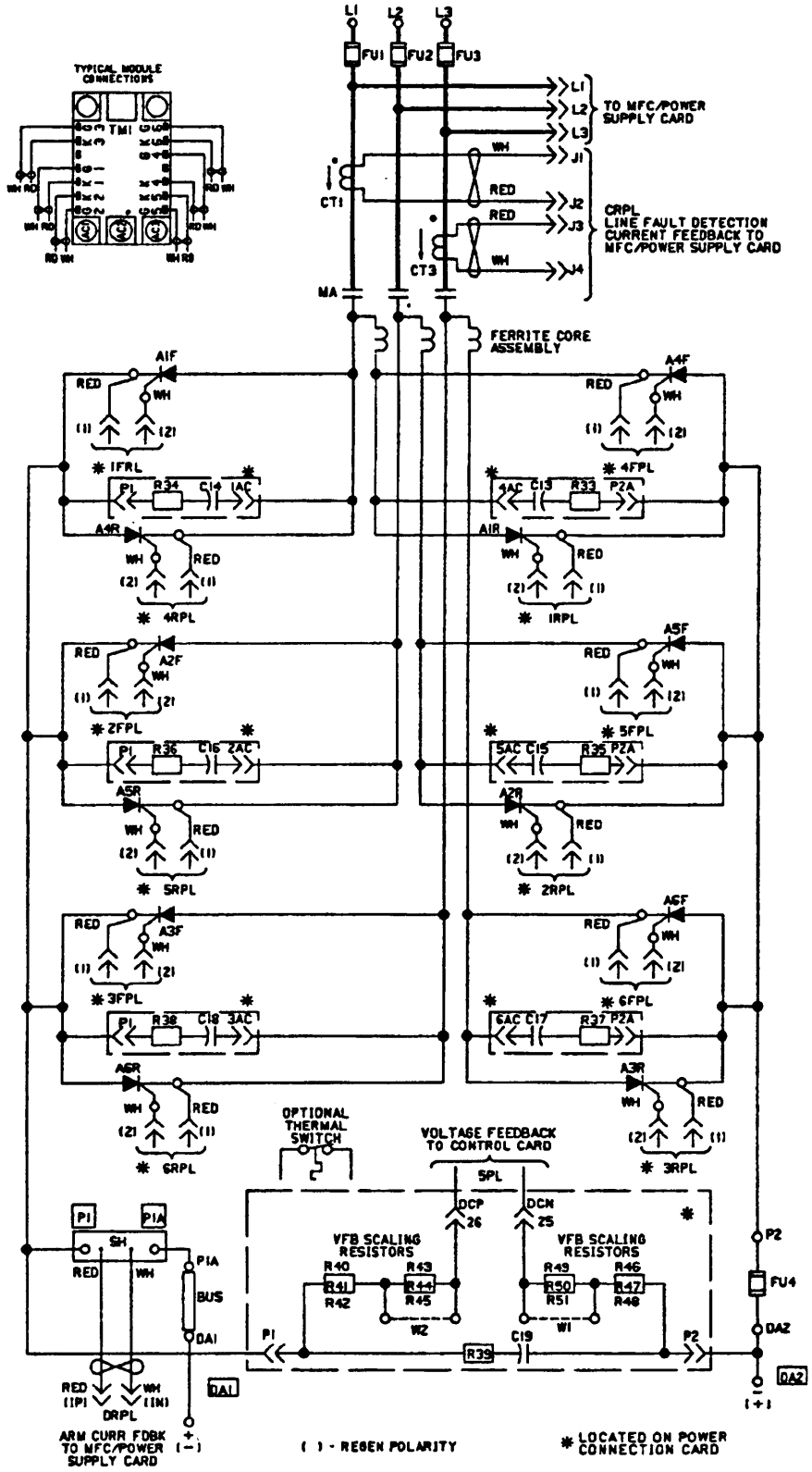


Figure 4-6. SCR POWER CONVERSION BRIDGE REGENERATIVE (A FRAME*) (Pre 1/87 SCR Module)

** "A" and "B" Frames no longer available as of 12/93 (replaced by "G" Frame).*

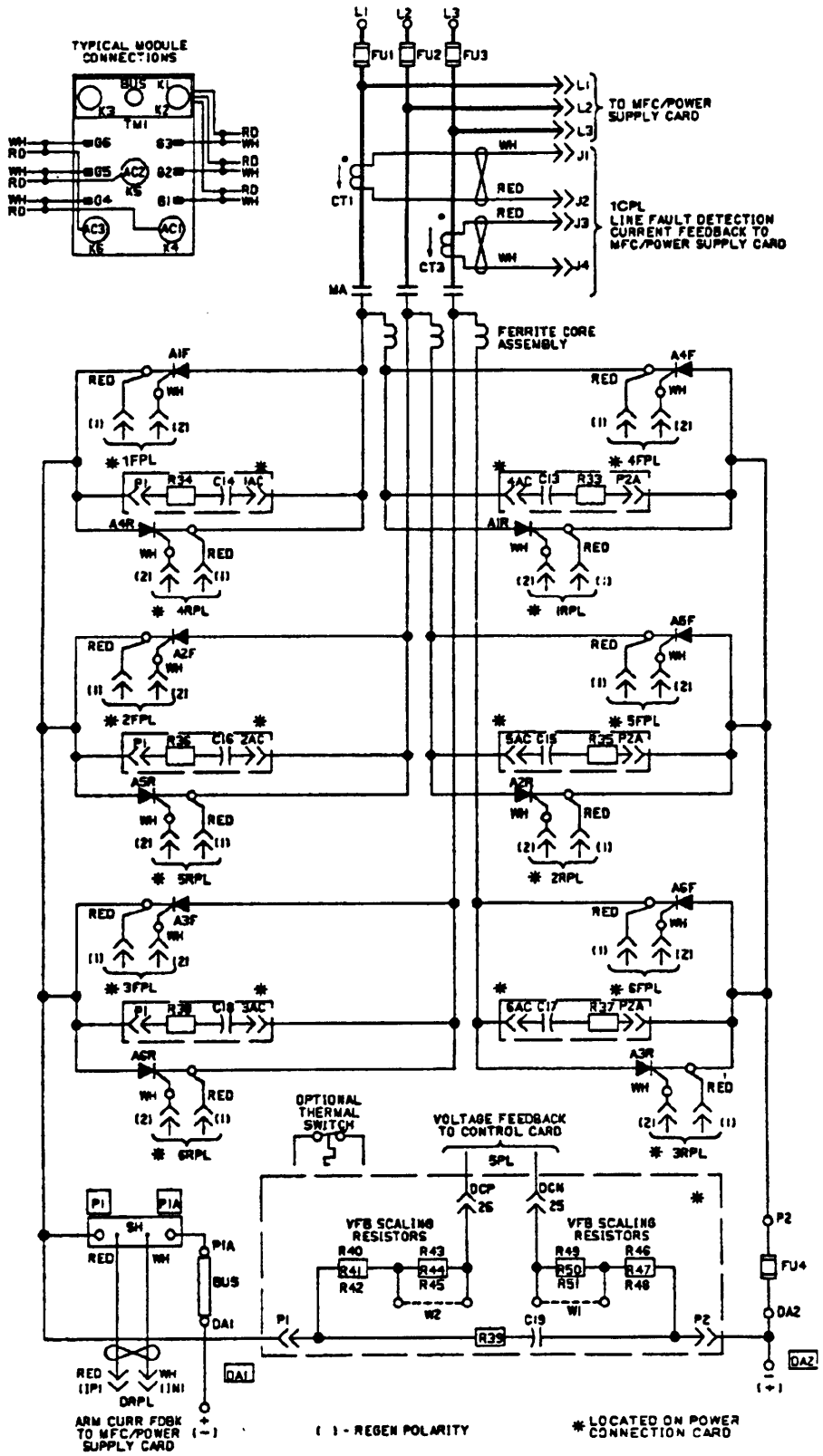


Figure 4-7. SCR POWER CONVERSION BRIDGE REGENERATIVE (A FRAME*) (1/87 and Later SCR Module)

** "A" and "B" Frames no longer available as of 12/93 (replaced by "G" Frame).*

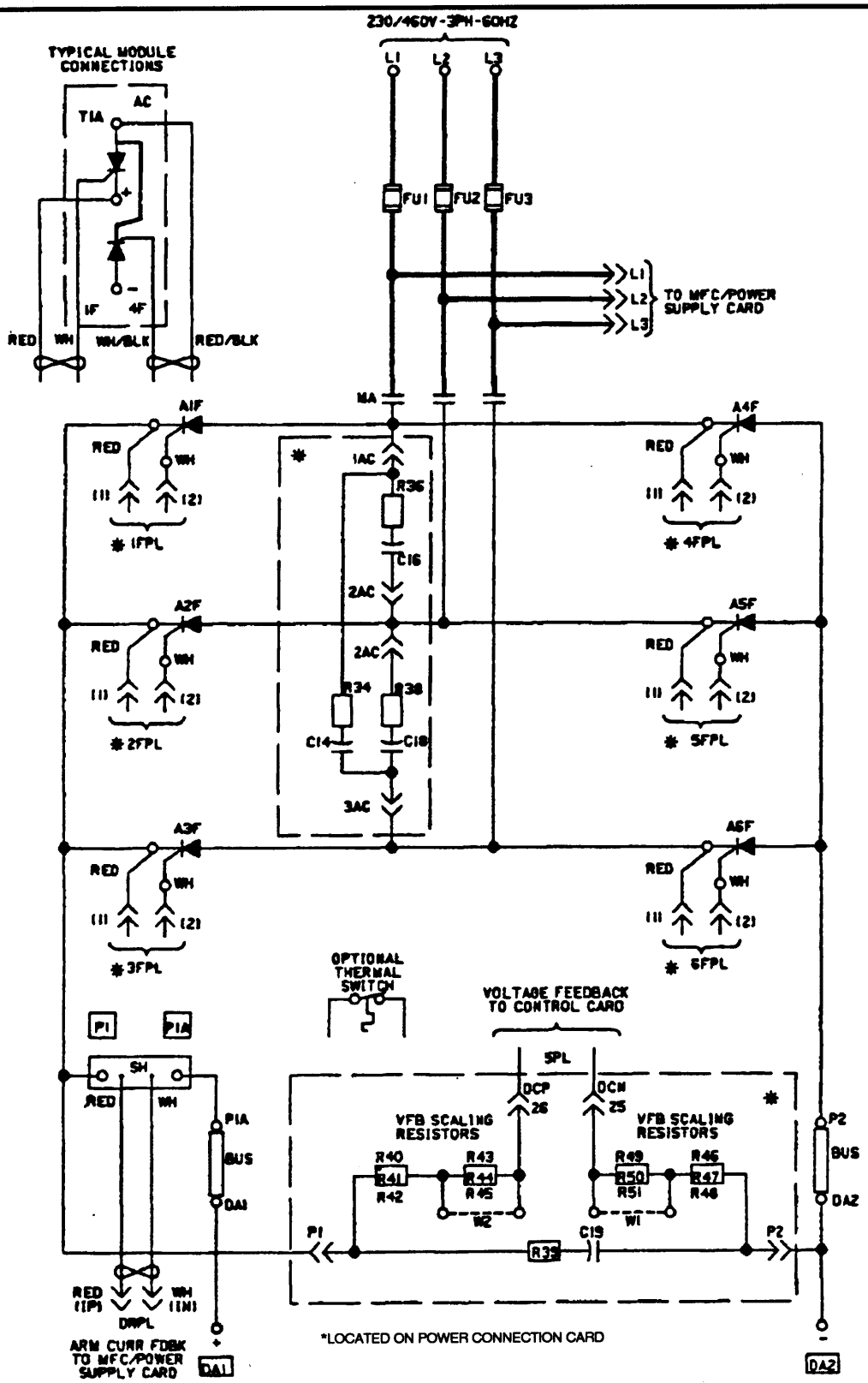


Figure 4-8. SCR POWER CONVERSION BRIDGE NON-REGENERATIVE (B*, G, & C FRAMES)

** "A" and "B" Frames no longer available as of 12/93 (replaced by "G" Frame).*

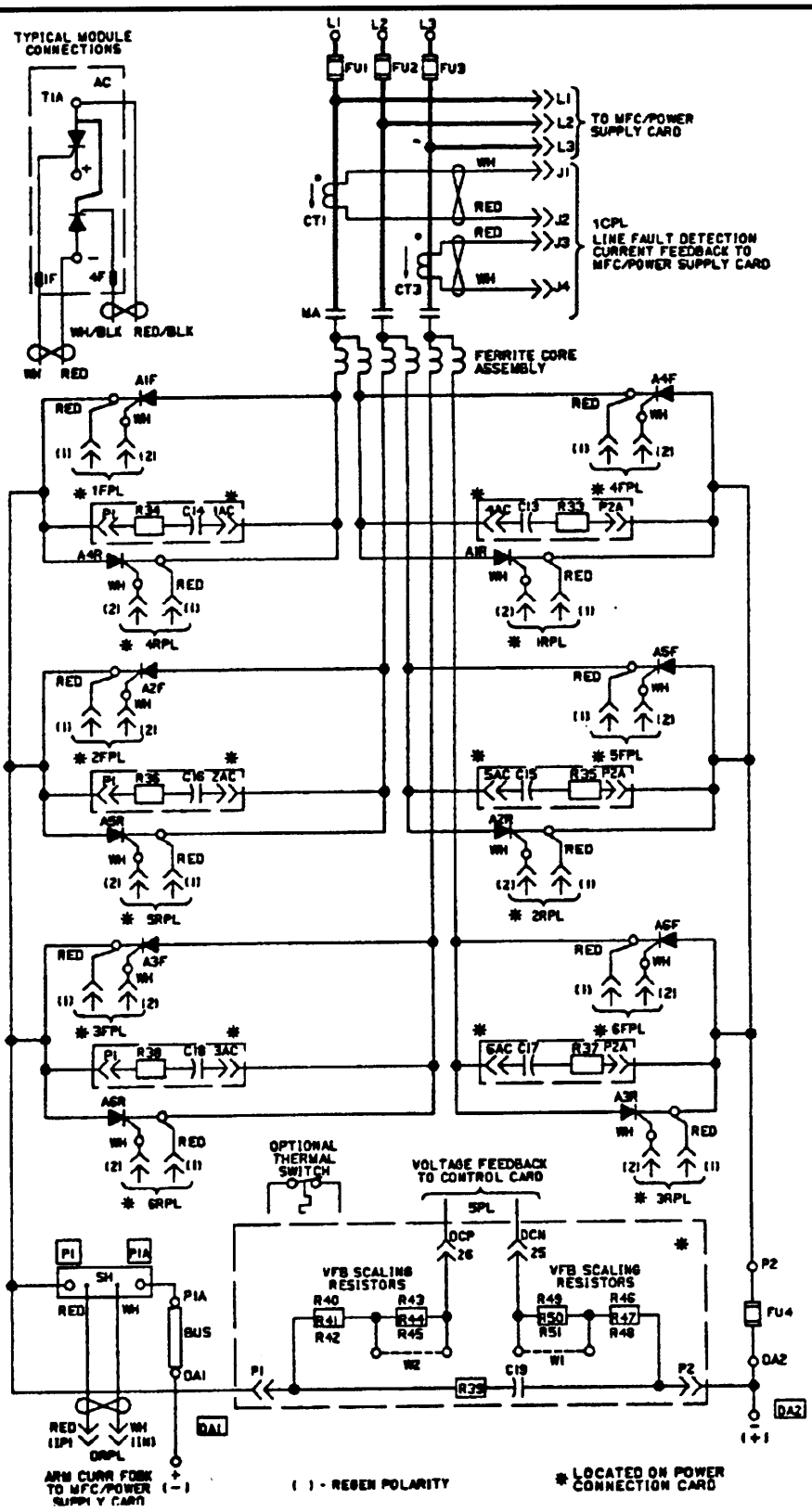


Figure 4-9. SCR POWER CONVERSION BRIDGE REGENERATIVE (B*, G, & C FRAMES)

** "A" and "B" Frames no longer available as of 12/93 (replaced by "G" Frame).*

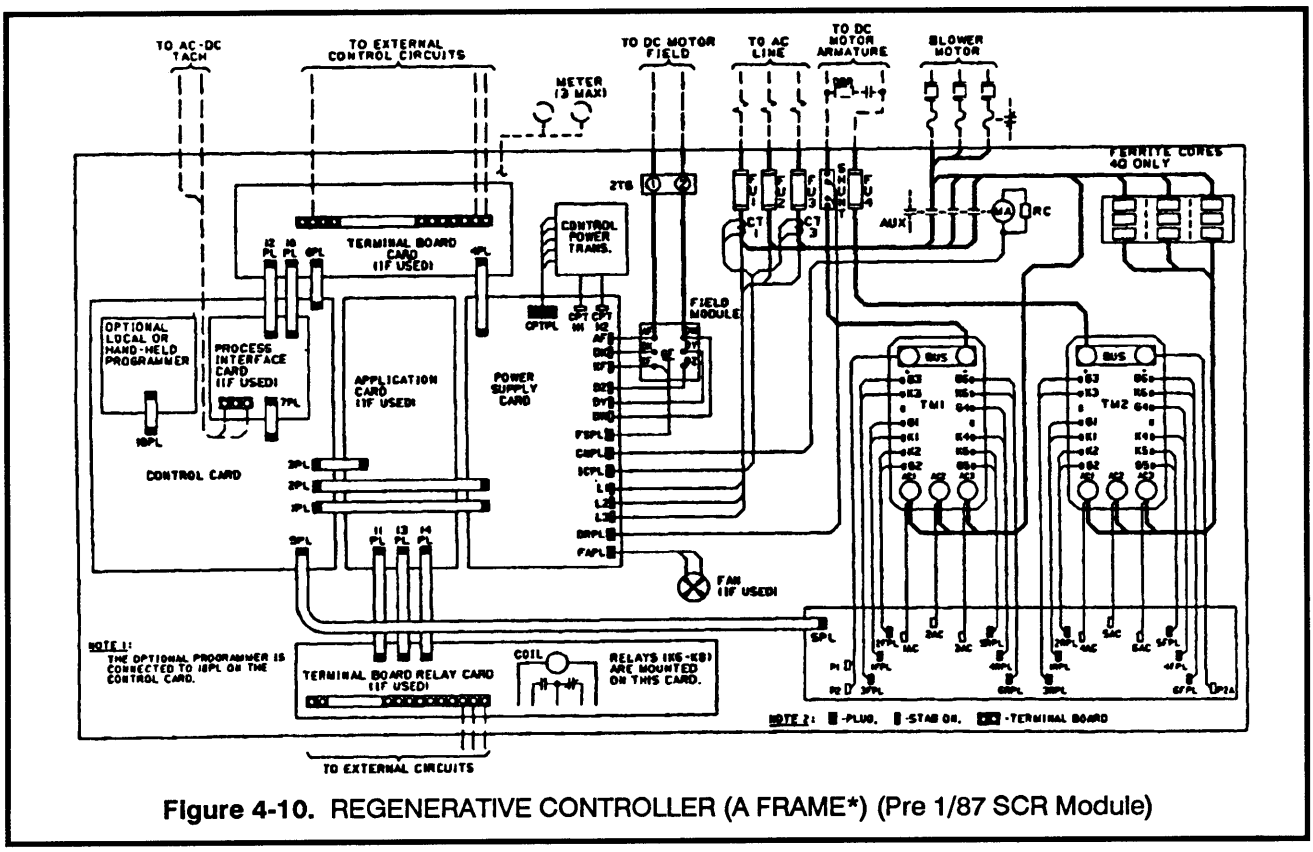


Figure 4-10. REGENERATIVE CONTROLLER (A FRAME*) (Pre 1/87 SCR Module)

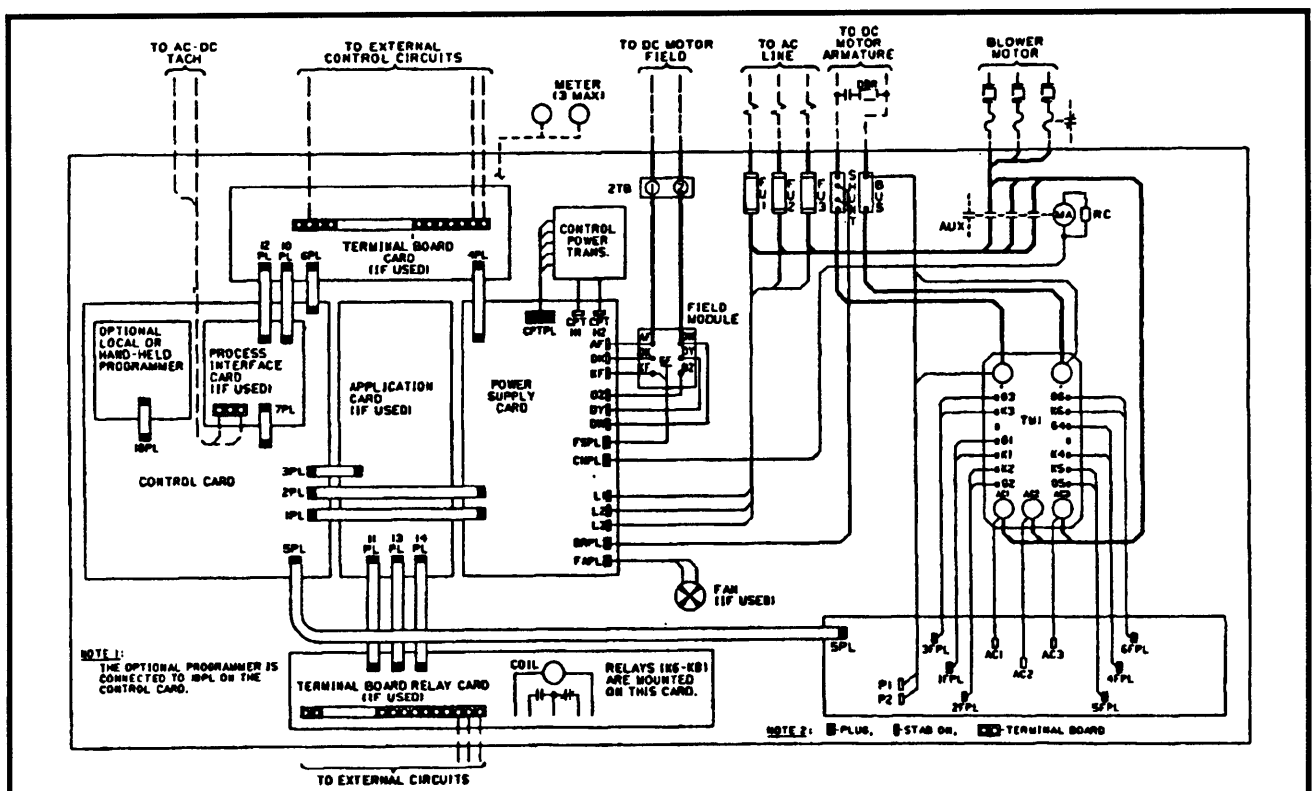


Figure 4-11. NON-REGENERATIVE CONTROLLER (A FRAME*) (Pre 1/87 SCR Module)

** "A" and "B" Frames no longer available as of 12/93 (replaced by "G" Frame).*

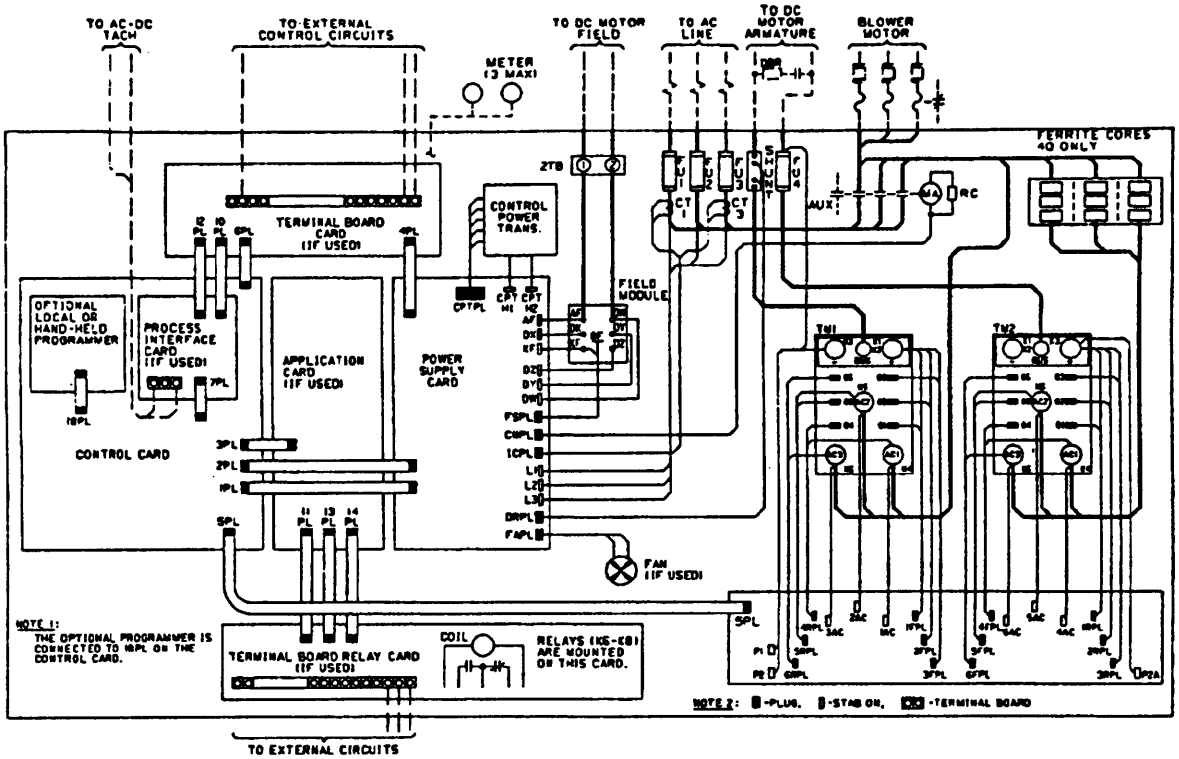


Figure 4-12. REGENERATIVE CONTROLLER (A FRAME*) (1/87 and Later SCR Module)

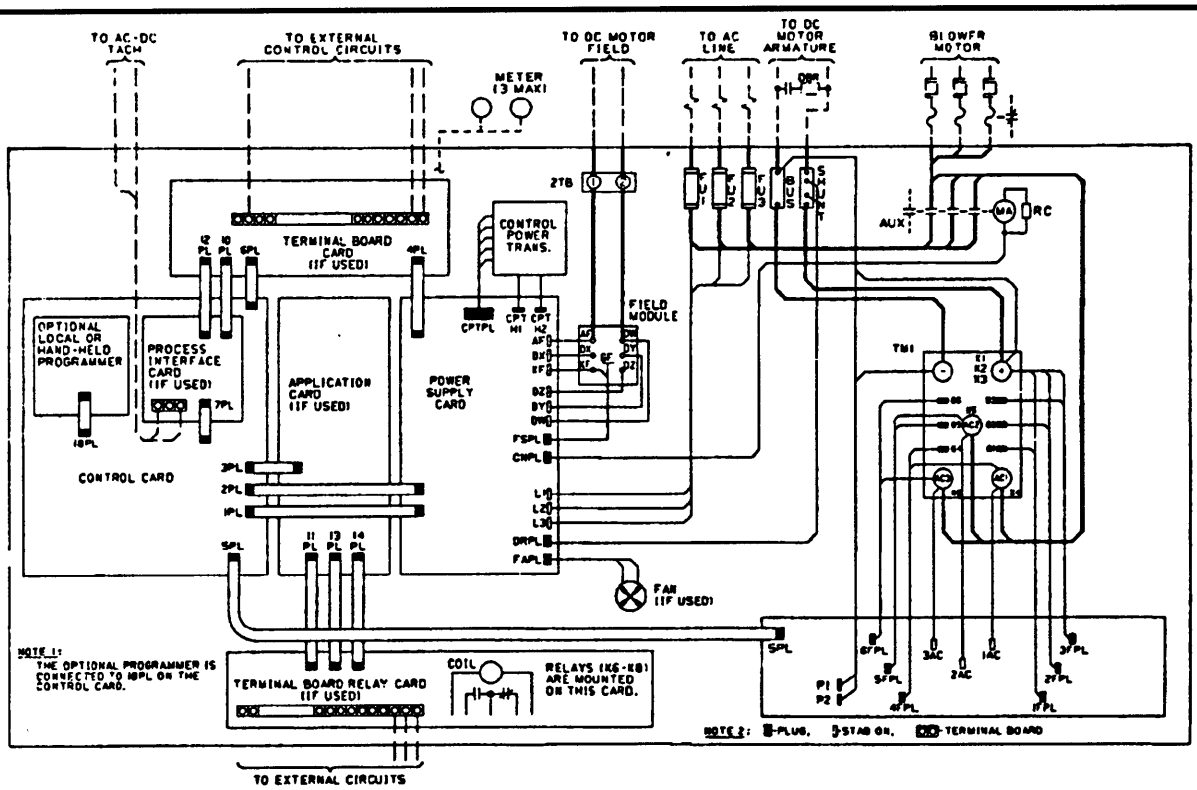


Figure 4-13. NON-REGENERATIVE CONTROLLER (A FRAME*) (1/87 and Later SCR Module)

** "A" and "B" Frames no longer available as of 12/93 (replaced by "G" Frame).*

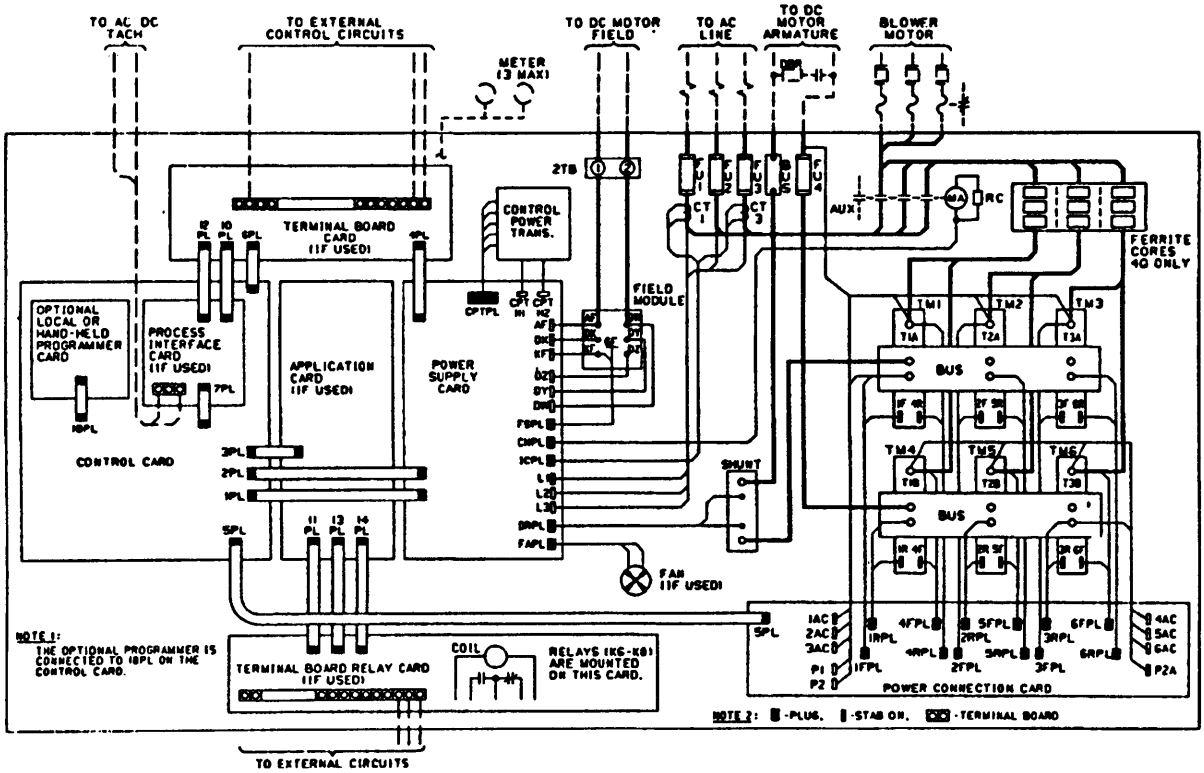


Figure 4-14. REGENERATIVE CONTROLLER (B*, G, & C FRAME)

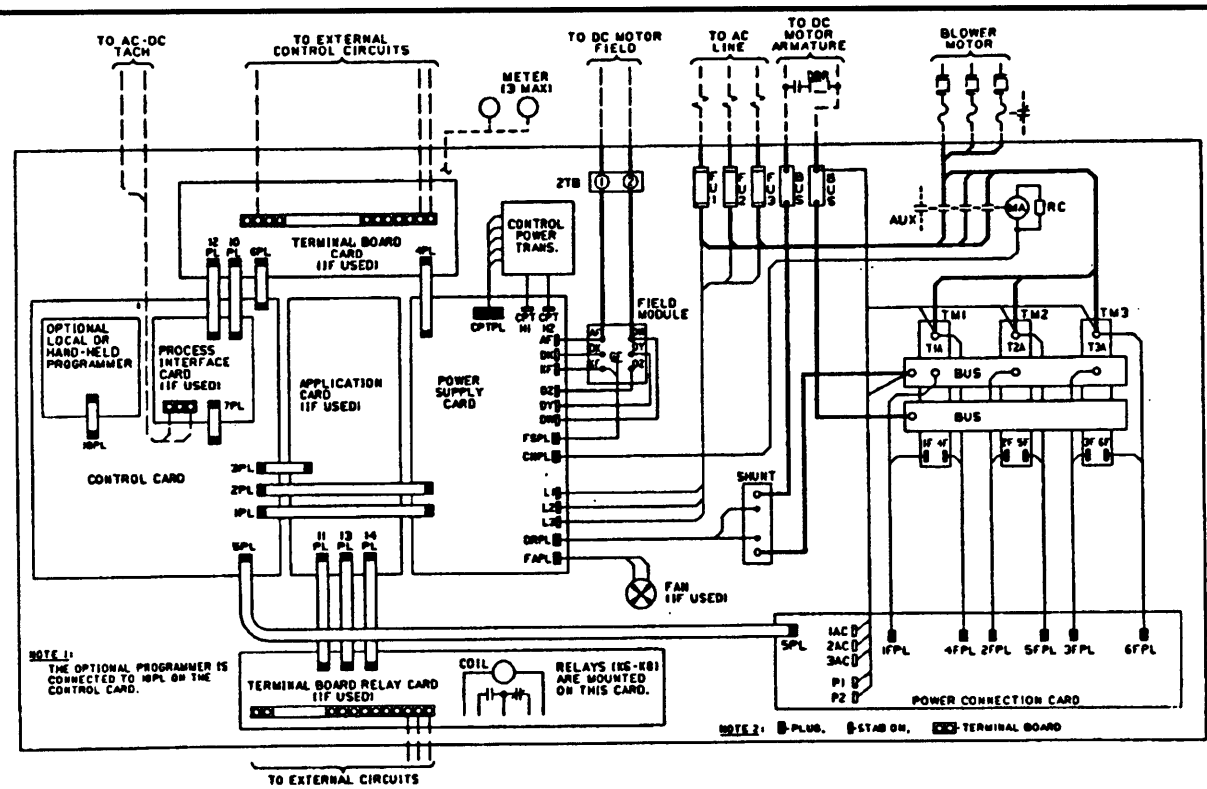


Figure 4-15. NON-REGENERATIVE CONTROLLER (B*, G, & C FRAME)

** "A" and "B" Frames no longer available as of 12/93 (replaced by "G" Frame).*

NOTES:

5. ELECTRICAL CONNECTOR PIN CHARTS

This section contains electrical connector pin charts for all connectors in the controller. The major connectors are listed in numerical order and any additional smaller connectors are listed at the end of the section under the heading **"ADDITIONAL CONNECTORS"**.

CONNECTOR 1PL -- BETWEEN CONTROL CARD AND POWER SUPPLY CARD (PSC)

Pin	Nomen-clature	Description
1	DCOM	Digital Common.
2	FFC	Field firing control--drives field gate pulse transformer.
3	DCOM	
4	FC	Field current feedback from PSC.
5	DCOM	
6	RT1	AC phase 1 input from PSC.
7	RT2	AC phase 2 input from PSC.
8	RT3	AC phase 3 input from PSC (RT1, RT2, & RT3 are resistance isolated).
9	FRC	Fault relay control - Drives the 24 volt fault relay on PSC.
10	-24V	Unregulated -24V return path for FRC and RP.
11	RP	MA contactor pilot relay driver (MAX - K2 on PSC).
12	RUN	Drive run signal from RUN (K1) relay contact on PSC.
13	REF24	Supplies +/-24V to RUN (K1) relay on PSC depending on setting of jumper JP20.
14	W5V	Watchdog Reference - Provides 5V reference from PSC to undervoltage detector on Control Card.
15	J1	Input from line phase 1 CT.
16	J2	Input from line phase 1 CT.
17	J3	Input from line phase 3 CT.
18	J4	Input from line phase 3 CT.
19	DCOM	
20	CFB	Armature current feedback input from PSC.

CONNECTOR 2PL -- BETWEEN CONTROL CARD AND POWER SUPPLY CARD (PSC)

Pin	Nomen-clature	Description
1	ACOM	Analog signal return for +/-15V.
2	-15V	-15V regulated analog power supply input.
3	+15V	+15V regulated power supply input.
4	+24V	+24V unregulated raw power supply input.
5	-24V	-24V unregulated raw power supply input.

CONNECTOR 2PL -- BETWEEN CONTROL CARD AND POWER SUPPLY CARD (PSC) (continued)

Pin	Nomen-clature	Description
6	+5V	+5V regulated digital power supply input.
7	DCOM	Digital signal return for +5V.
8	+5V	
9	PCOM	Raw power supply return for +/-24V.

CONNECTOR 3PL -- BETWEEN CONTROL CARD AND OPTIONAL APPLICATION CARD

Pin	Nomen-clature	Description
1	D0	U4 data bus bit 0 (least significant bit).
2	D1	U4 data bus bit 1.
3	DCOM	Digital common.
4	D2	U4 data bus bit 2.
5	D3	U4 data bus bit 3.
6	D4	U4 data bus bit 4.
7	DCOM	
8	D5	U4 data bus bit 5.
9	D6	U4 data bus bit 6.
10	D7	U4 data bus bit 7.
11	DCOM	
12	A8	U4 address bus bit 8.
13	A9	U4 address bus bit 9.
14	A10	U4 address bus bit 10.
15	DCOM	
16	A11	U4 address bus bit 11.
17	SEL	Signal from U3. Used for Microapplication Card synchronization.
18	ALE	Address latch from U4.
19	DCOM	
20	/RD	Read control line from U4.
21	/WR	Write control line from U4.
22	A15	U4 address bus bit 15, used for Application Card I/O select.
23	RST	Card reset signal for Application Card.
24	+5V	+5V regulated digital supply to Application Card.
25	+5V	
26	+5V	
27	PCOM	+/-24V power supply return.
28	+24V	+24V non-regulated raw power supply to Application Card.
29	-24V	-24V non-regulated power supply to Application Card.
30	DM4	Provides "dummy" connection to Application Card.
31	+15V	+15V regulated analog supply to Application Card.
32	-15V	-15V regulated analog supply to Application Card.
33	ACOM	Analog signal return for +/-15V.
34	OPTA	Input to A-D converter from Application Card multiplexer.

CONNECTOR 4PL -- BETWEEN POWER SUPPLY CARD (PSC) AND OPTIONAL 3TB

Pin	3TB Term. #	Nomen-clature	Description
1	46	SPRNO	Fault Relay Normally Open Contact
2	47	SPRCM	Fault Relay Common Connection
3	48	SPRNC	Fault Relay Normally Closed Contact
4	49	START	Connection Point for Start PB to "RUN" relay (K1)
5	50	PBCM	Common Point for Start and Stop Pushbuttons
6	51	STOP	Connection for Stop Pushbutton
7	52	MAXCM	MA Auxiliary Relay Common Connection
8	53	MAXNO	MA Auxiliary Relay Normally Open Contact
9	54	MAXNC	MA Auxiliary Relay Normally Closed Contact
			NOTE:*Contact rating K2 relay is 1.2 amp at 115 volts AC or 28 volts DC. Contact rating K3 relay is 1 amp at 115 volts AC or 28 volts DC.*

CONNECTOR 5PL -- BETWEEN CONTROL CARD AND POWER CONNECTION CARD

Pin	Nomen-clature	Description
1	A6F	Drives cell 6F gate pulse transformer.
2	DFP	Delayed firing power for A6F and A5F.
3	A5F	Drives cell 5F gate pulse transformer.
4	PCOM	Shield wire.
5	A4F	Drives cell 4F gate pulse transformer.
6	DFP	Delayed firing power for A4F and A3F.
7	A3F	Drives cell 3F gate pulse transformer.
8	PCOM	Shield wire.
9	A2F	Drives cell 2F gate pulse transformer.
10	DFP	Delayed firing power for A2F and A1F.
11	A1F	Drives cell 1F gate pulse transformer.
12	PCOM	Shield wire.
13	A1R	Drives cell 1R gate pulse transformer.
14	DFP	Delayed firing power for A1R and A2R.
15	A2R	Drives cell 2R gate pulse transformer.
16	PCOM	Shield wire.
17	A3R	Drives cell 3R gate pulse transformer.
18	DFP	Delayed firing power for A3R and A4R.
19	A4R	Drives cell 4R gate pulse transformer.
20	PCOM	Shield wire.
21	A5R	Drives cell 5R gate pulse transformer.
22	DFP	Delayed firing power for A5R and A6R.
23	A6R	Drives cell 6R gate pulse transformer.
24	PCOM	Shield wire.
25	DCN	Resistance isolated armature voltage.
26	DCP	Resistance isolated armature voltage.

CONNECTOR 6PL -- BETWEEN CONTROL CARD AND OPTIONAL 3TB

Pin	3TB Term. #	Nomen-clature	Description
1	69	RSET	A hard reset of drive is initiated when RSET is connected to +15V. Must be open to run. RSET stops firing of all SCRs, removes delayed firing power, drops out the MA contactor and fault relay, and holds all microprocessors in a reset state until RSET is opened. The drive will not automatically restart if reset while running. Do not apply RSET while running.
2	68	+5V	Not for use external to drive.
3	67	ACOM	Not for use external to drive.
4	66	+15V	Signal voltage for System Reference pot, RSET circuit and others.
5	65	MSR	Manual system reference input to drive, ±15V max. Positive for forward motoring with POL at zero or open.
6	64	-15V	Signal voltage for System Reference pot and others.
7	63	JOGR	Jog reference input (can be selected to bypass linear time), ±14V = Top speed. Programmable to 7V = top speed for fast update rate. Refer to Custom Software Description.
8	62	CLA	Current limit adjust - Adds to software setting of current limit. +10V increases current limit by 100% of rated, -10V decreases it by 100%.
9	61	MUP1	Multiuse point 1 - connects to connection test point MUP1, which may be jumpered to another connection point. **Example:** DA0 may be jumpered to MUP1 for output on 3TB.
10	60	MUP2	Multiuse point 2 - connects to connection test point MUP2, which may be jumpered to another connection point.
11	59	MUP3	Multiuse point 3 - connects to connection test point MUP3, which may be jumpered to another connection point.
12	58	RUN	Alternate run command input point in place of "RUN" relay. Do not pull high if run relay is also being used. Drive runs when connected to 3TB56 (6PL-14), REF24. With JP19 and JP20 on the Control Card in the 1-2 and 3-4 positions respectively, input is active (controller runs) when RUN is pulled down to -24V DC. Input is inactive (controller stops) when RUN is open or driven to +24 VDC. With JP19 and JP20 in the 1-3 and 2-4 positions respectively, input is active (controller runs) when RUN is driven to +24V DC. Input is inactive (controller stops) when RUN is open or pulled down to -24 VDC.
13	57	JOG	Jog command input point. Drive jogs when connected to 3TB56 (6PL-14), REF24. Polarity controlled same as RUN above.
14	56	REF24	Selectable + or -24V supply for RUN, JOG, POL or XTSP input. Polarity set by JP19 and JP20 on Control Card.

CONNECTOR 6PL -- BETWEEN CONTROL CARD AND OPTIONAL 3TB (contd.)

Pin	3TB Term. #	Nomen-clature	Description
15	55	POL	Control input which reverses the direction of the drive for any applied reference; e.g. activating POL low is equivalent to reversing the polarity of the input at MSR. Activated/deactivated same as RUN above. Programmable for multiple functions. Refer to Custom Software Description and elementaries if used.
16	1	XSTP	Must be activated to enable drive. Activated/deactivated same as RUN described previously. Causes a normal, coast, quick or trip stop when open (see ADR004-STPJP in the Custom Software Description).
17	2	PCOM	Not for use external drive.
18	3	CTLN1	Control On - Must be connected to CTLN2 to allow the drive to run. Opening this connection unconditionally drops MA contactor after a 50 millisecond attempt by microprocessor to precondition the drive.
19	4	CTLN2	See CTLN1.
20	5	IMET	Analog load meter output proportional to armature current, adjustable for 1 to 8V DC output at rated load current. See P2 and JP15 on the Control card (TABLE 1) for details. Load not to exceed 1000 ohms/volt.
21	6	REF24	Selectable $\pm$24V supply for RUN, JOG, POL or XSTP input. Polarity set by JP19 and JP20 on the Control card.
22	7	INTR	Programmable 24V drive from Control card to the optional signal level detector SLD5 on 3TB terminal board. Controls contacts at 3TB70, 71 and 72. See JP1 on the 3TB Terminal Board card (TABLE 1) for details.
23	8	VMET	Analog armature voltmeter output. This output is bipolar, normally 4 volts at rated armature voltage output. (See P8 on the Control Card, TABLE 1, for details.) Normal load is 1 milliamp.
24	9	FCMET	Analog field current meter output, 1 to 5 VDC output. See P5 on the Control Card (TABLE 1) for details. Load not to exceed 1000 ohms/volt.
--	26	K5A	Optional SLD5 control input from REF24 for external control of SLD5. Controls contacts at 3TB70, 71, and 72. See JP1 on the 3TB Terminal Board Card (Table 1) for details.
--	70	S5NO	Optional SLD5 normally open contact.
--	71	S5C	Optional SLD5 common contact.
--	72	S5NC	Optional SLD5 normally closed contact.

CONNECTOR 7PL -- BETWEEN CONTROL CARD AND OPTIONAL PROCESS INTERFACE CARD

Pin	Nomen-clature	Description
1	ACOM	Analog signal return.
2	DVM	Output from digital voltmeter.
3	SFB	Output from analog tach circuit.
4	ASFB	Amplified signal from analog tach circuit, used for improved resolution at low speed operation.
5	ACOM	
6	-15V	-15V supply to analog circuits on Process Interface (PI) Card.
7	+15V	+15V supply to analog circuits on PI Card.
8	+5V	+5V supply to logic circuits on PI Card.
9	+5V	
10	TRS	RS422 driver tri-state command to the PI Card.
11	DCOM	Digital signal return.
12	DIR	Signal from digital tach interface signalling direction of motor rotation when a quadrature track tach is used.
13	TPN	Net pulse train signal from digital tach or AC analog tach interface used to determine tach speed.
14	PRF	Pulse train from digital reference interface used to determine desired reference when used.
15	DCOM	
16	DCOM	
17	RXD	Input from either the RS422 or RS232C serial link receiver from the PI Card.
18	TXD	Output to the RS422 and RS232C serial link drivers from the PI Card.
19	/DTR	Data terminal ready - Output indicating the PI Card is ready to receive a byte on the serial link. Low when ready, high when not.
20	/CTS	Clear to send - Input from serial link signalling when the receiver on the other end of the link is ready to receive a byte. When low, drive may transmit. When high, drive must not send anything out on TXD.
21	RST	Reset line from Control Card to PI Card, active high.
22	+5V	
23	+5V	
24	DCOM	
25	DCOM	
26	-24V	-24V raw power to P.I. card.

CONNECTOR 10PL -- BETWEEN ANALOG PROCESS INTERFACE CARD AND OPTIONAL 3TB

Pin	3TB Term. #	Nomen-clature	Description
1	10	PCN	Optional negative input for process follower current (1-5, 4-20, 5-50, or 5-65mA.) or voltage (0-10 VDC) reference.
2	11	PCP	Optional positive input for process follower reference.
3	12	MR1	Optional reference for manual system reference potentiometer to allow coordinated max speed scaling when PFRF is connected to Control card reference input via auto/manual selection.
4	13	PFRF	Optional process follower interface output (0-10V) htat connects to a speed reference or other analog process input.
5	14	DTB	Optional square wave tach quadrature input (track B high). Operating Voltage: -0.5V to +0.8V is logic 0 and +2.4V to +15V is logic 1.
6	15	/DTB	Optional inverted square wave differential tach quadrature input (track B low).
7	16	DTA	Optional square wave tach quadrature input (track A high). Operating Voltage: -0.5V to +0.8V is logic 0 and +2.4V to +15V is logic 1.
8	17	/DTA	Optional inverted square wave differential tach quadrature input (track A low).
9	18	PREF	Optional pulse reference frequency positive input, 0 to 6KHz Max.
10	19	PRFC	Optional high impedance common for pulse reference input.
11	20	SMET	Analog speed meter output, 10 VDC. See P2 on the P.I. card (TABLE 1).
12	21	ACOM	Common for optional RS232C communications.
13	22	DCOM	Not for use external drive.
14	23	R232	Optional RS232C receive line.
15	24	T232	Optional RS232C transmit line.
16	25	DM5	Spare connection tied to undedicated point DM5.

NOTE: / = NOT; **Example:** /C = Not C

CONNECTOR 11PL -- BETWEEN OPTIONAL APPLICATION CARD AND OPTIONAL 4TB

Pin	4TB Term. #	Nomen-clature	Description
1	30	EBR	No connection (used with Microapplication Card).
2	29	/EBR	No connection (used with Microapplication Card).
3	85	MAP3	Micro-application multiuse point 3 - connects to connection point MAP3, which may be jumpered to another connection point.
4	25	PSB	Strobe for optional 16 bit parallel input on 16PL. See Application Card JP7 (TABLE 1). Inputs must be set 50 microseconds before strobe, and strobe held for 100 microseconds.
5	26	B03	Optional programmable 24V bit output #3 (5V TTL on version APG Card).
6	27	B02	Optional programmable 24V bit output #2 (5V TTL on version APG Card).
7	28	B01	Optional programmable 24V bit output #1 (5V TTL on version APG Card).
8	32	EBA	No connection (used with Microapplication Card).
9	31	/EBA	No connection (used with Microapplication Card).
10	86	EBB	No connection (used with Microapplication Card).
11	87	/EBB	No connection (used with Microapplication Card).
12	33	ENCR	Optional non-inverting differential or single-ended encoder marker pulse input.
13	34	/ENCR	Optional inverting differential encoder marker pulse input.
14	35	ENCB	Optional non-inverting differential or single-ended encoder Track B input.
15	36	/ENCB	Optional inverting differential encoder Track B input.
16	37	ENCA	Optional non-inverting differential or single-ended encoder Track A input.
17	38	/ENCA	Optional inverting differential encoder Track A input.
18	39	FC6	Bit 6 of optional fault code 24V output (5V TTL on version APG Card).
19	40	FC0	Bit 0 of optional fault code 24V output (5V TTL on version APG Card).
20	41	FC1	Bit 1 of optional fault code 24V output (5V TTL on version APG Card).
21	42	FC2	Bit 2 of optional fault code 24V output (5V TTL on version APG Card).
22	43	FC3	Bit 3 of optional fault code 24V output (5V TTL on version APG Card).
23	44	FC4	Bit 4 of optional fault code 24V output (5V TTL on version APG Card).
24	45	FC5	Bit 5 of optional fault code 24V output (5V TTL on version APG Card).

CONNECTOR 12PL -- BETWEEN OPTIONAL PROCESS INTERFACE CARD AND OPTIONAL 3TB

Pin	3TB Term. #	Nomen-clature	Description
1	--	--	No connection.
2	--	--	No connection.
3	--	--	No connection.
4	--	--	No connection.
5	--	--	No connection.
6	--	--	No connection.
7	32	CTSA	Optional inverted RS422 clear-to-send input from serial link signaling when the Control Card may transmit a byte, active high. (EIA standard nomenclature is CTSA.)
8	29	DTRA	Optional inverted RS422 data-terminal-ready output from Control Card signaling when it is ready to receive a byte on the serial link, active high. (EIA standard nomenclature is DTRA.)
9	34	TXA	Optional inverted RS422 transmitted data signal from Control Card to serial link. (EIA standard nomenclature is TXA.)
10	27	RXA	Optional inverted RS422 received data signal from serial link to Control Card. (EIA standard nomenclature is RXA.)
11	36	RXB	Optional non-inverted RS422 received data signal from serial link to Control Card. (EIA standard nomenclature is RXB.)
12	43	TXB	Optional non-inverted RS422 transmitted data signal from Control Card to serial link. (EIA standard nomenclature is TXB.)
13	38	DTRB	Optional non-inverted RS422 data-terminal-ready output from Control Card. (EIA standard nomenclature is DTRB.)
14	41	CTSB	Optional non-inverted RS422 clear-to-send input from serial link. (EIA standard nomenclature is CTSB.)
15	40	RST	Reset signal from Control Card, active high, not for use external to drive.
16	39	+5V	Not for use external to drive.
17	42	+5V	Not for use external to drive.
18	37	DCOM	Not for use external to drive.
19	44	DCOM	Not for use external to drive.
20	45	-24V	Not for use external to drive.

CONNECTOR 13PL -- BETWEEN OPTIONAL APPLICATION CARD AND OPTIONAL 4TB

NOTE: *Refer to the Custom Instruction Book, software jumper description, for definitions of I/O through 13PL for particular application.*

Pin	4TB Term. #	Nomen-clature	Description
1	6	MODE 0	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
2	5	MODE 1	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
3	4	MODE 2	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
4	3	MODE 3	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
5	2	MODE 4	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
6	1	MODE 5	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
7	7	MODE 6	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
8	8	MODE 7	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
9	9	CI0	No connection (used with Microapplication Card).
10	10	CI1	No connection (used with Microapplication Card).
11	11	MAP1	Microapplication multiuse point 1 - connects to connection point MAP1, which may be jumpered to another connection point.
12	12	MAP2	Microapplication multiuse point 2 - connects to connection point MAP2, which may be jumpered to another connection point.
13	13	AN2	No connection (used with Microapplication Card).
14	14	DA12	Optional analog output from 12-bit D-A converter. Range = -10V to +10V, 1 MA, with 450 ohms output impedance. Update rate is 30 to 40 times/second.
15	15	SJ1	Optional analog input to summing junction #1. Operating range = $\pm$10V; maximum input = $\pm$15V.
16	16	SJ2	Optional analog input to summing junction #2. Operating range = $\pm$10V; maximum input = $\pm$15V.
17	17	PSREF	Optional analog position or process reference input ($\pm$10V).
18	18	LTA	Optional analog linear time adjust. Positive voltages increase timing, negative voltages decrease timing. $\pm$10 volts maximum. 10 volts is equal to 66 second adjust.
19	19	DA8	Optional analog output from 8-bit D-A converter. See DA12 for specifications. Normal update rate of 180 times/sec. Reduced to 30 to 40 times/sec if fast update selected for DA0.

**Functions of MODE 0 - MODE 7 are determined by Addresses 000-015 (ADR000 - ADR015) in the software description.*

CONNECTOR 13PL -- BETWEEN OPTIONAL APPLICATION CARD AND OPTIONAL 4TB (continued)

Pin	4TB Term. #	Nomen-clature	Description
20	20	FRA	Optional analog field reference adjust. Positive voltages up to +10V increase field current. Negative voltages down to -10V decrease field current. See Application Card JP32 and P2 (TABLE 1).
21	21	ASP1	Optional spare analog input #1 (programmable).
22	22	ASP2	Optional spare analog input #2 (programmable).
23	23	ASR	Optional auto speed reference input to drive, ±15V maximum, positive for forward cell firing while motoring. ASR is the active speed reference in the AUTO Mode when analog auto reference is selected.
24	24	AN1	No connection (used with Microapplication Card).

CONNECTOR 14PL -- BETWEEN OPTIONAL APPLICATION CARD AND OPTIONAL 4TB

Pin	4TB Term. #	Nomen-clature	Description
1	62	S0NO	Optional SLD #0 Normally Open Contact.
2	63	S0NC	Optional SLD #0 Normally Closed Contact.
3	64	S0C	Optional SLD #0 Common Contact.
4	65	S1NO	Optional SLD #1 Normally Open Contact.
5	66	S1NC	Optional SLD #1 Normally Closed Contact.
6	67	S1C	Optional SLD #1 Common Contact.
7	68	S2NO	Optional SLD #2 Normally Open Contact.
8	69	S2NC	Optional SLD #2 Normally Closed Contact.
9	70	S2C	Optional SLD #2 Common Contact.
10	71	S3NO	Optional SLD #3 Normally Open Contact.
11	72	S3NC	Optional SLD #3 Normally Closed Contact.
12	73	S3C	Optional SLD #3 Common Contact.
13	74	S4NO	Optional SLD #4 Normally Open Contact.
14	75	S4NC	Optional SLD #4 Normally Closed Contact.
15	76	S4C	Optional SLD #4 Common Contact.

CONNECTOR 16PL -- LOCATED ON OPTIONAL APPLICATION CARD

Pin	Nomen-clature	Description
1	DCOM	Digital common.
2	+5V	Not used.
3	PI0	Optional parallel word input least significant binary bit input. For all inputs refer to TABLE 1, JP8 description for voltage level select. Input must be set for 50 microseconds prior to strobe.
4	PI1	Optional parallel binary bit input 1.
5	PI2	Optional parallel binary bit input 2.
6	PI3	Optional parallel binary bit input 3 (or PI0 - PI3 = BCD units digit).
7	PI4	Optional parallel binary bit input 4.
8	PI5	Optional parallel binary bit input 5.
9	PI6	Optional parallel binary bit input 6.
10	PI7	Optional parallel binary bit input 7 (or PI4 - PI7 = BCD tens digit).
11	PI8	Optional parallel binary bit input 8.
12	PI9	Optional parallel binary bit input 9.
13	PI10	Optional parallel binary bit input 10.
14	PI11	Optional parallel binary bit input 11 (or PI8 - PI11 = BCD hundreds digit).
15	PI12	Optional parallel binary bit input 12.
16	PI13	Optional parallel binary bit input 13.
17	PI14	Optional parallel binary bit input 14 (or PI12 - PI14 = BCD thousands digit).
18	PI15	Optional parallel sign bit input.
19	DCOM	Digital common.
20	DCOM	Digital common.

CONNECTOR 18PL -- BETWEEN CONTROL CARD AND OPTIONAL PROGRAMMER CARD

Pin	Nomen-clature	Description
1	----	No Connection.
2	----	No Connection.
3	----	No Connection.
4	----	No Connection.
5	----	No Connection.
6	----	No Connection.
7	----	No Connection.
8	----	No Connection.
9	----	No Connection.
10	----	No Connection.
11	RXT	Input from the Programmer Card serial link driver to Control Card.
12	TXT	Output from the Control Card to the Programmer Card serial link receiver.
13	/DTRD	Data Terminal Ready - Output from Control Card indicating that it is ready to receive a byte from the Programmer Card. Signal goes low when ready, high when not.
14	/CTSD	Clear To Send - Input from Programmer Card signalling when it is ready to receive a byte. Signal low when ready, high when not.
15	RST	Reset line from Control Card to the Programmer Card, active high.
16	+5V	+5V regulated power supply to the Programmer Card.
17	+5V	
18	DCOM	Signal and power return for the Programmer Card.
19	DCOM	
20	-24V	-24V unregulated power supply to Programmer Card.

TERMINAL BOARD 1TB LOCATED ON OPTIONAL PROCESS INTERFACE CARD

Pin	Nomen-clature	Description
1	TKN	Negative tachometer feedback signal. Maximum input is 310 VDC or 344 VAC.
3	TKP	Positive tachometer feedback signal.

TERMINAL BOARD LANTB LOCATED ON OPTIONAL MICROAPPLICATION CARD

Pin	Nomen-clature	Description
1	TX	Non-inverting transmitter output and receiver input in the half-duplex (LAN) mode. Transmitter output in the single-ended mode.
2	/RX	Inverting transmitter output and receiver input in the half-duplex (LAN) mode. For non-serial applications, may be used as a general purpose MUP input for application software. To use this mode, place jumpers JP16, JP18, JP19 in the 2-3 position and leave JP17 open.
3	--	NO CONNECTION - Used to ground the shield at one end of the link or to daisy chain the shield at intermediate drops.

NOTES:

ADDITIONAL CONNECTORS

CONNECTOR 1CPL -- BETWEEN POWER SUPPLY CARD AND AC LINE CTs (CT1 AND CT3)
Regenerative Controllers Only

Pin	Nomenclature	Description
1	J1	Input from line phase 1 CT.
2	J2	Input from line phase 1 CT.
3	J3	Input from line phase 3 CT.
4	J4	Input from line phase 3 CT.

CONNECTOR CNPL -- BETWEEN POWER SUPPLY CARD AND MA CONTACTOR

Pin	Nomenclature	Description
1	MAP	MA contactor control output.
2	X2	MA contactor control output.

CONNECTOR CPTPL -- BETWEEN POWER SUPPLY CARD AND CONTROL POWER TRANSFORMER

Pin	Nomenclature	Description
1	--	38 VAC, center tapped for controller power supplies.
2	--	38 VAC, center tapped for controller power supplies.
3	--	Center tap point for 38 VAC.
4	--	Not used.
5	--	115 VAC for controller cooling fan (if supplied).
6	--	115 VAC for controller cooling fan (if supplied).

CONNECTOR DRPL -- BETWEEN POWER SUPPLY CARD AND SHUNT

Pin	Nomenclature	Description
RED	IP	Armature current feedback from shunt, 0-100mV. Positive when motoring with the forward power circuit, Red wire.
WH	IN	Armature current feedback from shunt. Negative when motoring with the forward power circuit, White wire.

CONNECTOR FAPL -- BETWEEN POWER SUPPLY CARD AND CONTROLLER COOLING FAN (IF USED)

Pin	Nomenclature	Description
1	X1	115 VAC for controller cooling fan.
2	X2	115 VAC for controller cooling fan.

CONNECTOR FCPL -- BETWEEN POWER SUPPLY CARD AND EXTERNAL MOTOR FIELD CONTROL ASSEMBLY

Pin	Nomenclature	Description
1	ACOM	Analog common.
2	--	External motor field control feedback. Used only with external motor field control assembly.

CONNECTOR FSPL -- BETWEEN POWER SUPPLY CARD AND FIELD POWER MODULE*

Pin	Nomenclature	Description
1	FLC	Gate pulse transformer output to field power module SCR, Red wire.
2	FLG	Gate pulse transformer output to field power module SCR, White wire.

CONNECTOR OPTPL -- LOCATED ON POWER SUPPLY CARD

Pin	Nomenclature	Description
1	FX1	Fused 115 VAC from Control Power Transformer.
2	X2	115 VAC return from Control Power Transformer.

CONNECTOR PSPL -- LOCATED ON POWER SUPPLY CARD

Pin	Nomenclature	Description
1	+5V	+5V regulated digital supply (not for use external to drive).
2	DCOM	Digital signal return for +5V (not for use external to drive).

CONNECTOR COMPL -- OPTIONAL R232C SERIAL COMMUNICATIONS CONNECTOR ON OPTIONAL 3TB

Pin	Nomenclature	Description
1	GND	Ground.
2	RXA	For Data Communications Equipment (DCE), dumb terminal mode, pin 2 is the
	or	RS232C receive line (RXA).
	TXA	For Data Terminal Equipment (DTE), modem or PDS system, pin 2 is the RS232C transmit line (TXA).
3	TXA	For DCE, pin 3 is the TXA line.
	or	*or*
	RXA	For DTE, pin 3 is the RXA line.
4	----	Not used.
5	DTRA	For DCE, pin 5 is the data-terminal-ready (DTRA) line.
	or	*or*
	CTSA	For DTE, pin 5 is the clear-to-send (CTSA) line.
6	DSR	Data set ready, no connection in DCE or DTE mode.
7	DCOM	Pin 7 is the communication system common.
8 *thru* 19	----	Not used.
20	CTSA	For DCE, pin 20 is the clear-to-send (CTSA) line.
	or	*or*
	DTRA	For DTE, pin 20 is the data-terminal-ready (DTRA) line.
21 thru 25	----	Not used.

NOTE: *Refer to 3TB SW1 and SW2 description in TABLE 1, for setting DCE and DTE modes.*

Not Used with Motor Field Remote Supply; See* **GEK-24979 *for additional information.*

6. STARTUP PROCEDURE

This startup procedure is meant as a general guide for the initial checkout and startup of a DC-300 drive. It references some general hardware and software checks relative to most basic drive applications. Refer to the sections on "Hardware Jumpers and Pot Adjustments" and "Key Parameter Adjustments" and the accompanying "Parameter Map" (Figure 8-1) for more complete information on adjustments available on the DC-300 drive. Refer to the Custom Software Description and Elementaries for information on wiring and adjustments specific to your drive application.

NOTE: *Prior to beginning actual drive startup, read this entire manual and be aware of all pertinent information. If an optional Programmer was supplied with the drive, review the "Use of Optional Programmer Module" and "Diagnostic Mode for Drive Setup" Sections. Special diagnostic functions are available which will greatly aid drive startup.*

BEFORE APPLYING AC POWER

WARNING: Review warning and caution notices at the beginning of this instruction book before proceeding.

Every DC-300 DC SCR drive controller has been factory tested and is ready to operate provided the power and control connections have been properly made and the following steps are performed:

1. Verify that all incoming wiring is in agreement with the elementary drawings supplied with the drive, and conforms to approved wiring practices (wire size, separation of power from control, etc.).

2. Verify that incoming 3-phase voltage is in accordance with nameplate requirements (+10%, -5%), unless otherwise specified.

3. Confirm that the following jumpers (JP) and switch positions (SW) are in accordance with the jumper selection chart supplied with the drive in the drive door pocket and Custom Software Description. (Refer to the card outlines, Figures 7-1 thru 7-15, for the location of jumpers.)

A. Control Card: **See Figure 7-1**

Jumper	Description/Position
JP1	Reference voltage range for maximum reference **1-2:** 8 - 15 VDC maximum reference. **2-3:** 5 - 8 VDC maximum reference.
JP15	Loadmeter Select for percent load output **1-2:** Bipolar (zero center scale) **2-3:** Absoluted (zero left scale)

B. Power Supply Card: Version G1, See Figure 7-2

Jumper	Description/Position
JP1, JP2	Set maximum motor field current range (amps). JP1 JP2 **2-3: 2-3:** 0.64 to 1.6 Amps with gain factor (GF) of 0.4 **1-2: 2-3:** 1.6 to 4.0 Amps with gain factor (GF) of 1.0 **1-2: 1-2:** 4.0 to 10.0 Amps with gain factor (GF) of 2.5
JP3, JP4	Armature current feedback signal. **1-2:** Normal operating mode.

B. Power Supply Card: (continued)

Version G2, See Figure 7-2

Jumper	Description/Position
JP1, JP2	Set maximum motor field current range (amps).
	JP1 JP2
	2-3: 2-3: 1.5 to 6.0 Amps with gain factor (GF) of 1.5
	1-2: 2-3: 6.0 to 12.0 Amps with gain factor (GF) of 3.0
	1-2: 1-2: 12.0 to 24.0 Amps with gain factor (GF) of 6.0
	NOTE: *Gain Factor required in "Apply AC Power" paragraph of start-up procedure.*
JP3, JP4	Armature current feedback signal.
	1-2: Normal operating mode.

NOTE: *For field current requirements up to 72 Amps, refer to* **GEK-85791.**

C. Process Interface Card: **See Figure 7-7**

SW1, Tach Scale	Refer to SW1 setting chart in "Hardware Jumpers and Pot Adjustments" section, TABLE 1.

4. Verify that all connections are tight.

5. Verify that the DC motor has been properly installed and that the shaft is free to rotate.

APPLY AC POWER

1. Check for proper direction of rotation of all three-phase blower motors on both the power unit(s) and the DC motor(s).

2. Verify that the Diagnostic LEDs are blinking one at a time from right to left. If the optional Programmer is supplied, confirm that it displays the word **OPERATE.**

NOTE: *If at any time the LEDs remain locked, indicating a fault message, the Programmer displays a fault code, or either display is blank, refer to the "Troubleshooting" section for corrective procedures.*

3. Check the field for proper voltage. Since the drive may be in field economy until a **"START"** is called for, check the setting in **ADR050-FLDEC** in the Custom Software Descriptiion to determine the proper field voltage.

NOTE: *If an optional programmer is supplied with the drive, proceed to the following step (see Figure 6-1 for Programmer). If no Programmer is available, skip to step 14.*

4. Place the EEROM write device on the Control card in the "ENABLE" position to allow programming.

NOTE: *The EEROM write protect device is JP13 on the Control card. To enable programming move JP13 to the 2-3 position.*

CAUTION: Do not perform a "Hard Reset" or turn power OFF or ON with the EEROM write device in the "ENABLE" position.

5. Access the Parameter mode by pressing **[SET] [DRV#] [7] [7]** and **[ENTER]** on the keypad (the number or function encased in brackets **[ ]** indicates the Programmer key to be pressed).
   - -- The word **PARAMETER** should be displayed. The **BLACK** keypad functions are now usable. To access a parameter address, press all three numbers on the keypad corresponding to the three-digit number of the desired address. The name of the address and its present value will be displayed.
   - -- To access a new parameter, press **[CLEAR/ MODE]** and the three digits corresponding to the next parameter desired.

6. Check first 14 parameters and the two checksums to confirm that the drive parameters agree with the "as shipped" parameter list inside the drive door or in the Custom Software Description prior to making any adjustments. Document any changes.

7. In order to allow programming of new parameter values, first verify that the parameter address #003, DGNJP (**ADR003-DGNJP**) contains an odd number (1, 3, 9, etc.) by pressing **[0] [0] [3]**. If this address contains an **even** number, add a "1" to this number and enter the new value.

> **Example:** **ADR003** contains the number 8.
> -- Add 1 to obtain 9.
> -- Press **[9]** and **[ENTER]**.
> -- Press **[CLEAR/MODE]** and the three digits for the next desired parameter.

8. Adjust maximum field current (**ADR076-FDLMX**) to the motor nameplate value per the following formula (must be in **PARAMETER** mode):

   NOTE: *Obtain the gain factor (GF) from step 3b, under "BEFORE APPLYING POWER".*

   MAX FLD SETTING = MAX N.P. FLD. CUR. x 2048 *divided by* GAIN FACTOR x 5.

> **Example:** Maximum Nameplate Field Current = 4.5A, G1 Power Supply Card, GF = 2.5
> 1. SETTING = (4.5 x 2048) *divided by* (2.5 x 5) = 737
> 2. Call up **ADR076 - FLDMX** by pressing: **[0] [7] [6]**
> 3. Enter setting by pressing: **[7] [3] [7] [ENTER]**
> 4. Press **[CLEAR/MODE]**

9. Similarly adjust minimum field by calling up **ADR077-FLDMN.** (Minimum field is defined at 60% of the lowest running field current.)

   NOTE: *Field current is at the field economy setting ADR050-FLDEC until a "START" is called for (START, RUN, JOG, etc.).*

   MIN FIELD SETTING = MIN N.P. FLD. CURRENT x 0.6 x 2048 *divided by* GAIN FACTOR x 5.

> **Example:** Minimum Nameplate Field Current = 2.0 amps; other parameters same as previous:
> 1. SETTING = (2.0 x 0.6 x 2048) *divided by* (2.5 x 5) = 196.6
> 2. Enter by pressing: **[1] [9] [7] [ENTER]**
> 3. Press **[CLEAR/MODE]**

10. Adjust acceleration and deceleration time by use of **ADR057-ACELT** and **ADR058-DECELT.** Setting can be any number between 0 and 9999 (999.9 seconds). Some drives may have an optional extended linear time or optional "S-Curve" capability. Refer to the Custom Software Description for information.

11. Call up **ADR070-RPMSF** and enter the desired motor top speed in RPM (**ADR068-REFSF** should contain the same value).

> **Example:** Motor top speed is to be 1500 RPM.
> Enter 1500 in both **ADR068** and **ADR070.**

12. **Armature Voltage Settings** - With the motor armature at rated voltage and test point VFB equal to 4V, the values for **ADR079-CROSS** or **ADR080-CMFLM** can be obtained from the following:
    -- A setting of 1638 in **ADR079-CROSS** or **ADR080-CMFLM** is equal to rated armature volts.
    -- DESIRED PARAMETER VALUE = Required Volts x 1638 *divided by* Rated Volts

NOTE: *Typically,* **ADR080-CMFLM** *is set at approximately 108% for non-regenerative, or 105% for regenerative drives.*

CAUTION: **ADR080-CMFLM** is set at approximately 105% of rated armature voltage for regenerative drives. Damage may occur to fuses and SCRs during regeneration from top speed if **CMFLM** is set too high.

Example:
Rated Armature Volts = 500 VDC
Desired CEMF Limit = 520 VDC
ADR080-CMFLM = 520 x 1638 *divided by* 500
ADR080-CMFLM = 1704

13. **Current Feedback** - The current feedback circuit is factory adjusted to give 1 VDC at **CFB** for the rated current of the motor in the armature circuit.
    - -- All current parameter adjustments can be set in percent (100 = 100% shunt rated current).
    - -- Check **P2-CFB ZERO** on the Power Supply card and adjust so that test point "**CFB**" is as close to zero volts as possible with no current flowing in the armature circuit (see TABLE 1 for **CFB** adjustments).
    - -- If low current limits are required, check setting of **ADR040-ILMMN**. No current limits from any source may be lower than this setting. In other words, set this slightly lower than the lowest current limit commanded from other sources internal or external to the drive.
    - -- Parameter **ADR129-IRATD** may be set to 100 if it is desired to read % amps instead of amps on the Programmer, Diagnostic **Test 5**.

14. Exit the **PARAMETER** mode by pressing **[CLEAR/ MODE] [CLEAR/MODE] [CLEAR/MODE]**.
    - -- **"OPERATE"** should appear. The **RED** keypad functions will now control the drive.

15. Check all incoming signals for proper value and correct connection.
    - -- Refer to the "Use of Optional Programmer Module" section, **Test 20** - Electronic Wire Check.

NOTE: *The speed reference referred to in the following steps may be preprogrammed in the drive software. Refer to the Custom Software Description and elementaries to determine the reference source.*

16. Set the speed reference (speed pot, N/C reference, etc.) at its **minimum** setting. Start the drive (press **"START"** on an operator station or call for a **"RUN"** from the control). Verify that the **MA** contactor picks up and the field current is of the correct value with an amp probe or the direct reading field current panel meter if supplied.

17. Ensure that all **"STOP"** circuits are functional, dropping out **MA**. **"STOP"** from an operator station (115 VAC **START/STOP** circuit - **3TB49, 50 & 51**) or a **"RUN"** input at **3TB58** will not function if the motor is started from the Programmer keypad.

18. Restart the drive and slowly increase the speed reference until the motor starts to rotate. If the motor does not rotate, or rotates in the wrong direction, check the Diagnostic LEDs or the Programmer display. If DC tach feedback is provided, a tach polarity fault may be indicated. Refer to the fault codes in the Troubleshooting section for corrective action.

NOTE: *If an optional Programmer and Process Interface card are included in the drive, an on-board digital voltmeter (DVM) and electronic wire check are available to verify wiring and make the following adjustments. Refer to the "Use of Optional Programmer Module" section for instructions.*

19. With the drive stopped, increase the speed reference to 100% speed.
    - -- Adjust **P1-REF SCALE** on the Control card, to obtain 4 VDC at test point **SRS** (Manual Speed Reference only; Jumper **JP1** on the Control Card selects the manual speed reference input voltage range).
    - -- Measure with volt-ohmmeter (or the **DVM** diagnostic test as mentioned above) at the appropriate test point and **PCOMX**. (If Auto System Reference [**ASR**] is used on the optional Application Card, adjust **P1-ASR SCALE** for 4 VDC at test point **ASR**, also located on the optional Application Card.)
    - -- Set the speed reference to 0% speed.

NOTE: *If no tach or a digital tach is used, SFB will be "0" VDC in the next step.*

20. Start the drive and gradually increase speed reference to maximum.
    - -- Check for 4 VDC at test points **SRS** and **SFB** on the Control and Process Interface cards. (If **ASR** is used, check test point **ASR** on the Application Card.)
    - -- Adjust **P1-TACH SCALE** (requires optional Process Interface Card) if necessary to reach the correct top speed (only on drives with an AC or DC tach).
    - -- Confirm rated armature voltage.

WARNING - ELECTRICAL SHOCK HAZARD: Exercise caution as high voltage may be present.

21. Confirm good motor commutation (absence of sparking between brushes and commutator) under load.

22. Slowly decrease the speed. Verify stability through the full speed range. If the drive is **NOT** stable, refer to the Stability Adjustment procedure in the Troubleshooting section.

23. After the drive is stable throughout the full speed range, confirm that the armature current does not exceed the nameplate rating.
    -- The above list of adjustments should result in a properly running drive with desired readouts at any diagnostic test on the Programmer.
    -- Similarly, other parameters as sold and shipped with the drive can be accessed and changed as necessary. (A complete list of pertinent parameters is contained in the Custom Software Description shipped with the drive.)
    -- A partial listing of "KEY" Parameters, their descriptions, and a Parameter Map are included in Section 8.

NOTE: *Final responsibility for parameter settings resides with the customer. Document all changes to parameter settings in the space provided in the Custom Software Description.*

24. Further accidental reprogramming should be prevented by subtracting a 1 from the value presently in **ADR003-DGNJP** (refer to step 4 of this section), and entering the new value; then place the EEROM write device in the **"PROTECT"** position.

25. Exit the Parameter mode by pressing **[CLEAR/MODE] [CLEAR/MODE] [CLEAR/MODE].**
    -- **OPERATE** should appear.
    -- The **RED** keypad functions will now control the drive.

The drive is now ready for operation.

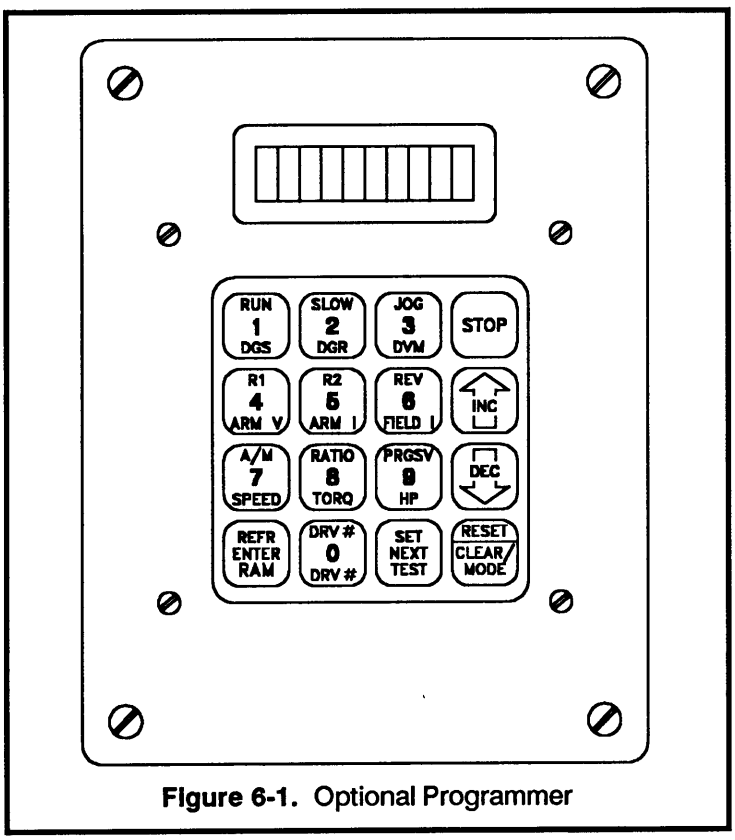


Figure 6-1. Optional Programmer

NOTES:

7. HARDWARE ADJUSTMENTS

The following section details adjustments available in hardware and software. TABLE 1 describes the location and function of all jumpers and potentiometers on the DC-300 controller cards. Software information and key software "parameter" adjustments available are furnished in Section 8. A "Parameter Map" that relates the adjustments to electrical hardware equivalents is also in Section 8 (Figure 8-1).

HARDWARE JUMPER AND POT ADJUSTMENTS

Most of the jumper selections and pot adjustments have been factory set. When one of the cards has to be changed, make sure the replacement card switches, jumpers and pots are set to be the same as on the original card. For a full description of the jumper and pot adjustments, see TABLE 1.

NOTE: *If the Control Card is replaced, the EEROM (U12) from the original card should be put into the replacement card. If the same failure symptoms still exist, the new EEROM, shipped with the replacement card, can be reinstalled but must be reprogrammed per the customer software jumper and adjustment values supplied in the Custom Software Description or parameter list in the drive door pocket.*

Hardware jumpers and pots are located on the Control card, Power Supply card, optional Process Interface card, optional Application card, Diagnostic Readout card (if used), and optional Programmer card. Figures 7-1 thru 7-16 show the locations of pots, jumpers, test points, and other key components.

Most of the pots and jumpers are used in the interface circuits of the microprocessor software control circuit. Some of the jumpers are only for manufacturer's use. The test data sheets supplied with each controller (in the drive door pocket) indicate the position of these adjustments as they were set in the factory. Adjustments not set in the factory can easily be made by using the startup procedure in this book or in the Startup Guide (GEJ-7152) and by using TABLE 1.

NOTE: *Only high impedance digital voltmeters or the optional "on-board" DVM (Test 3) should be used to make indicated measurements.*

NOTE: *Before changing any jumper positions, compare actual jumper positions with "as shipped" jumper positions through use of the Custom Software Description.*

TABLE 1. JUMPER AND POT ADJUSTMENTS

Location	Jumper or Pot	Number	Nomenclature	Description
Control Card 531X300 CCH_ _M_	Pot	P1	MAN REF	To adjust test point SRS to four (4) volts when top speed reference is at MSR, 3TB65 (6PL-5; connector 6PL, pin number 5). Used in conjunction with jumper JP1.
		P2	IMET	This pot is used to adjust the output of the analog load meter circuit. Adjust the output voltage to be from 1 to 8 volts at rated load. Maximum current load on this signal is 1000 ohms/volt.

Location	Jumper or Pot	Number	Nomenclature	Description
Control Card 531X300 CCH_ _M_ (continued)	Pot	P3	HP SEL	Adjusted by the manufacturer. To adjust in the field, set to make it visually the same on replacement cards.
		P4	OFFSET	Adjusted by the manufacturer, no need to be adjusted in the field unless directed to do so by fault codes. Refer to adjustment procedure in "Use of the Optional Programmer Module" section if a Programmer is available.
		P5	FCMET	This pot is used to adjust the output of the analog field current voltmeter circuit. Adjust the output from 1 to 5 volts at rated maximum field current. Maximum current load on this signal is 1000 ohms/volt.
		P6	DA0 SCL	This pot is used to adjust the gain of the 12-bit digital-to-analog converter (D A0) output from the Control Card. CCW sets the gain at 0.25, full CW sets the gain at 1.25. The output saturates at 12 VDC.
		P7	DAMP	This pot adjusts the amount of current feedback fed into the voltage loop integrator. Full CCW is the default setting (no damping).
		P8	VMET	This pot is used to adjust the output of the analog armature voltage meter circuit. Output is adjustable from 0 to 8 volts at rated voltage. Normal current load on this signal is 1mA.
	Jumper	JP1	MAN REF	This jumper selects the maximum voltage range for the manual speed reference input. **1-2:** 8 to 15 volts **2-3:** 5 to 8 volts
		JP2 JP4 JP5 JP6 JP7		JP2 - JP7 should be set as follows: **1-2:** Set in this position for proper operation.
		JP8 JP9 JP10 JP11		These jumpers are set in accordance with the HP of the drive per the following Chart: **Horsepower** **230 VAC / 460 VAC / JP8 / JP9 / JP10 / JP11** 1 / ---- / 1-2 / 2-3 / 2-3 / 2-3 1.5 / 3 / 1-2 / 2-3 / 2-3 / 2-3 2 / 5 / 1-2 / 2-3 / 2-3 / 2-3 3 / 7.5 / 1-2 / 2-3 / 2-3 / 2-3 5 / 10 / 1-2 / 2-3 / 2-3 / 1-2 7.5 / 15 / 1-2 / 2-3 / 2-3 / 1-2

Location	Jumper or Pot	Number	Nomenclature	Description
Control Card 531X300 CCH_ _M_ (continued)	Jumper	JP10 JP11		**Horsepower** **230 VAC / 460 VAC / JP8 / JP9 / JP10 / JP11** 10 / 20 / 1-2 / 2-3 / 1-2 / 2-3 12.5 / 25 / 1-2 / 2-3 / 1-2 / 2-3 15 / 30 / 1-2 / 2-3 / 1-2 / 1-2 20 / 40 / 1-2 / 1-2 / 2-3 / 2-3 25 / 50 / 1-2 / 1-2 / 2-3 / 1-2 30 / 60 / 1-2 / 1-2 / 1-2 / 2-3 ---- / 75 / 1-2 / 1-2 / 1-2 / 1-2 40 / ---- / 1-2 / 1-2 / 2-3 / 2-3 50 / 100 / 1-2 / 1-2 / 2-3 / 1-2 60 / 125 / 1-2 / 1-2 / 1-2 / 2-3 75 / 150 / 1-2 / 1-2 / 1-2 / 1-2 100 / 200 / 1-2 / 1-2 / 2-3 / 2-3 125 / 250 / 1-2 / 1-2 / 2-3 / 1-2 150 / 300 / 1-2 / 1-2 / 1-2 / 2-3
		JP12		Allows SCR test without Programmer **1-2:** Disables SCR test (normal operation). **2-3:** SCR test before power-up. An SCR test will be executed the first time a RUN is commanded. Subsequent RUN commands will execute normally. (Not used with elevators.)
		JP13		This jumper provides a hardware write protection for the software jumpers and adjustments. **1-2:** Protected (should be in this position for maximum security and reliability of the drive). **2-3:** Enables reprogramming. Do not perform a "Hard" reset or turn power ON or OFF in this position.
		JP14		This jumper selects proper line voltage. **1-2:** 440 to 480 VAC **2-3:** 380 to 415 VAC **2-5:** 310 to 340 VAC **3-4:** 220 to 240 VAC
		JP15		This jumper selects whether the load meter driver output is to be bipolar or absoluted. **1-2:** Absoluted (always positive) **2-3:** Bipolar (zero center)
		JP16 JP17 JP18		JP16 - JP18 should be set as follows: Set by factory - refer to jumper listing in Custom Instruction Book or drive door pocket for proper setting.

Location	Jumper or Pot	Number	Nomenclature	Description
Control Card 531X300 CCH_ _M_ (continued)	Jumper	JP19 JP20		JP19 & JP20 set the polarity of RUN, JOG, POL, and XSTOP. **1-2 & 3-4:** Inputs are **active** when pulled down between -24 to 0.4 VDC. Inputs are **inactive** when open or tied to +3.5 to +24 VDC. **1-3 & 2-4:** Inputs are **active** when tied between +3.5 to +24 VDC. Inputs are **inactive** when open or held at -24 to 1.5 VDC.
		JP21		Allows interactive card test. **1-2:** Disables card test (normal operation). **2-3:** Initiates card test if drive is not running. To exit card test, move JP21 back to the 1-2 position and Hard reset drive (push SW2).
		JP22 JP23 JP24 JP25 JP26 JP27 JP28 JP30 JP31		JP22-JP28, 30 and 31 set as follows: Set by factory - refer to jumper listing in Custom Instruction Book in drive door pocket for proper setting.
		JP29		This jumper selects whether the DA0 output has a fixed gain or adjustable gain. **1-2:** Fixed gain (see TABLE 3, RAM Variable Scaling). **2-3:** Gain adjustable by P6-DA0 SCL.
	Switch	SW2	Hardware Reset	This switch causes a hardware reset ("Hard" reset of the controller) when pressed. Used to clear some drive faults (refer to TABLE 5, Specific Fault Troubleshooting). The power supplies are toggled "OFF" and "ON" and the drive will return to the initial power up condition.
MFC/Power Supply Card 531X111 PSH_ _G1/G2	Pot	P1	FIELD SCALE	P1 is used to calibrate the field current feedback using a known field current. This pot is set and sealed by the manufacturer.
		P2	CFB ZERO	P2 is used to adjust offset of the current isolator circuit so circuit so that test point CFB (on Control Card) is zero.

Location	Jumper or Pot	Number	Nomenclature	Description
MFC/Power Supply Card 531X111 PSH_ _G1/G2 (continued)	Pot	P3	CFB ADJUST	**CAUTION:** The drive must not be running when making this adjustment. P3 is used to adjust the gain of the current isolator circuit so that CFB equals the value indicated on the label inside the right side of the card compartment (in front of the Power Supply Card holder). Pots P2 and P3 are interdependent and should be adjusted alternately. **WARNING - ELECTRICAL SHOCK HAZARD:** **High voltage is present on the Power Supply card. Remove power from the drive before handling the plug and jumpers in this procedure.** P3 can be set by disconnecting the current feedback signal (DRPL) and moving jumpers JP3 and JP4 to the 2-3 position (will cause a fault 46; no concern). This applies a calibration voltage (approx. 0.1V) between pin IP and IN of DRPL. P2 should be adjusted prior to moving JP3 and JP4. Adjust P3 until test point CFB on the Control Card measures the value indicated on the drive calibration label. After setting P3, P2 should be rechecked. Return JP3 and JP4 to the 1-2 positions and reconnect DRPL. **CAUTION:** If JP3 and JP4 are left in the 2-3 position, erratic operation and card failures can occur.
	Jumper	JP1 JP2		JP1 and JP2 select field current feedback signal (FC) gain. Version G1 MFC/Power Supply Card (Figure 7-2). **JP1 JP2 Gain Factor Field AMP Range** **2-3: 2-3:** 0.4 0.64 to 1.6 AMPS **1-2: 2-3:** 1.0 1.6 to 4.0 AMPS **1-2: 1-2:** 2.5 4.0 to 10.0 AMPS Version G2 MFC/PSC (Figure 7-2) and optional Remote Field Module (Figure 4-3). **JP1 JP2 Gain Factor Field AMP Range** **2-3: 2-3:** 1.5 1.5 to 6.0 AMPS **1-2: 2-3:** 3.0 6.0 to 12.0 AMPS **1-2: 1-2:** 6.0 12.0 to 24.0 AMPS
		JP3 JP4		JP3 and JP4 are used for calibration of the armature current feedback signal. **JP3 JP4 Function** **1-2: 1-2:** Normal Operating Mode **2-3: 2-3:** CFB calibration (See description at pot P3.)

Location	Jumper or Pot	Number	Nomenclature	Description
MFC/Power Supply Card 531X111 PSH_ _G1/G2 (continued)	Jumper	JP5		K2 coil suppression capacitor selection jumper: 1-2: Capacitor in circuit (normal). 2-3: Capacitor out of circuit.
Power Connect Card 531X121 PCR_ _G_	Jumper	JP1 JP2		JP1 and JP2 select DC armature voltage range and must be stabbed on to the appropriate posts, P3 - P10. **NOTE:** *Scaling range is dependent on which group is being used.* **Jumper Position** JP1 JP2 / **G1/G4 Voltage** / **G2 Voltage** / **G3 Voltage** OPEN OPEN 500 700 240 P4-P8 P5-P9 450 630 210 P3-P7 P6-P1 400 550 170 P4-P7 P5-P10 340 570 150 P7-P8 P9-P10 290 500 120 P3-P4 P5-P6 240 420 80 P3-P8 P6-P9 190 350 50
Power Connect Card 531X122 PCN_ _G_	Jumper	JP1 JP2		JP1 and JP2 select DC armature voltage range and must be stabbed on to the appropriate posts, P3 - P6. **NOTE:** *Scaling range is dependent on which group is being used.* **Jumper Position** JP1 JP2 / **G1 Voltage** / **G2 Voltage** OPEN OPEN 500 700 P3-P4 P5-P6 240 550
Power Connect Card 531X308 PCS_ _G1/G2	Jumper	JP1 JP2		JP1 and JP2 select DC armature voltage range and must be stabbed on to the appropriate posts, P3 - P10. **NOTE:** *Scaling ranges are dependent on which group is being used.* **Jumper Position** JP1 JP2 / **G1/G3 Voltage** / **G2/G4 Voltage** OPEN OPEN 630 700 P4-P8 P5-P9 580 595 P3-P7 P6-P10 570 550 P4-P7 P5-P10 550 500 P7-P8 P9-P10 500 390 P3-P4 P5-P6 290 350 P3-P8 P6-P9 240 240

<table>
<tr><th>Location</th><th>Jumper or Pot</th><th>Num-ber</th><th>Nomen-clature</th><th>Description</th></tr>
<tr><td rowspan="2">Power Connect Card 531X123 PCH_ _G_</td><td>Jumper</td><td>JP1</td><td></td><td>This selects the SCR firing power enable.
1-2: Controlled by drive's delayed firing power (DFP).
2-3: Always enabled (firing power = +24 volt source)
3-4: Always disabled (for test purposes only)</td></tr>
<tr><td></td><td>JP2
JP3</td><td></td><td>JP2 and JP3 select DC armature voltage range and must be stabbed on to the appropriate posts, P3 - P10.
NOTE: Scaling range is dependent on which group is being used.
<table>
<tr><th colspan="2">Jumper Position</th><th>G1 Voltage</th><th>G2 Voltage</th></tr>
<tr><th>JP1</th><th>JP2</th><th></th><th></th></tr>
<tr><td>0PEN</td><td>OPEN</td><td>630</td><td>700</td></tr>
<tr><td>P6-P8</td><td>P5-P7</td><td>530</td><td>570</td></tr>
<tr><td>P8-P10</td><td>P7-P9</td><td>500</td><td>630</td></tr>
<tr><td>P4-P6</td><td>P3-P5</td><td>470</td><td>550</td></tr>
<tr><td>P6-P10</td><td>P5-P9</td><td>400</td><td>500</td></tr>
<tr><td>P4-P8</td><td>P3-P7</td><td>370</td><td>420</td></tr>
<tr><td>P4-P10</td><td>P3-P9</td><td>240</td><td>350</td></tr>
</table></td></tr>
<tr><td>3TB Terminal Board 531X170 TBS_ _G_</td><td>Jumper</td><td>JP1</td><td></td><td>JP1 selects whether optional SLD5 mounted on 3TB is controlled by the Control Card or by an external contact closure.
1-2: Control card control of SLD5, programmed by ADR085, 086 and 087.
2-3: External control of SLD5 by connecting 3TB26 K5A to 3TB56, REF24.</td></tr>
<tr><td></td><td>Switch</td><td>SW1
SW2</td><td></td><td>SW1 and SW2 select the mode of serial link communication through COMPL on 3TB (depress switch as shown by dot to set for up or down position).
For use with Data Terminal Equipment (DTE):
<table>
<tr><th>Switch</th><th>Set Position</th></tr>
<tr><td>SW1-1,2, 3:</td><td>Up</td></tr>
<tr><td>SW1-4:</td><td>Down</td></tr>
<tr><td>SW2-1, 3, 4:</td><td>Down</td></tr>
<tr><td>SW2-2:</td><td>Up</td></tr>
</table>
For use with Data Communication Equipment (DCE), set all switches in down position.</td></tr>
</table>

Location	Jumper or Pot	Number	Nomenclature	Description
Process Interface Card 531X133 PRU_ _G_ (Optional)	Switch	SW1	Tach Range Select	These switches provide coarse scaling of the tach circuit as follows: DIP switch positions are selected for the legend of the side depressed. To select position 3 ON, depress the switch at position 3 next to the ON legend. **SW1 Position No. "Closed" or "ON"** / **Tach Volt At Maximum Speed DC Tach** / **AC Tach** All "open" or "off" — 10 - 12.5V — 11.1 - 13.9V 1 — 12.5 - 15.5V — 13.9 - 17.2V 2 — 15.5 - 19V — 17.2 - 21.1V 1, 2 — 19 - 22V — 21.1 - 24.4V 3 — 22 - 25V — 24.4 - 27.8V 1, 3 — 25 - 30V — 27.8 - 33.3V 1, 2, 3 — 30 - 37.5V — 33.3 - 41.7V 4 — 37.5 - 46V — 41.7 - 51.1V 1, 2, 4 — 46 - 56V — 51.1 - 62.2V 1, 2, 3, 4 — 56 - 70V — 62.2 - 77.8V 5 — 70 - 87V — 77.8 - 96.7V 2, 3, 5 — 87 - 105V — 96.7 - 117V 1, 2, 4, 5 — 105 - 120V — 117 - 133V 1, 2, 3, 4, 5 — 120 - 144V — 133 - 160V 6 — 144 - 180V — 160 - 200V 1, 2, 4, 6 — 180 - 205V — 200 - 228V 5, 6 — 205 - 250V — 228 - 278V 1, 2, 3, 4, 5, 6 — 250 - 310V — 278 - 344V **NOTE:** *SW1 postion 7 must be OFF (or open) for DC tachs or ON (or closed) for AC tachs. Maximum voltages less than 10 volts are not to be used.*
	Pot	P1	TACH SCALE	Used to scale the tach circuit to obtain 4.0 volts at test point SFB at maximum speed.
		P2	SMET	Used to adjust the analog output signal at 3TB20 of the speed metering circuit for 10 volts at maximum motor speed. Maximum current load on this signal is 2mA. **NOTE:** *A "Digital Tach System" will not provide an SMET output of speed.*
		P3	DVM TRIM	With DVM test point connected to +5V and optional Programmer set to DVM operation, adjust P3 until the Programmer displays 5.00.

Location	Jumper or Pot	Number	Nomenclature	Description
Process Interface Card 531X133 PRU_ _G_ (Optional) (continued)	Pot	P4 P5	PF GAIN PF OFF	These pots allow trimming the gain and offset of the circuit that provides up to 10 volts from 1 to 5, 4 to 20, 5 to 50, or 5 to 65mA transducer inputs. To adjust, start with PF gain at the full CCW position and proceed as follows: -- Apply the current reference corresponding to minimum speed. -- Adjust PF OFF for 0 volts at test point PFRF. -- Adjust the current reference corresponding to maximum speed. -- Adjust PF GAIN for 10.0 volts at PFRF. -- Repeat as necessary **NOTE:** *To prevent reversing when the current reference is less than the minimum current, the drive should be set up in a non-reversing mode.*
		P6	STP SCALE	Set Point Scale - Used to adjust scaling of definable input at test point STPRF. Refer to elementaries and engineering notes if used.
		P7	SFB ZERO	This pot adjusts test point SFB to be zero with zero tach feedback. Set speed reference to zero (jumper MSR or ASR to PCOMX if used) and adjust P7 for zero motor speed when a run is commanded.
	Jumper	JP1		JP1 selects circuitry for the serial link communication mode. **1-2:** Enables RS422 Reception. **2-3:** Enables RS232C Reception.
		JP2 JP3		JP2 and JP3 select the process follower mode per the following table: **JP2** **JP3** **Value** 1-2 1-2 5-65 mA current loop follower. 1-2 2-3 5-50 mA current loop follower. 2-3 1-2 4-20 mA current loop follower. 2-3 2-3 1-5 mA current loop follower or 10V voltage follower.
		JP4		JP4 selects the speed meter (SMET) circuit mode. **1-2:** Unipolar speed meter driver mode. **2-3:** Bipolar meter drive mode (DC tach only).
		JP5		JP5 selects digital tach quadrature mode **1-2:** Quadruples frequency of digital tach input (only when JP6 is in doubler mode). **2-3:** Disables quadrature tach mode.

Location	Jumper or Pot	Number	Nomenclature	Description
Process Interface Card 531X133 PRU_ _G_ (Optional) (continued)	Jumper	JP6		JP6 selects digital tach doubler mode **1-2:** Doubles frequency of digital tach input at 3TB16. **2-3:** Disables doubler tach mode.
		JP7		JP7 selects type of digital tach. **1-2:** AN tach linearization mode or reluctance digital tach. **2-3:** Square wave digital tach.
		JP8		JP8 selects analog tach filter mode. **1-2:** Enables 4MS filter in analog tach feedback (required for AC tach operation). **2-3:** Disables 4MS filter - Required for speed regulator performance above 20 radians (DC or Digital Tach only).
		JP9 JP10		JP9 and JP10 select the digtal tach mode. **1-2:** Differential RS422 compatible with terminating resistor. **2-3:** Single-Ended input at 3TB14 and 3TB16. **Open:** Differential operation with no terminating resistor. **NOTE:** *Jumpers JP9 and JP10 have been replaced by jumpers JP15 and JP16 on later version cards.*
		JP11		TXD Mode Select **1-2:** 12PL-9, 12PL-12 RS422 transmitted data mode. **2-3:** 12PL-9 RS232 transmitted data mode.
		JP12		DTR Mode Select **1-2:** 12PL-8, 12PL-13 RS422 DTR handshake mode. **2-3:** 12PL-8 RS232 handshake mode.
		JP13		DTR RS232 Logic Polarity **1-2:** 12PL-8 high = drive ready to receive **2-3:** 12PL-8 low = drive ready to receive
		JP14		CTS Mode Select **1-2:** For RS422 handshaking **2-3:** For RS232 handshaking **NOTE:** *For Series Six interfaces using the CCM Card, set JP1 and JP11 in the 1-2 position and JP12, JP13, and JP14 in the 2-3 position. 12PL-7 should be tied to +5V. (Not applicable to AAG1, ABG1, or ACG1 version cards.)*
		JP15		DTA - /DTA maximum input voltage select (3TB16, 17) (See **NOTE** for JP16.) **3-4:** 1.5 VDC **2-3:** 5.0 VDC **1-2:** 10 VDC **1-4:** 15 VDC

Location	Jumper or Pot	Number	Nomenclature	Description
Process Interface Card 531X133 PRU_ _G_ (Optional) (continued)	Jumper	JP16		DTB - /DTB maximum input voltage select (3TB14, 15) **3-4:** 1.5 VDC **2-3:** 5.0 VDC **1-2:** 10 VDC **1-4:** 15 VDC **NOTE:** *When replacing **"AJ"** or earlier version card with **"AK"** or later version card, there is **not** a complete circuit for single-ended tachometers. With inputs at DTA (3TB16) and DTB (3TB14) only, /DTA (3TB17) and /DTB (3TB15) **must** be connected to PSPL (3TB28) to complete the circuit through the optical isolator LED. Also, common and +15V of tachometer should be twisted pair wiring routed through balun core with common tied to PSPL.*
		JP17		Voltage Follower Enable **1-2:** Voltage Follower Disabled **2-3:** Voltage Follower Enabled
Application Card 531X139 APM_ _G2 (Optional)	Pot	P1	ASR SCALE	P1 is used in conjunction with JP1 to trim test point ASR to four (4) volts for top speed reference applied to 13PL-23 (4TB23).
		P2	FRA SCALE	P2 is used to scale the field reference adjust input at 13PL-20 (4TB20). One volt amplitude at test point FRA increases/decreases the FLDMX parameter setting by 204 counts.
		P7	AN6 SCALE	P7 is used to provide a variable input lead for general purpose amplifier #1. Refer to elementaries and operating notes for further information, if used (version APM Card, Figure 7-10).
		P8	AN7 SCALE	P8 is used to provide a variable input lead for general purpose amplifier #2. Refer to elementaries and operating notes for further information, if used (version APM Card, Figure 7-10).
	Jumper	JP1		JP1 selects the voltage range for the auto speed reference input (ASR) at 4TB23 (13PL-23). **1-2:** 8 to 15 volts. **2-3:** 5 to 8 volts.
		JP2		JP2 selects whether the encoder Track A input at 11PL-16 (4TB37) is single-ended or differential. **1-2:** RS422 differential. **2-3:** TTL compatible single-ended.

Location	Jumper or Pot	Number	Nomenclature	Description
Application Card 531X139 APM_ _G2 (Optional) (continued)	Jumper	JP3		JP3 selects whether the encoder Track B input at 11PL-14 (4TB35) is single-ended or differential. **1-2:** RS422 differential. **2-3:** TTL compatible single-ended.
		JP4		JP4 selects whether the encoder marker track input at 11PL-12 (4TB33) is single-ended or differential. **1-2:** RS422 differential. **2-3:** TTL compatible single-ended.
		JP5 JP6		JP5 and JP6 select polarity of the eight logic inputs, MODE 0-MODE 7, at 13PL-1 thru 8 (4TB1 thru 8). **1-2 & 3-4:** The control is active low. Inputs are active when pulled below 0.8 volts. **1-3 & 2-4:** The control is active high. Inputs are active when pulled above 3.5 volts or driven by TTL high output.
		JP7		JP7 selects whether the sixteen-bit parallel input is latched on the high or low level of the strobe. **1-2:** Data latched on low to high transition of PSB. **2-3:** Data latched on low level of PSB.
		JP8		JP8 selects whether the sixteen-bit parallel input and strobe signals are driven by an active high or low driver. **1-2:** Lines internally pulled up to +5V. Must be driven by active pull-down to common driver. **2-3:** Lines internally pulled down to DCOM. Must be driven by an active pull-up (above 3.5 volt) driver.
		JP14		JP14 set as follows: **1-2:** Should be in this position for proper operation.
		JP24 JP25 JP26 JP27 JP28 JP29 JP30 JP31		JP24 thru JP31 define functions for the on-board general purpose amplifiers (GP AMP #1 and GP AMP #2). Refer to drive elementaries, operating notes and Custom Software Description for more information (if used).
		JP32		JP32 selects the scaling range of P2-FRA SCALE. **1-2:** Normal range. **2-3:** Extended range.
		JP36 JP37		JP36 & JP37 set as follows: **1-2:** Should be in this position for proper operation. **Open:** Only for manufacturer's use.

Location	Jumper or Pot	Number	Nomenclature	Description
Application Card 531X139 APM_ _G2 (Optional) (continued)	Jumper	JP40 JP41		JP40 & JP41 select whether the encoder input is used as a feedback or a pulse reference. JP40 JP41 **1-2: 1-2:** Normal encoder operation. **2-3: 1-2:** Enables encoder tract A as a pulse reference frequency input at 3TB37 (11PL-16, ENCA), 40 kHz maximum. **2-4: --** Not used.
		JP45		JP45 selects whether P7-AN6 SCALE is in or out of the GMP AMP #1 circuit: **1-2:** Connects P7 into amplifier circuit. **2-3:** Disconnects P7 from amplifier circuit.
		JP46		JP46 selects whether P8-AN7 SCALE is in or out of the GMP AMP #1 circuit: **1-2:** Connects P8 into amplifier circuit. **2-3:** Disconnects P8 from amplifier circuit.
Programmer Card 531X135 PRG_ _G_ (Optional)	Jumper	JP1 JP3 JP4 JP5		For use with Control Card 18PL: JP1, JP3, JP4 in 1-2 position and JP5 in 2-3 position. For use with Process Interface Card 12PL: JP1 in 1-2 position and JP3, JP4, JP5 in 2-3 position. **NOTE:** *Older version programmers may have JP2 instead of JP5. Set JP2 in 1-2 position when provided.*

Location	Jumper or Pot	Number	Nomenclature	Description
Micro-application Card 531X139 APM_ _M_ (Optional)	Pot	P1	ASR SCALE	P1 is used in conjunction with JP1 to trim test point ASR to four (4) volts for top speed reference applied to 13PL-23 (4TB23).
		P2	FRA SCALE	P2 is used to scale the field reference adjust input at 13PL-20 (4TB20). One volt amplitude at test point FRA increases/decreases the FLDMX parameter setting by 204 counts.
		P3	OFFSET	Used to trim the zero offset of the MUP Bus A-D converter.
		P4	AN1 SCALE	Used to scale the analog input AN1 at 13PL-24. Gain of this input is 0 bits/volt in the full CCW position and 204.8 bits/volt in the full CW position. Full scale 12-bit conversion is +2047/-2048 bits.
		P5	AN2 SCALE	Used to scale the analog input AN2 at 13PL-23. Range is the same as AN1 (P4).
		P6	AN3 SCALE	Used to scale the analog input AN3 that is available through MAP3 (11PL-3). See JP28 and JP29. Range is the same as AN1 (P4).
		P7	AN6 SCALE	P7 is used to provide a variable input lead for general purpose amplifier #1. Refer to elementaries and operating notes for further information, if used.
		P8	AN7 SCALE	P8 is used to provide a variable input lead for general purpose amplifier #2. Refer to elementaries and operating notes for further information, if used.

Location	Jumper or Pot	Number	Nomenclature	Description	
Micro-application Card 531X139 APM_ _M_ (Optional) (continued)	Jumper	JP1		**1-2:** (Def) **2-3:**	Auto max ref. volts greater than 8 VDC. Auto max ref. volts less than or equal to 8 VDC.
		JP2		**1-2:** **2-3:** (Def) **Open:**	Jumpers JP2, JP3, JP4 are set by the type of encoder signal or pulse reference to be used at the "shared bus encoder" input. **NOTE:** *This jumper has been replaced by jumpers 54, 55, and 56 on later version cards.*
		JP3		**1-2:** **2-3:** (Def) **Open:**	See JP2 description. **NOTE:** *This jumper has been replaced by jumpers JP55 and JP56 on later version cards.*
		JP4		**1-2:** **2-3:** (Def) **Open:**	See JP2 description. **NOTE:** *This jumper has been replaced by jumpers 54, 55, and 56 on later version cards.*
		JP5		**2-4:** **3-4:** (Def)	+24 VDC logic level. -24 VDC logic level.
		JP6		**1-2:** (Def) **1-3:**	-24 VDC logic level. +24 VDC logic level.
		JP7		**1-2:** (Def) **2-3:**	16 bit parallel data input latched on **falling** edge of strobe. 16 bit parallel data input latched on **rising** edge of strobe.
		JP8		**1-2:** (Def) **2-3:**	16 bit parallel input and strobe driven by active **low** driver. 16 bit parallel input and strobe driven by active **high** driver.
		JP9		**1-2:** **2-3:** (Def) **Open:**	Jumpers JP9, JP10, JP11 are set by the type of encoder signal or pulse reference to be used at the "MUP encoder" input.
		JP10		**1-2:** **2-3:** (Def) **Open:**	See JP9 description.

Location	Jumper or Pot	Number	Nomenclature	Description	
Micro-application Card 531X139 APM_ _M_ (Optional) (continued)	Jumper	JP11		**1-2:** **2-3:** (Def) **Open:**	See JP9 description.
		JP12		**1-2:** (Def) **Open:**	Always Never
		JP13		**1-2:** **2-3:** (Def)	LAN Used. MUP encoder marker pulse detection.
		JP14		**1-2:** (Def) **Open:**	Always Never
		JP15		**1-2:** (Def) **2-3:**	Always Never
		JP16		**1-2:** (Def) **2-3:**	Always Never
		JP17		**1-2:** (Def) **2-3:** **Open:**	Always Never Never
		JP18		**1-2:** **2-3:** (Def)	Last drop on LAN (termination resistor connected). Not on LAN or not last drop in LAN (termination resistor not connected).
		JP19		**1-2:** **2-3:** (Def)	Last drop on LAN (termination resistor connected). Not on LAN or not last drop in LAN (termination resistor not connected).
		JP20		**1-2:** (Def) **2-3:**	128K EPROM on Microapplication Card. 256K or 512K EPROM on Microapplication Card. **NOTE:** *128K EPROM: U33 = 409X106AB106 through AD106 and 409X111AA106 through AC106.* *256K EPROM: U33 = 409X113_ _ _ _ _ or 409X114_ _ _.*
		JP21		**1-2:** (Def) **2-3:**	Always Never
		JP22		**1-2:** (Def) **2-3:**	MUP encoder marker non-inverting input. Bring in LAN TX line via 11PL. **NOTE:** *JP39 must also be in 2-3 position.*

Location	Jumper or Pot	Number	Nomenclature	Description	
Micro-application Card 531X139 APM_ _M_ (Optional) (continued)	Jumper	JP23		**1-2:** (Def) **2-3:**	MUP encoder marker inverting input. Bring in LAN/RX line via 11PL. **NOTE:** *JP38 must be also in 2-3 position.*
		JP24		**1-2:** (Def) **2-3:** **2-4:**	GP amp #1 inverting input tied to common. GP amp #1 inverting input tied to pin 5 of JP29. (If JP29 in 2-5 position, then MAP3 tied to GP amp #1 inverting input.) GP amp #1 inverting input tied to pin 4 of JP30. (If JP30 in 2-4 position, then MAP2 tied to GP amp #1 inverting input.)
		JP25		**1-2:** (Def) **1-3:** **1-4:**	GP amp #1 non-inverting input tied to common. GP amp #1 non-inverting input tied to pin 5 of JP29. (If JP29 in 2-5 position, then MAP3 tied to GP amp #1 non-inverting input.) GP amp #1 non-inverting input tied to pin 4 of JP31. (If JP31 in 2-4 position, then MAP1 tied to GP amp #1 non-inverting input.)
		JP26		**1-2:** (Def) **2-3:** **2-4:**	GP amp #2 inverting input tied to common. GP amp #2 inverting input tied to DM4. GP amp #2 inverting input tied to pin 4 of JP31. (If JP31 in 2-4 position, then MAP1 tied to GP amp #2 inverting input.)
		JP27		**1-2:** (Def) **1-3:** **1-4:**	GP amp #2 non-inverting input tied to common. GP amp #2 non-inverting input tied DM4. GP amp #2 non-inverting input tied to pin 4 of JP30. (If JP30 in 2-4 position, then MAP2 tied to GP amp #2 non-inverting input.)
		JP28		**1-2:** (Def) **2-3:** **2-4:**	GP amp #1 output (AN6) tied to pin 4 of JP29. (If JP29 in 2-4 position, then AN6 output tied to MAP3.) GP amp #2 output (AN7) tied to pin 4 of JP29. (If JP29 2-4 position, then AN7 output tied to MAP3.) Analog input AN3 tied to pin 4 of JP29. (If JP29 in 2-4 position, then AN3 input tied to MAP3.)

Location	Jumper or Pot	Number	Nomenclature	Description	
Micro-application Card 531X139 APM_ _M_ (Optional) (continued)	Jumper	JP29		**1-2:** (Def)	MAP3 is digital control input C14.
				2-3:	MAP3 is tied to digital output bit B06.
				2-3:	MAP3 is tied to pin 2 of the JP28. (JP28 position then determines function of MAP3.)
				2-5:	MAP3 is tied to pin 3 of JP24/JP25. (If JP24 in 2-3 position, then MAP3 is the inverting input to GP amp #1.) (If JP25 in 1-3 position, then MAP3 is the non-inverting input to GP amp #1.) **NOTE:** *This jumper has been replaced by jumpers 29A, 29B, and 29C on later version cards.*
		JP29A		**6-7:**	Park position.
		JP29B		**1-6:**	Input L1-5 is digital control input C14.
				1-2:	MAP3 tied to digital control input C14.
				2-3:	MAP3 tied to GP amp outputs (JP28).
				2-5:	MAP3 tied to pin 3 of JP24/JP25.
				2-3:	MAP3 tied to digital output B06.
		JP29C		**3-8:**	K17 (29PL-23) to B06 output.
		JP30		**1-2:** (Def)	MAP2 is digital control input C13.
				2-3:	MAP2 is tied to digital output bit B05.
				2-4:	MAP2 is tied to pin 4 of JP24 and pin 4 of JP27. (If JP24 in 2-4 position, then MAP2 is the inverting input to GP amp #1; if JP27 in 1-4 position, then MAP2 is the non-inverting input to GP amp #2.)
				2-5:	MAP2 is analog input AN5. **NOTE:** *This jumper has been replaced by jumpers 30A, 30B, and 30C on later version cards.*
		JP30A		**6-7:**	Park position.
		JP30B		**1-6:**	Input L1-4 is digital control input C13.
				1-2:	MAP2 tied to digital control input C13.
				2-4:	MAP2 tied to GP amps (JP27, JP24).
				2-5:	MAP2 tied to AN5.
				2-3:	MAP2 tied to digital output B05.
		JP30C		**3-8:**	K16 (29PL-22) to B05 output.

Location	Jumper or Pot	Number	Nomenclature	Description	
Micro-application Card 531X139 APM_ _M_ (Optional) (continued)	Jumper	JP31		**1-2:** (Def) **2-3:** **2-4:** **2-5:**	MAP1 is digital control input C12. MAP1 is tied to digital output bit B04. MAP1 is tied to pin 4 of JP25 and pin 4 of JP26. (If JP25 in 1-4 position, then MAP1 is the non-inverting input to GP amp #1; if JP26 in 2-4 position, then MAP1 is the inverting input to GP amp #2.) MAP1 is analog input AN4. **NOTE:** *This jumper has been replaced by jumpers 31A, 31B, and 31C on later version cards.*
		JP31A JP31B JP31C		**6-7:** **1-6:** **1-2:** **2-4:** **2-5:** **2-3:** **3-8:**	Park position. Input L1-3 is digital control C12. MAP1 tied to digital control input C12. MAP1 tied to GP amps (JP25, JP26). MAP1 tied to AN4. MAP1 tied to digital output B04. K15 (29PL-21) to B04 output.
		JP32		**1-2:** (Def) **2-3:**	Normal FRA pot range. Extended FRA pot range.
		JP33		**1-2:** (Def) **Open:**	Always Never
		JP34		**1-2:** (Def) **Open:**	Always Never
		JP35		**1-2:** (Def) **Open:**	Always Never
		JP36		**1-2:** (Def) **Open:**	Never Never
		JP37		**1-2:** (Def) **Open:**	Always Never
		JP38		**1-2:** (Def) **2-3:**	Normal operation - LAN /RX via 11PL disabled. Bring in LAN /RX line via 11PL. **NOTE:** *JP23 must also be in 2-3 position.*
		JP39		**1-2:** (Def) **2-3:**	Normal operation - LAN TX via 11PL disabled. Bring in LAN TX line via 11PL. **NOTE:** *JP22 must also be in 2-3 position.*

Location	Jumper or Pot	Number	Nomenclature	Description	
Micro-application Card 531X139 APM_ _M_ (Optional) (continued)	Jumper	JP40		**1-2:** (Def) **2-3:** **2-4:**	RUP encoder input used as encoder input with direction controlled by encoder rotation. RUP encoder input used as pulse reference. RUP encoder input used as encoder input with direction controlled by input level at ENCB.
		JP41		**1-2:** (Def) **2-3:**	RUP encoder input used as encoder input. RUP encoder input used as pulse reference.
		JP42		**1-2:** (Def) **2-3:** **2-4:**	Normal operation - no special MUP timer or external interrupt selected. External interrupt from EBR with a LAN and without MUP encoder. External pulse counting capability from EBA when MUP encoder not used.
		JP43		**1-2:** (Def) **2-3:**	Microprocessor U30, port 1.3 use - If an 8032 processor is used, this is a spare I/O port. Microprocessor U30, port 1.3 use - If an 80C252 processor is used, this port is used to support simultaneous LAN and MUP encoder operation.
		JP44		**1-2:** (Def) **2-3:**	Microprocessor U30, port 1.4 use - Spare I/O port. Microprocessor U30, port 1.4 use - Enables this microprocessor to directly access digital control input C10.
		JP45		**1-2:** (Def) **2-3:**	Enables potentiometer P7 in GP amp #1 circuitry. Disables potentiometer P7 in GP amp #2 circuitry.
		JP46		**1-2:** (Def) **2-3:**	Enables potentiometer P8 in the GP amp #2 circuitry. Disables potentiometer P8 in GP amp #2 circuitry.

Location	Jumper or Pot	Number	Nomenclature	Description
Micro-application Card 531X139 APM_ _M_ (Optional) (continued)	Jumper	JP47		**3-4:** (Def) Microprocessor U30 is type 8032. **4-5:** Microprocessor U30 is type 80C252 - Allows simultaneous LAN and MUP encoder operation. **NOTE:** *This jumper has been replaced by jumpers 47A, 47B, and 47C on later version cards.*
		JP47A		**1-2:** (Def) Normal position. **2-3:** For use with 80C252 microprocessor.
		JP47B		**1-2:** Not connected (for future use). **2-3:** Enable L1-14 as logic input.
		JP47C		**1-2:** Not connected (for future use). **2-3:** Not connected (for future use).
		JP48		**1-2:** For U33 type 27256 (JP20 must be 2-3). **2-3:** For U33 type 27512 (JP20 must be 2-3). **NOTE:** *For U33 part numbers 409X113AA107 through 409X113AB107, 409X114AA107 and 409X114BA107 set JP48 in 1-2 position. Otherwise set in 2-3 position.*
		JP51		**1-2:** (Def) L1-13 non-inverting MUP interrupt. **2-3:** L1-13 inverting MUP interrupt.
		JP52		**1-2:** Enables 13PL-9 as C10 input. **2-3:** Enables 29PL-1 as L1-1 input.
		JP53		**1-2:** Enables 13PL-11 as C11 input. **2-3:** Enables 29PL-2 as L1-2 input.
		JP54		Jumpers set max input voltage to optical isolators from ENCR (11PL-12). **Position** — **Max. Volts** **1-2:** 10 volts **2-3:** 5 volts **3-4:** 1.5 volts **Open:** 15 volts
		JP55		Jumpers set max input voltage to optical isolators from ENCB (11PL-14). **Position** — **Max. Volts** **1-2:** 10 volts **2-3:** 5 volts **3-4:** 1.5 volts **Open:** 15 volts

Location	Jumper or Pot	Number	Nomenclature	Description
Micro-application Card 531X139 APM_ _M_ (Optional) (continued)	Jumper	JP56		Jumpers set max voltage input to optional isolators from ENCA (11PL-16). **Position** — **Max. Volts** **1-2:** 10 volts **2-3:** 5 volts **3-4:** 1.5 volts **Open:** 15 volts
		JP57		**NOT USED** - Set in 1-2 position.
		JP58		**1-2:** (Def) Standard LAN selected. **2-3:** Optically isolated LAN selected. **NOTE:** *JP58, JP59, JP60 must all be set either 1-2 or 2-3.*
		JP59		**1-2:** (Def) Standard LAN selected. **2-3:** Optically isolated LAN selected.
		JP60		**1-2:** (Def) Standard LAN selected. **2-3:** Optically isolated LAN selected.
Encoder Process Interface Card 531X134 PRU_ _G1 (Optional)	Switch	SW1	Tach Range Select	These switches provide coarse scaling of the tach circuit as follows: DIP switch positions are selected for the legend of the side depressed. To select position 3 ON, depress the switch at position 3 next to the ON legend.

SW1 Position No. "Closed" or "ON"	Tach Volt At Maximum Speed DC Tach	AC Tach
All "open" or "off"	10 - 12.5V	11.1 - 13.9V
1	12.5 - 15.5V	13.9 - 17.2V
2	15.5 - 19V	17.2 - 21.1V
1, 2	19 - 22V	21.1 - 24.4V
3	22 - 25V	24.4 - 27.8V
1, 3	25 - 30V	27.8 - 33.3V
1, 2, 3	30 - 37.5V	33.3 - 41.7V
4	37.5 - 46V	41.7 - 51.1V
1, 2, 4	46 - 56V	51.1 - 62.2V
1, 2, 3, 4	56 - 70V	62.2 - 77.8V
5	70 - 87V	77.8 - 96.7V
2, 3, 5	87 - 105V	96.7 - 117V
1, 2, 4, 5	105 - 120V	117 - 133V
1, 2, 3, 4, 5	120 - 144V	133 - 160V
6	144 - 180V	160 - 200V
1, 2, 4, 6	180 - 205V	200 - 228V
5, 6	205 - 250V	228 - 278V
1, 2, 3, 4, 5, 6	250 - 310V	278 - 344V

NOTE: *SW1 postion 7 must be OFF (or open) for DC tachs or ON (or closed) for AC tachs. Maximum voltages less than 10 volts are not to be used.*

Location	Jumper or Pot	Number	Nomenclature	Description
Encoder Process Interface Card 531X134 EPR_ _G1 (Optional) (continued)	Pot	P1	TACH SCALE	Used to scale the tach circuit to obtain 4.0 volts at test point SFB at maximum speed.
		P2	SMET	Used to adjust the analog output signal at 3TB20 of the speed metering circuit for 10 volts at maximum motor speed. Maximum current load on this signal is 2mA. **NOTE:** *A "Digital Tach System" will not provide an SMET output of speed.*
		P3	DVM TRIM	With DVM test point connected to +5V and optional Programmer set to DVM operation, adjust P3 until the Programmer displays 5.00.
	Jumper	JP1		RXD MODE SELECT **1-2:** Enables RS422 reception on the serial link data line. **2-3:** Enables RS232C reception.
		JP2		5-50 Milliamp Select (see JP3)
		JP3		4-20 Milliamp Select - JP2 and JP3 select the process follower mode per the following table: **JP2 JP3 Range** **1-2: 1-2:** 5-65ma current loop follower **1-2: 1-2:** 5-50ma current loop follower **2-3: 1-2:** 4-20ma current loop follower **2-3: 2-3:** 1-5ma current loop follower or 10V voltage follower
		JP4		SMET TYPE SELECT **1-2:** Unipolar speed meter driver mode **2-3:** Bipolar meter driver mode (DC tach only)
		JP5		MARKER PULSE MODE SELECT **1-2:** Differential RS422 compatible, with 0.33 ohm termination **2-3:** Single ended input at PCP
		JP7		MARKER PULSE ENABLE **1-2:** Not used **2-3:** Enables marker pulse to TUP

Location	Jumper or Pot	Number	Nomenclature	Description
Encoder Process Interface Card 531X134 EPR_ _G1 (Optional) (continued)	Jumper	JP8		ANALOG TACH FILTER SELECT **1-2:** Enables 4ms filter in analog tach feedback; required for AN tach operation **2-3:** Disables 4ms filter; required for speed regulator performance above 20 radians
		JP9 & JP10		DIGITAL TACH OR ENCODER MODE SELECT **1-2:** Differential RS422 compatable with termination resistor **2-3:** Single-ended input at 10PL-5 and 10PL-7 **Open:** Differential operation with no terminating resistor.
		JP11		TXD MODE SELECT **1-2:** 12PL-9, 12PL-12 RS422 transmitted data mode **2-3:** 12PL-9 RS232 transmitted data mode
		JP12		DTR MODE SELECT **1-2:** 12PL-8, 12PL-13 RS422 DTR handshake mode **2-3:** 12PL-8 RS232 DTR handshake mode
		JP13		DTR RS232 LOGIC POLARITY **1-2:** 12PL-8 high=drive ready to receive **2-3:** 12PL-8 low=drive ready to receive
		JP14		CTS MODE SELECT **1-2:** For RS422 CTS handshaking **2-3:** For RS232 CTS handshaking **NOTE:** *For Series 6 interfaces using the CCM Card, set JP1 and JP11 to 1-2 and JP12-JP14 to 2-3. 12PL-7 should be tied to +5V.*
		JP15		ENCODER COUNTER DIRECTION CONTROL **1-2:** Normal control by encoder direction of rotation. **2-3:** Up only
		JP16		ENCODER A QUAD B ENABLE **1-2:** Normal A quad B encoder operation **2-3:** Count on leading edge of track A only
		JP17		LUP MARKER PULSE SELECT **1-2:** Marker pulse not connected to 9PL-14 (normal operation) **2-3:** Marker pulse circuit output passed to 9PL-14 for use by LUP
		JP18		LUP DM5 SELECT **1-2:** DM5 not connected to 9PL-13 (normal operation) **2-3:** DM5 connected to 9PL-13 for use by LUP

TABLE 1A. TEST POINTS

As a further aid to troubleshooting the DC-300 drive controller, a series of test points have been provided to allow for specific signal observation and/or measurements. Test points not defined in the following listing are for manufacturer's use only and should not have instrumentation connected to them.

CONTROL CARD TEST POINTS

Test Point Name	Description	Nominal Value
ACOM	Common for analog circuit power supplies. (Not for use external to the drive.)	
CFB	Armature current feedback signal - positive when forward SCRs are conducting. Maximum external loading is 5 mA.	Calibration label voltage level at rated drive current (approx. 1.0 VDC)
DA0	Scaled output from 12-bit digital-to-analog converter (DAC). **NOTE:** *DA0 is only available on version 300 Control Cards.*	
DCOM	Common for digital circuit power supply. (Not for use external to the drive.)	
DFP	Delayed firing power for the SCR gate circuits. Maximum external loading is 25 mA. Enabled only when armature SCR firing is allowed.	+24V when enabled or 0 volts when disabled
MUP1 (DM1)	Multiuse point 1, connects to 6PL-9. Multiuse points may be used to jumper signals from other test points to the associated connector for custom applications. Refer to elementaries supplied with drive.	
MPU2 (DM2)	Multiuse point 2, connects to 6PL-10.	
MPU3 (DM3)	Multiuse point 3, connects to 6PL-11.	
FC	Field current feedback from Power Supply Card. Maximum external loading is 1 mA. FC = Max. rated motor field Amps X 4 VDC *divided by* Max. field current set by JP1and JP2 **NOTE:** *See TABLE 1, Power Supply Card for maximum field current set by JP1 and 2.*	
JOGR	Jog reference input at 6PL-7.	
-15V	-15 VDC power supply.	-14.25 to -15.75 VDC
+15V	+15 VDC power supply.	+14.25 to +15.75 VDC
+5V	+5 VDC power supply.	+4.95 to +5.05 VDC

CONTROL CARD TEST POINTS (contd.)

Test Point Name	Description	Nominal Value
R/F	High when reverse SCRs being fired Low when forward SCRs being fired.	High = 2 to 5 V Low = 0 to 2 V
SRS	Scaled speed reference signal.	4 volts at top speed reference
VFBB or VFB	Armature voltage feedback signal. Positive when forward motoring. Maximum external loading is .1 mA. Not for external use. **NOTE:** *VFBB is affected by P8-VMET.*	4 volts at rated armature voltage (with P8-VMET at full CCW position only; otherwise 4-10 volts)

POWER SUPPLY CARD TEST POINTS

Test Point Name	Description	Nominal Value
TP1	Field current feedback signal (FC). FC=I_{field}/Gain Factor (See TABLE 1, MFC/ Power Supply Card JP1 and JP2 for Gain Factor setting.)	Maximum loading is 1 mA.

OPTIONAL PROCESS INTERFACE CARD TEST POINTS

Test Point Name	Description	Nominal Value
ACOM	Common of analog circuit power supplies.	
DCOM	Common of digital circuit power supply.	
+5V	+5 VDC power supply.	+4.95 to +5.05 VDC
DVM	Input probe connection for digital voltmeter. Requires optional Programmer.	
SFB	Analog speed feedback signal. No value when digital tach or no tach being used in drive system.	4 volts at top speed
PFRF	Output of the reference interface circuitry.	10 volts at top speed
DM5	Dummy point 5, connects to 10PL-16.	
TPN	Digital or AN tach pulse output.	

OPTIONAL APPLICATION CARD TEST POINTS

Test Point Name	Description	Nominal Value
ASR	Auto System Reference signal.	4 volts at top speed reference
DA8	Output from the 8-bit D-A converter. Only on version APG Cards.	±10 volts = maximum signal
DA12	Output from the 12-bit D-A converter. Only on version APG Cards.	±10 volts = maximum signal
FRA	Field reference adjust signal.	±10 volts = full adjust
RPU	Encoder interface marker track, one pulse per revolution.	
OSC	500 KHz TTL oscillator used by the shared bus encoder interface and by the real time clock on MUP.	
DM4	Dummy point 4, connects to 3PL-30. May be used to jumper signals from other test points to 3PL for custom applications, or to bring signals from the Control Card to the general purpose amplifier circuits.	
CLK	Not Used	
SPT	Not Used	

OPTIONAL ENCODER PROCESS INTERFACE CARD TEST POINTS

Test Point Name	Description	Nominal Value
ACOM	Common of analog circuit power supplies.	
DCOM	Common of digital circuit power supply.	
+5V	+5V DC power supply.	+4.95 to +5.05VDC
DVM	Input probe connection for digital voltmeter. Requires optional Programmer.	
SFB	Analog speed feedback signal. No value when digital tach or no tach being used in the drive system.	4 VDC at top speed
PFRF	Output of the reference interface circuitry.	10 VDC at top speed

OPTIONAL ENCODER PROCESS INTERFACE CARD TEST POINTS (continued)

Test Point Name	Description	Nominal Value
DM5	Dummy point 5, connects to 10PL-16 (3TB-25).	
TPN	Net digital or AN tach pulse output.	
STPRF	Access point to process follower amplifier. Used to bring in a setpoint controller reference by jumpering STPRF to DM5 and using 10PL-16 as the input.	
CLK	Net pulse input into the encoder up/down counter.	
RPU	Marker pulse input into the up/down counter reset circuitry.	

NOTE: *All analog and digital voltage signals should be referred to 3TB-PCOMX only (unless shown differently on system elementary diagrams).*

OPTIONAL MICROAPPLICATION CARD TEST POINTS

Test Point Name	Description	Nominal Value
ASR	Auto System Reference signal used to calibrate auto speed reference. Apply analog reference corresponding to top speed to 13PL-23 and adjust P1 (and JP1 if necessary) to obtain 4 volts.	4 volts at top speed reference
CLK	Output from shared bus encoder interface quadrature track combinatorial logic that generates one pulse at CLK for each transition at the encoder main track inputs ENCA and ENCB. For an "n" pulse per revolution encoder, the output at CLK will be 4n pulses per revolution. Used as input to an up/down counter that is read on the shared bus.	
FRA	Field reference adjust signal. Used to calibrate the analog input FRA (field forcing adjustment). One volt at this test pont increases/decreases the FLDMX parameter setting by 204 counts.	$\pm$10 volts = full adjust
OSC	460 KHz (G1) or 500 KHz (G2) TTL oscillator used by the shared bus encoder interface and by the real time clock on MUP.	

OPTIONAL MICROAPPLICATION CARD TEST POINTS (continued)

Test Point Name	Description	Nominal Value
RPU	Output from shared bus encoder interface marker track logic. Generates one pulse per revolution and is used as a reset pulse for the up/down counter.	
SPT	Spare TTL output from the shared bus. NOT USED	
STS	TTL output from MUP's A-D converter that goes high while conversions are in progress.	
TP4	Spare TTL input or output point on the shared bus.	
TP5	Spare TTL input or output point on the shared bus.	
TP6	Spare TTL output point from MUP.	

NOTES:

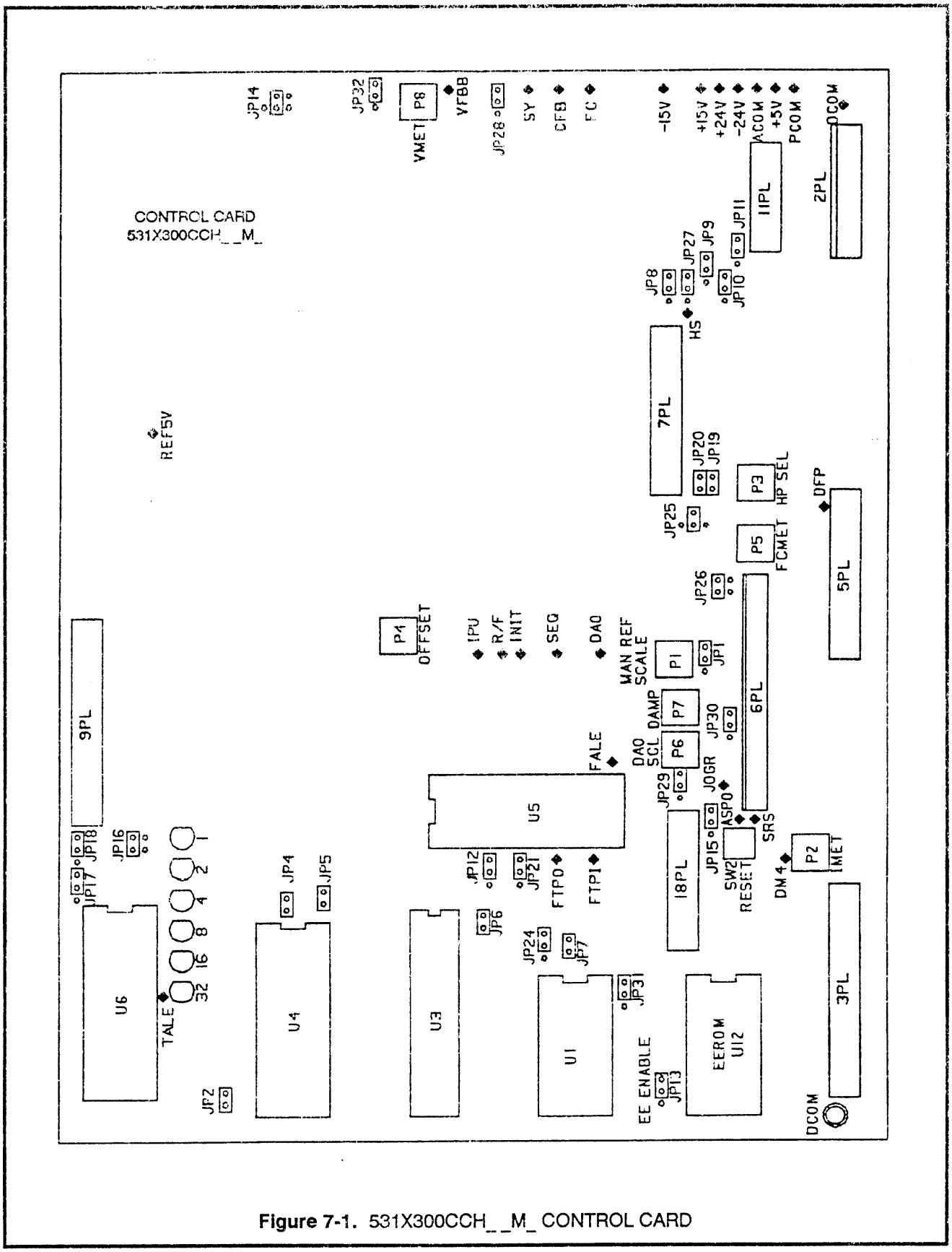


Figure 7-1. 531X300CCH__M_ CONTROL CARD

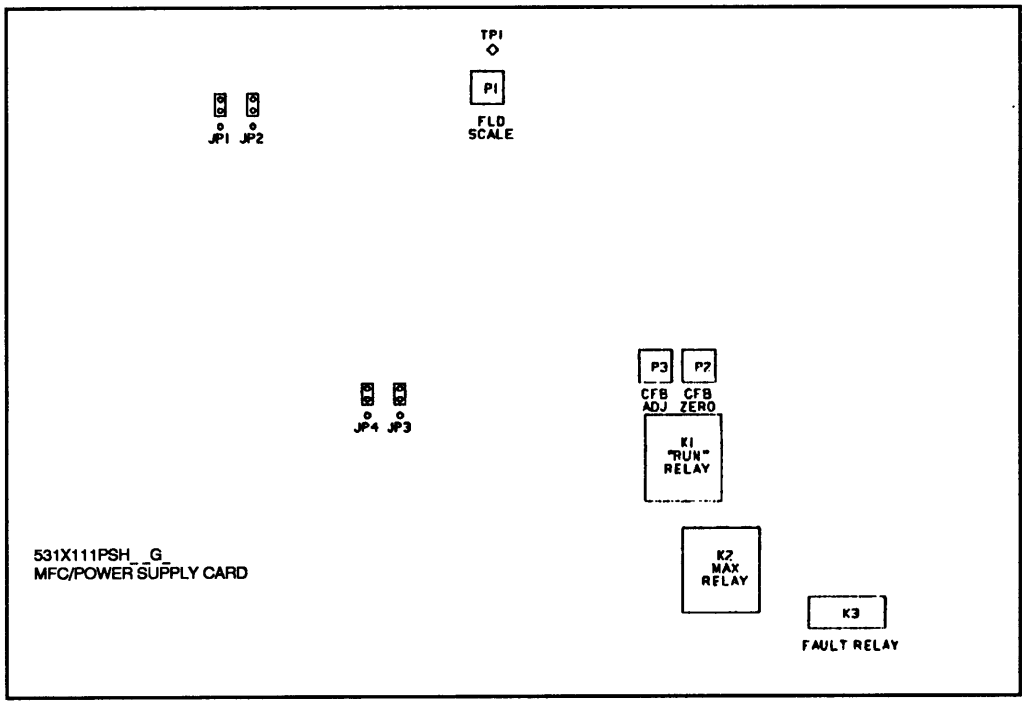


Figure 7-2. 531X111PSH_ _G1/G2 POWER SUPPLY CARD

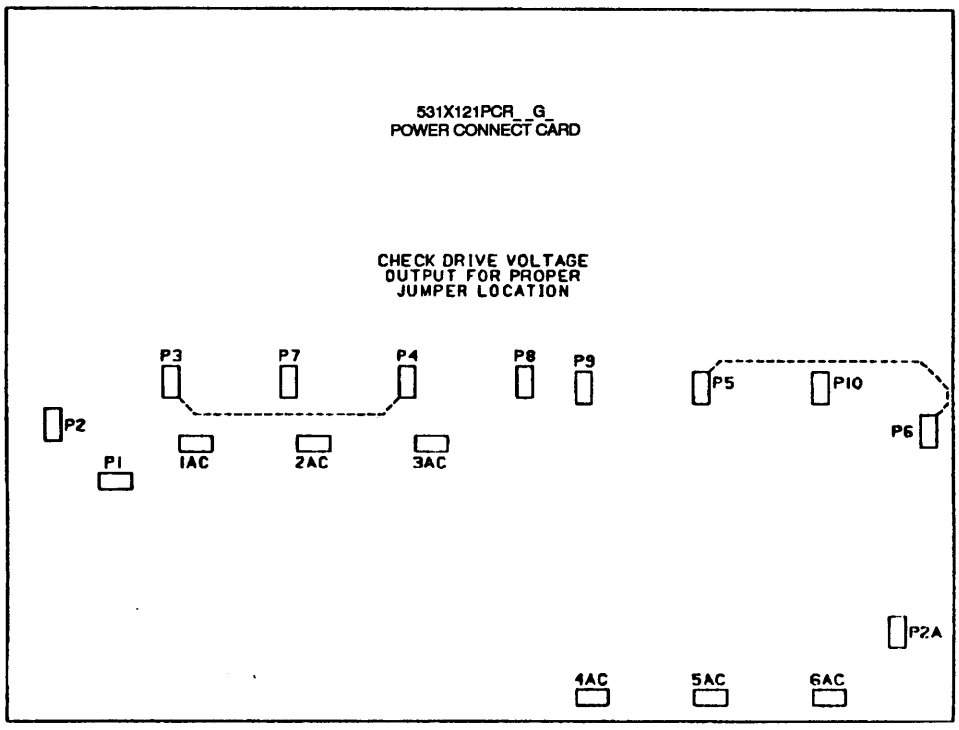


Figure 7-3. 531X121PCR_ _G_ POWER CONNECTION CARD (4 QUADRANT)

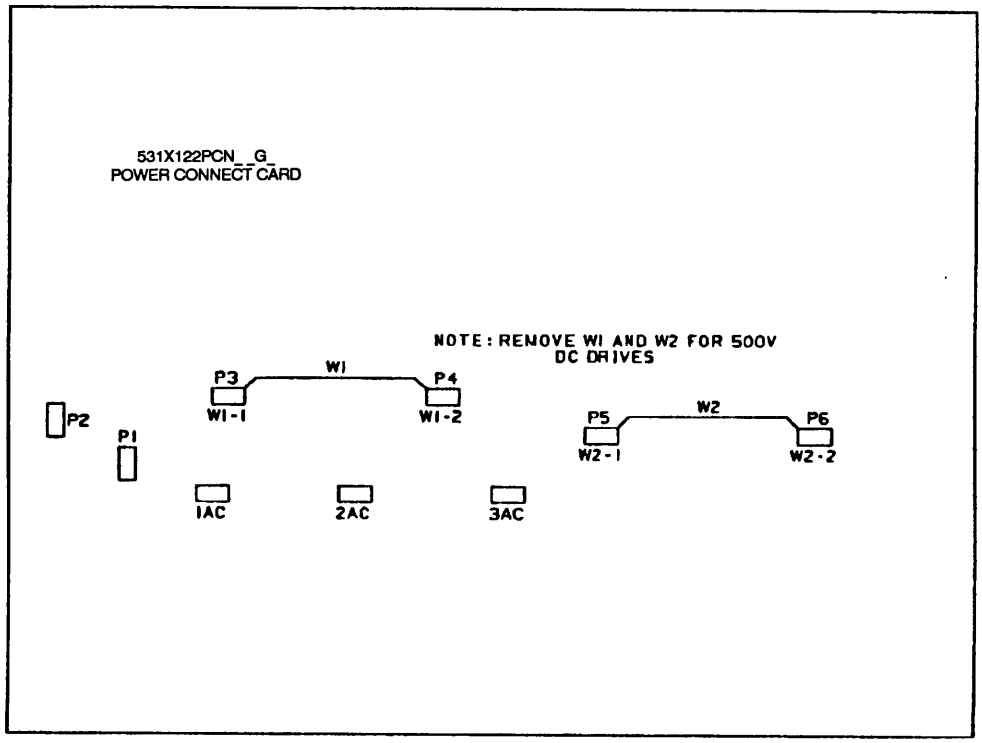


Figure 7-4. 531X122PCN__G_ POWER CONNECTION CARD (1 QUADRANT)

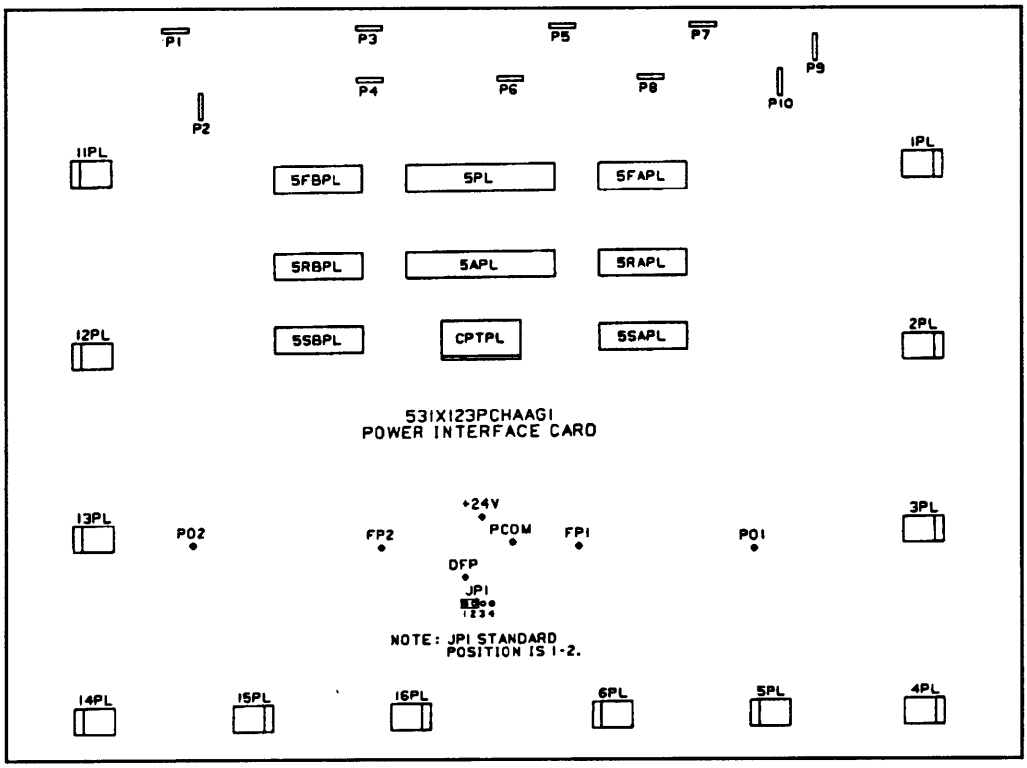


Figure 7-5. 531X123PCH__G_ POWER CONNECTION CARD

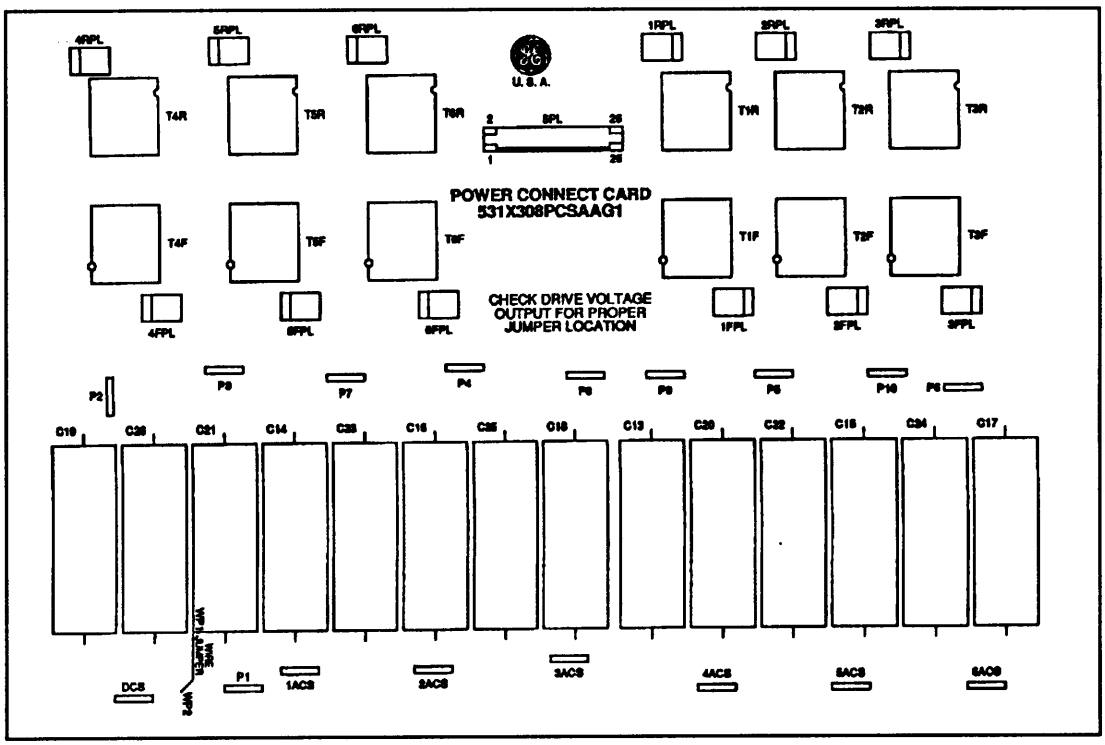


Figure 7-6. 531X308PCS_ _G1/G2 POWER CONNECTION CARD

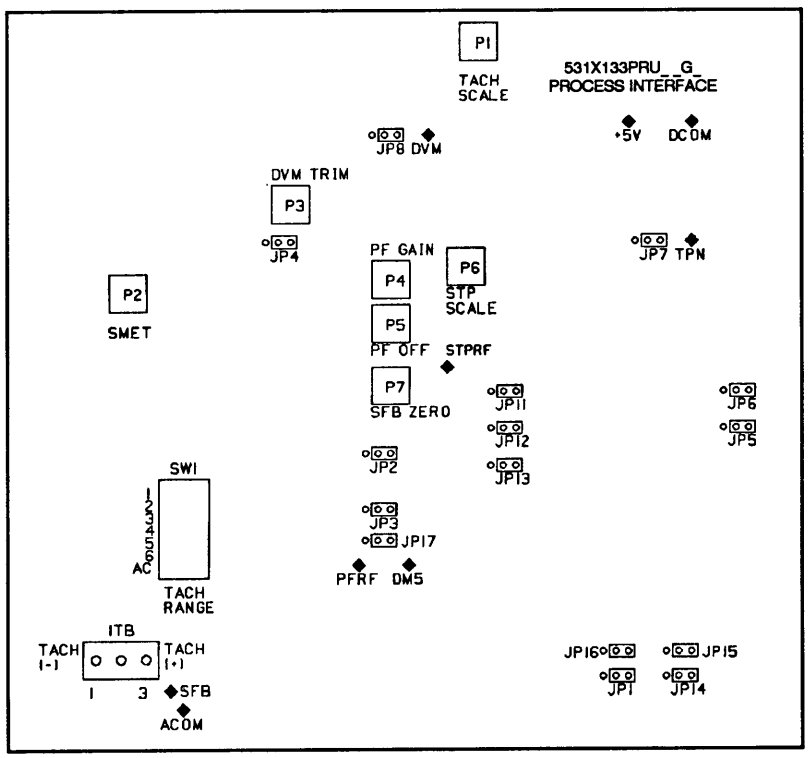


Figure 7-7. 531X133PRU_ _G_ PROCESS INTERFACE CARD

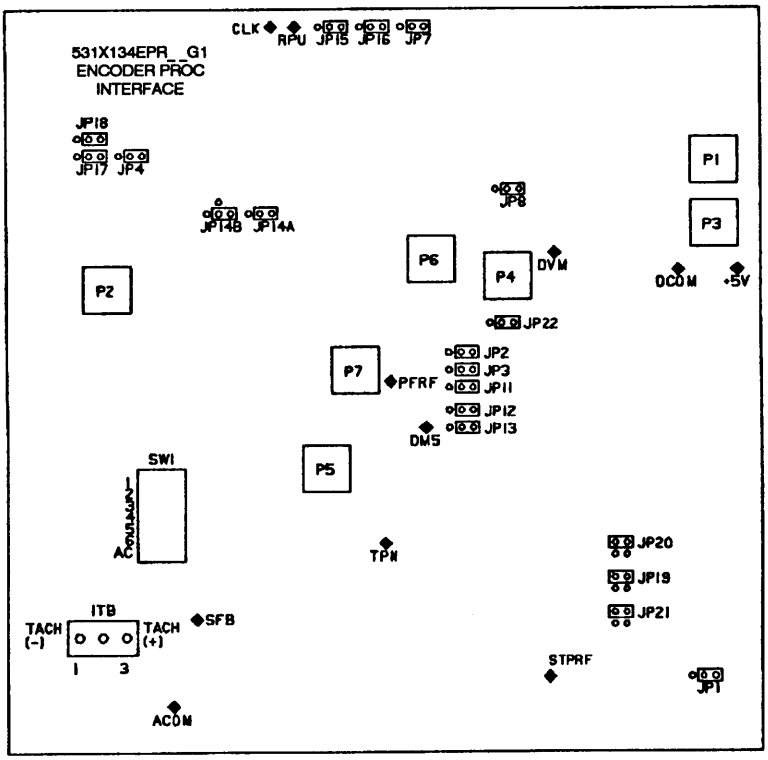
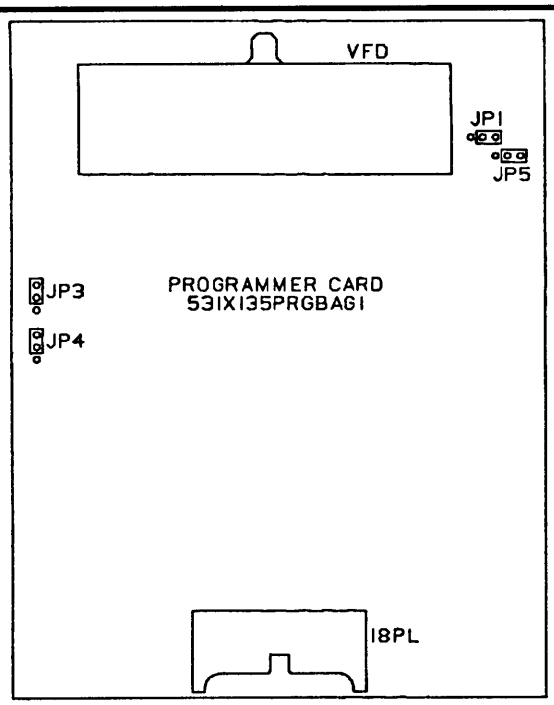


Figure 7-8. 531X134EPR_ _G1 ENCODER PROCESS INTERFACE CARD



NOTE: JP2 ON OLDER VERSION CARDS IS IN APPROX. SAME LOCATION AS JP5.

Figure 7-9. 531X135PRG_ _M_ PROGRAMMER CARD

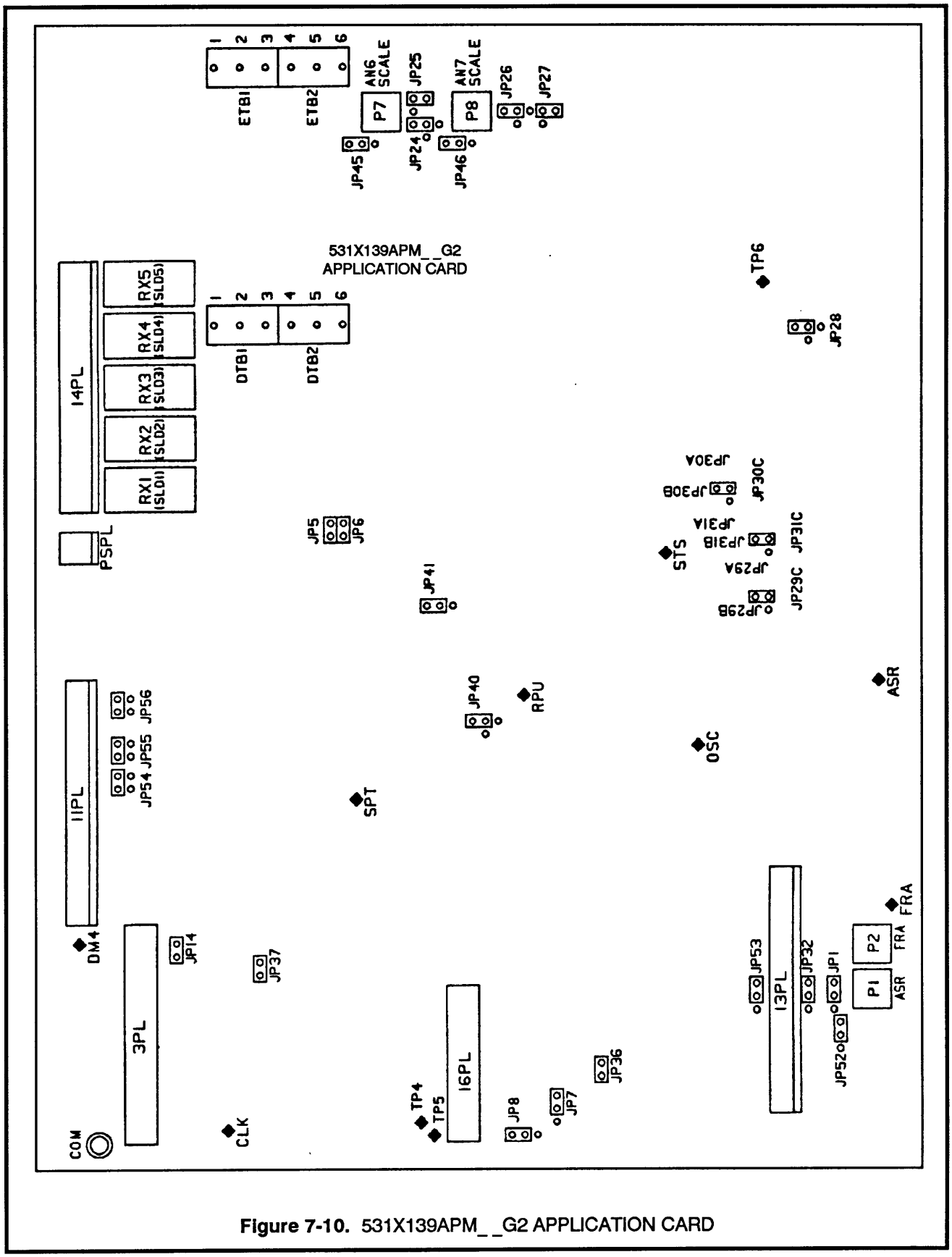


Figure 7-10. 531X139APM__G2 APPLICATION CARD

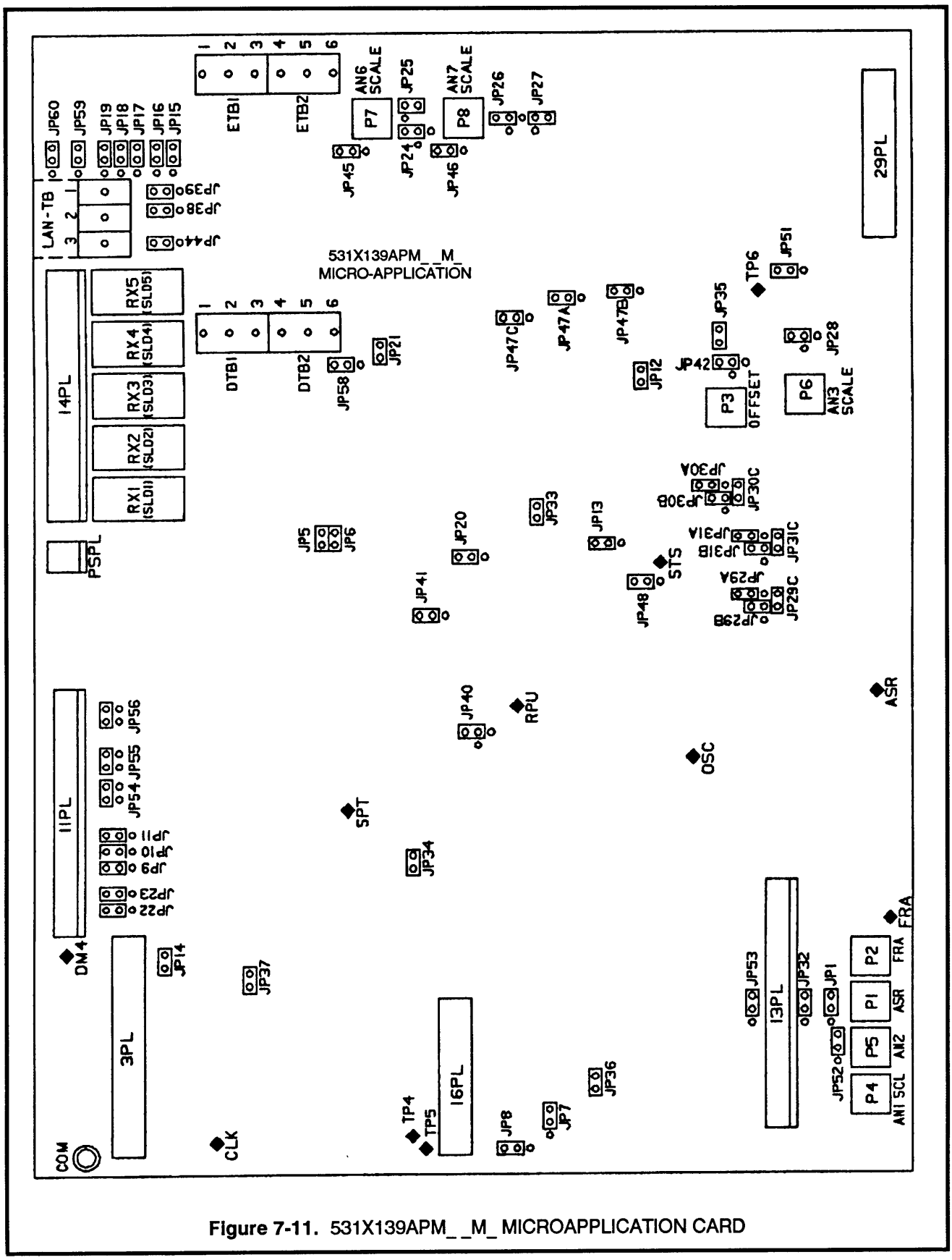


Figure 7-11. 531X139APM_ _M_ MICROAPPLICATION CARD

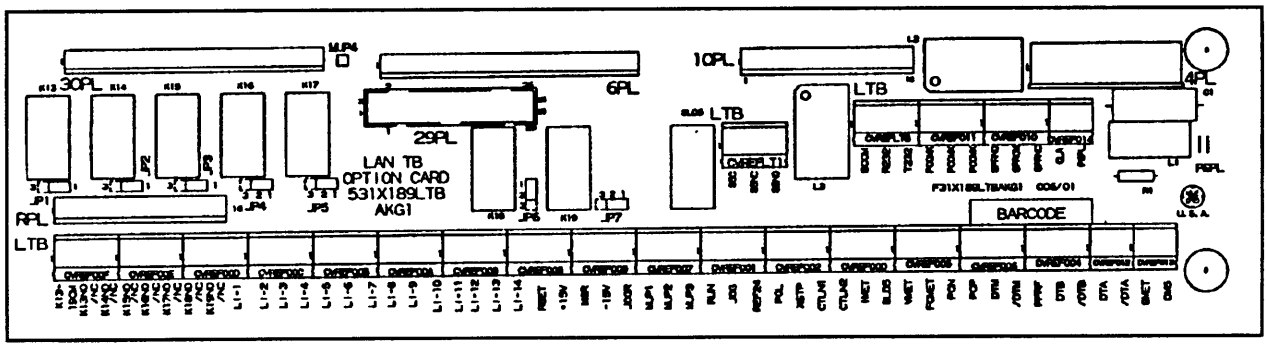


Figure 7-12. 531X189LTB

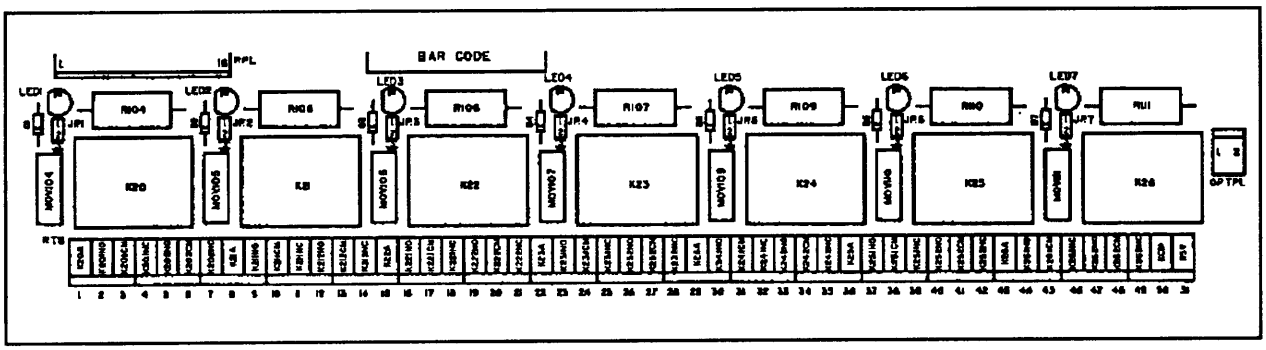


Figure 7-13. 531X191RTB

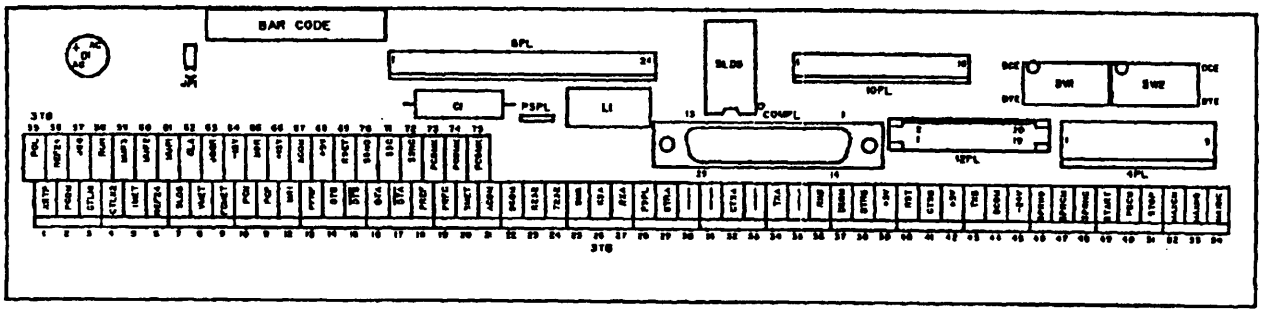


Figure 7-14. 531X170TBS_ _G_ TERMINAL BOARD CARD - 3TB

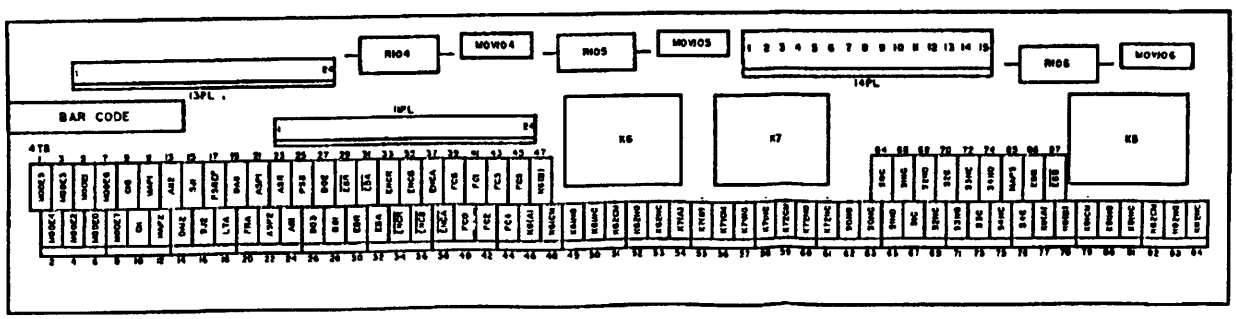


Figure 7-15. 531X171TMA_ _G_ MICROAPPLICATION TERMINAL BOARD CARD - 4TB

8. SOFTWARE ADJUSTMENTS

TABLE 2 on the following pages is a listing of "standard" DC-300 software adjustments (parameters) common in most DC-300 drive applications. Included is the **"DC-300 Parameter Map"** (equivalent analog representation of parameters), the **"Key Parameter Adjustments"**, and the **"RAM Variable Scaling"** table that defines the scaling of RAM addresses available for interrogation of the drive control "circuit". The parameters can be examined and adjusted with the optional Programmer (refer to **"Use of Optional Programmer Module"**) or via the serial link.

Additionally, a **Custom Software Description** is shipped with every drive. The **Custom Software Description** describes the setting and/or function of various programmable jumpers and adjustments (hardware and software) within the drives that are identified with the specific model number of that drive. Included in that book is a listing of specific programmer options, their descriptions, and a set of software adjustment definitions pertinent to the specific drive.

PARAMETER MAP (Figure 8-1)

The Parameter Map is a diagnostic aid designed to reference specific software adjustments and correlate them to their specific regulating functions. For example, **ADR015-BSLPG** is represented as a potentiometer feedback across the field current regulating amplifier, giving an effective linear gain adjustment of input to output.

Software adjustments are shown next to their symbol with the mnemonic and the parameter address in a box beside it. Hardware adjustments are represented by a symbol, the adjustment name, and the card it is located on (e.g.: **FCMET** is a potentiometer, **P5** on the Control Card). RAM addresses that can be used to examine outputs of certain stages of regulation through **TEST 11, TEST 21, TEST 22,** or **D-A Converters** are shown as a three-digit number in a circle connected to the regulator section it is associated with (e.g.: Speed loop output can be examined by using RAM address **053** in **Test 11, 21, 22,** or the **D-A** converters).

The optional Programmer display scaling adjustments are shown in the lower left section of the map as well as their relation to feedback.

KEY PARAMETER ADJUSTMENTS

The following parameter adjustments, listing Programmer mnemonic, address number, function name, and description, will prove helpful in setup and troubleshooting of the DC-300 drive. This is a partial list and should be used in conjunction with the **"Parameter Map"**. For a more complete list of parameters contained in a specific drive, refer to the Custom Software Description furnished with that drive.

The following is a description of column headers:

NAME	Mnemonic displayed by Programmer when interrogating parameter.
ADR	Parameter address.
UNITS	Unit designation of displayed number (example - RPM).
MIN	Minimum allowable value.
MAX	Maximum allowable value.
STEP	Amount by which value can be incremented or decremented.
DEFAULT	Standard value if no special setting is required.

NOTE: *Parameters listed in TABLE 2 are grouped functionally per the parameter map (Figure 8-1).*

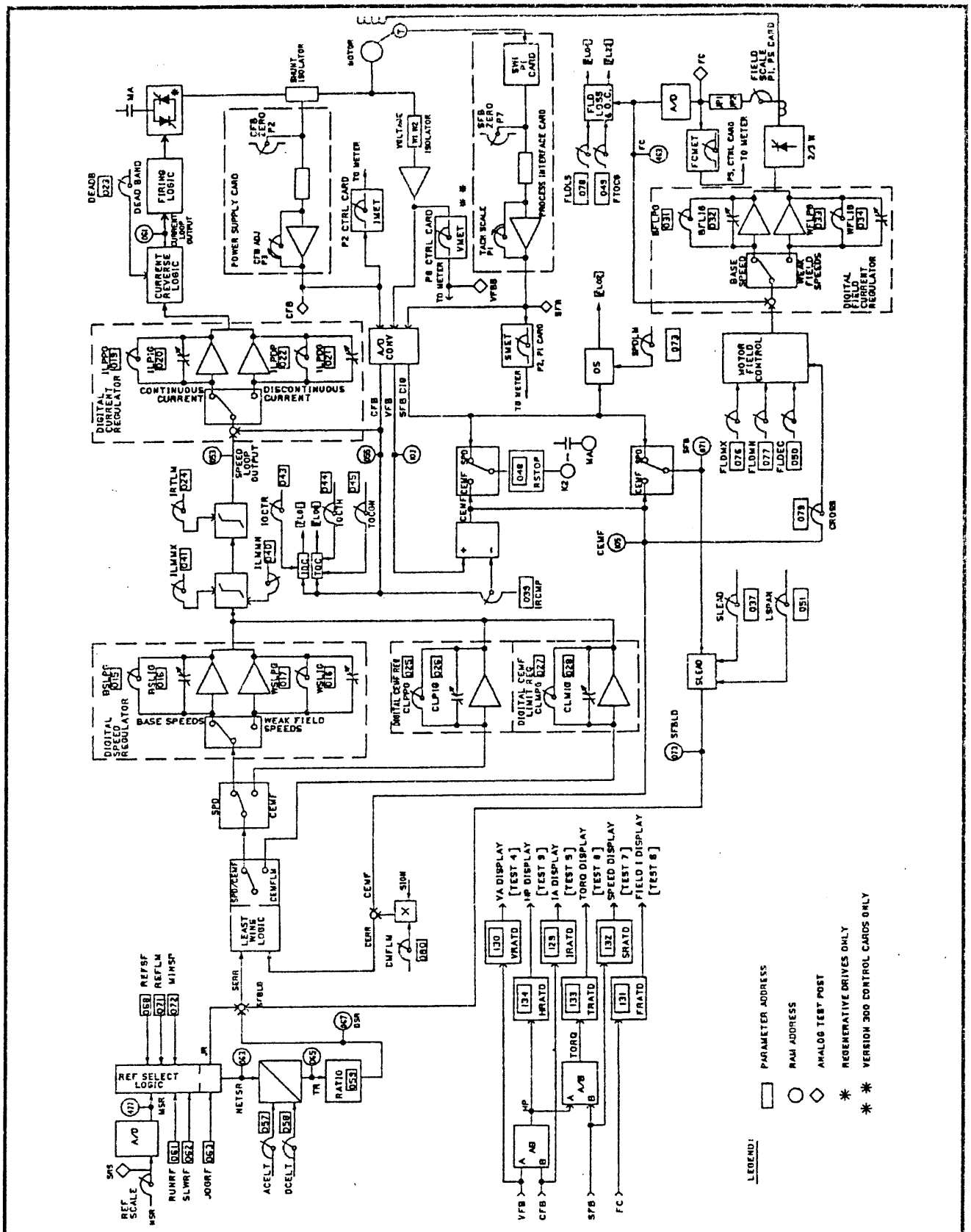


Figure 8-1. DC-300 PARAMETER MAP

TABLE 2: KEY PARAMETER ADJUSTMENTS

MOTOR FIELD

Name	Adr.	Units	Min.	Max.	Step	Default
FLDMX	076	None	0	2047	1	800
		Full Field Current -- sets the motor field current below base speed. Also is the voltage reference in the open loop mode of field operation. In the current regulated mode, 1638 corresponds to 4 volts at test point FC. The field current scaling jumpers on the Power Supply Card should be set to give the largest feedback at FC that does not exceed roughly 4.5 volts. In the open loop mode, 0 corresponds to fully retarded firing and 2047 corresponds to fully advanced firing. FLDMX = 2048 x If (max) *divided by* 5 x Gain Factor **NOTE:** *The gain factor is determined by the field current scaling jumpers on the Power Supply card (refer to Table 1).*				
FLDMN	077	None	0	2047	1	200
		Minimum Field Current -- sets the minimum field current level the drive will regulate. 1638 equals 4 volts at FC. FLDMN = 0.6 x 2048 x If (min) *divided by* 5 x Gain Factor. **NOTE:** *This equation varies for material handling applications.*				
FLDLS	078	None	0	2047	1	100
		Field Loss Trip Setpoint -- sets the level of field current below which the drive will trip. 1638 equals 4 volts at FC. FLDLS = 2048 x If (min) *divided by* 12.5 x Gain Factor. **NOTE:** *A setting of 60,000 disables the field loss trip function. This equation varies for material handling applications.*				
CROSS	079	None	0	2047	1	1850
		Field Crossover -- sets the CEMF level at which the field will begin to weaken. A setting of 1638 corresponds to rated motor voltage (4 volts at VFB). Determine % of Rated Armature volts desired at Crossover. Multiply that % by 1638 to obtain CROSS parameter setting.				
FLDEC	050	None	0	255	1	0
		Field Economy -- sets level of field current when drive is stopped. Determine % of FLDMX desired for field economy. Then FLDEC will equal that percent x 255. **EXAMPLE:** 70% of maximum field current desired for field economy. 0.70 x 255 = 178.5 Setting of FLDEC = 179. **NOTE:** *A setting of zero defaults FLDEC to 70% field current also.*				
BFLPG	031	None	0	255	1	40
		Base Field Loop Proportional Gain -- sets the proportional gain of the field current regulator below base speed.				
BFLIG	032	None	0	255	1	20
		Base Field Loop Integral Gain -- sets the integral gain of the field current regulator below base speed.				

MOTOR FIELD (continued)

Name	Adr.	Units	Min.	Max.	Step	Default
WFLPG	031	None	0	255	1	40
		Weak Field Loop Proportional Gain -- sets the proportional gain of the field current regulator above base speed.				
WFLIG	032	None	0	255	1	20
		Weak Field Loop Integral Gain -- sets the integral gain of the field current regulator above base speed.				
FTOCG	049	None	0	255	1	82
		Field TOC Gain -- sets the time constant of the field timed overcurrent integrator. Set to 0 to inhibit this feature. For field forcing of FF per unit over a setting of FLDMX, the drive will trip in 6990/(FLDMX * OFF * FTOCG) minutes. The default allows 0.1 P.U. forcing over a nominal FLDMX of 850 for 1 minute.				

LINEAR TIME

ACELT	057	Decisec	0	9999	1	20
		Acceleration Time -- sets the rate at which motor speed will increase in response to speed change command. For linear time, enter time in tenths of a second for accel from 0 to top speed. ACELT = Time (in sec.) x 10. Some drives may have an optional extended linear time range. Refer to the custom Software Description for information.				
DCELT	058	Decisec	0	9999	1	20
		Deceleration Time -- sets the rate at which motor speed will decrease in response to speed change command. For linear time, enter time in tenths of a second for decel from top speed to 0. DCELT = Time (in sec.) x 10. Some drives may have an optional extended linear time range. Refer to the custom Software Description for information.				

REFERENCES AND SCALING

RATIO	059	% x 100	0	19999	1	10000
		Ratio -- sets the ratio (draw) between the actual speed and the speed reference (NETSR). A ratio greater than 100% (RATIO set at 10000) causes the drive to run faster than the speed reference; below 100%, drive runs slower than the speed reference.				
REFSF	068	None	0	6553	1	6553
		Reference Scale Factor -- sets the factor by which references entered from the serial link and Programmer will be multiplied and by which speeds displayed on the Programmer and sent to the serial link will be divided. Enter the value of speed reference which corresponds to top speed (4 volts at SFB). Important - affects settings of RUNRF, SLWRF, JOGRF, R1RF, R2RF, SLKRF, parallel input scaling, and speed reference display. With Tach, REFSF = Top Speed RPM; for CEMF regulator, REFSF = Base Speed RPM at rated armature volts.				

REFERENCES AND SCALING (continued)

Name	Adr.	Units	Min.	Max.	Step	Default
RUNRF	061	RPM	0	6553	1	0
		"Run" Key Reference -- sets the preset reference for RUN commands from the serial link. Also may be used as the preset reference for manual start commands. Also serves as the top speed reference for motor operated rheostat (MRH) mode. May be a speed, torque or position depending on jumper setting in ADR006-RUNJP. Normally, RUNRF = Top Speed (RPM).				
SLWRF	062	RPM	0	6553	1	0
		"Slow" Key Reference -- sets the preset reference for SLOW commands from the serial link. Normally, SLWRF = Slow Speed (RPM).				
JOGRF	063	RPM	0	6553	1	10% of REFSF
		"Jog" Key Reference -- sets the preset reference for Jog commands from the serial link. Also may be used as the preset reference for manual jog commands. JOGRF = Jog Speed (RPM).				
RPMSF	070	RPM	0	6553	1	6553
		RPM Scale Factor -- sets the speed in RPM which corresponds to 4 volts at SFB. **NOTE:** *Change parameter only while the drive is not running, and toggle the reset switch afterwards to make the change fully effective.* **IMPORTANT:** *Affects setting of REFLM, MINSP, SPDLM, tapered current limit speed setting and digital tach scaling. Same as setting for REFSF.*				
REFLM	071	RPM	0	6656	1	6560
		Maximum Speed -- sets the level at which speed reference will be clamped if the input reference magnitude exceeds the level. Maximum value should not exceed 1.015 x RPMSF. Typically, REFLM = RUNRF + 100.				
MINSP	072	RPM	0	6553	1	0
		Minimum Speed -- sets the level at which speed reference will be clamped if the input reference magnitude is less than the level. MINSP = desired Minimum Speed (RPM). Maximum value should not exceed RMPSF.				
SPDLM	073	RPM	0	7936	1	7208
		Speed Limit Trip Setpoint -- sets the speed at which the drive will trip. SPDLM = RPM at drive overspeed trip. Should not exceed 1.2 x RPMSF.				

SPEED REGULATOR

BSLPG	015	None	0	255	1	7
		Base Speed Loop Proportional Gain -- sets the proportional gain of the outer speed regulator below base speed when tachometer feedback is used.				
BSLIG	016	None	0	255	1	8
		Base Speed Loop Integral Gain -- sets the integral gain of the outer speed regulator below base speed when tachometer feedback is used.				

SPEED REGULATOR (continued)

Name	Adr.	Units	Min.	Max.	Step	Default
WSLPG	015	None	0	255	1	7
		Weak Speed Loop Proportional Gain -- sets the proportional gain of the outer speed regulator above base speed.				
WSLIG	016	None	0	255	1	8
		Weak Speed Loop Integral Gain -- sets the integral gain of the outer speed regulator above base speed.				
SLEAD	037	None	0	255	1	10
		Speed Feedback Lead -- sets the lead time constant on tachometer feedback.				
LSPAN	051	None	0	15	1	0
		Speed Feedback Lead/Lag Span -- sets the span of the tachometer lead/lag network. A setting of 0 is treated as a two-to-one span.				
RSTOP	048	None	0	255	1	100
		Regenerative Stop Speed -- sets the speed at which the contactor opens during a regenerative stop. One Unit = 0.04% speed (for Speed Reg.). One Unit = 0.12% volts (for CEMF Reg.).				

CEMF REGULATOR

CMFLM	080	None	0	2047	1	1800
		CEMF Limit -- sets the CEMF level to which the drive will clamp. A setting of 1638 corresponds to rated motor voltage (4 volts at VFB). CMFLM = 1638 x (% motor rated voltage desired for CEMF limit). **CAUTION:** Do not set CMFLM above 1704 on regenerative drives. This may cause fuse and SCR failure to occur on regeneration.				
CLPPG	025	None	0	255	1	8
		(Tachless) CEMF Loop Proportional Gain -- sets the proportional gain of the outer CEMF regulator when CEMF is used as a feedback instead of a tachometer.				
CLPIG	026	None	0	255	1	10
		(Tachless) CEMF Loop Integral Gain -- sets the gain of the outer CEMF regulator in no tach mode.				
CLMPG	027	None	0	255	1	8
		CEMF Limit Proportional Gain -- sets the proportional gain of the spillover CEMF limit regulator.				
CLMIG	028	None	0	255	1	10
		CEMF Limit Integral Gain -- sets the integral gain of the spillover CEMF limit regulator.				

CEMF REGULATOR (continued)

Name	Adr.	Units	Min.	Max.	Step	Default
IRCMP	039	None	0	255	1	0
		IR Compensation -- sets positive or negative IR compensation for CEMF calculation.				

ARMATURE CURRENT REGULATOR

ILMMX	041	% Rated	10	510	2	150
		Maximum Current Limit -- sets the maximum DC current level allowed by the drive. One unit = 1% current.				
ILMMN	040	% Rated	10	510	2	10
		Minimum Current Limit -- sets the minimum allowable current limit level; must be set **below** the desired **maximum** current limit setting. Also acts as the third current limit setting when tapered current limit is used. One unit = 1% current.				
IOCTR	043	% Rated	10	510	2	400
		Instantaneous Overcurrent Trip Setpoint -- sets the current level at which the drive will trip. One unit = 1% current.				
TOCTH	044	% Rated	10	510	2	120
		Timed Overcurrent Sense Threshold -- sets the current level at which the TOC integrator begins to time-up. One unit = 1% current. Refer to the Custom Software Description if multi-motor function is used.				
TOCGN	045	None	0	255	1	24
		Timed Overcurrent Response -- sets the time constant of the TOC integrator. Refer to the Custom Software Description if multi-motor function is used. TOCGN = 66,666 *divided by* (Trip Time) (X-TOCTH) **NOTE:** *X = % of armature current corresponding to desired trip time.* **NOTE:** *The default values for TOCTH and TOCGN allow the drive to operate at 150% current for 90 seconds.*				
ILPPG	019	None	0	255	1	40
		Current Loop Proportional Gain -- sets the proportional gain of the inner current regulator in continuous current.				
ILPIG	020	None	0	255	1	20
		Current Loop Integral Gain -- sets the integral gain of the inner current regulator in continuous current.				
ILPDG	021	None	0	255	1	30
		Current Loop Integral Discontinuous Gain -- sets the integral gain of the inner current regulator in discontinuous current.				
ILPDP	022	None	0	255	1	0
		Current Loop Proportional Discontinuous Gain -- sets the proportional gain of the inner current regulator in discontinuous current.				

ARMATURE CURRENT REGULATOR (continued)

Name	Adr.	Units	Min.	Max.	Step	Default
DEADB	023	None	0	255	1	0

Current Regulator Deadband -- controls the sensitivity of the current loop reversal logic. A setting of 0 is maximum sensitivity (dither mode).

IRTLM	024	PU/S	1	255	1	50

Current Rate Limit -- controls the maximum rate of change of current in response to commanded changes.
Exact relation is: Current Rate = 0.9 (IRTLM) per unit per second.

OUTPUT SIGNALS

DAC0	090	None	0	65535	1	0

D-A #0 Output Control -- controls what internal variable is sent to the 12-bit D-A converter (DAC0) on the Control Card. Enter the desired RAM variable number into ADR090 to program DAC0.

RAM Adr.	Variable Name	RAM Adr.	Variable Name
053	Loop Output	071	SFB - Speed Feedback
055	CFB - Current Feedback	103	VFB - Voltage Feedback
061	Current Loop Output	105	CEMF
065	Timed Speed Reference	463	Field Current
069	Jog Reference	501	TOC Timer

NOTE: *Refer to the "RAM Variable Scaling", TABLE 3 at the end of this listing for relative value of RAM addresses through various outputs.*

DAC8	096	None	0	65535	1	0

D-A #1 Output Control -- controls what internal variable is sent to the 8-bit D-A converter (DAC8) on the Application Card. See DAC0 for programming information.

DAC12	097	None	0	65535	1	0

D-A #2 Output Control -- controls what internal variable is sent to the 12-bit D-A converter (DAC12) on the Application Card. See DAC0 for programming information.

BIT #1	098	None	0	65535	1	0

Bit#1 Output Control -- controls what internal bit is sent to bit output #1 (B01) on the Application Card. Possible outputs include:

RAM Adr.	Variable Name	RAM Adr.	Variable Name
061	Direction of Firing	148	Reference Enable
137	Fault	152	Precondition
140	Current Limit Flag	156	Zero Current Flag
141	CEMF Limit Flag		

NOTE: *In addition, the sign of all of the variables given in DAC8 output control may be used as outputs.*

BIT #2	099	None	0	65535	1	0

Bit#2 Output Control -- controls what internal bit is sent to bit output #2 (B02) on the Application Card. See Bit #1 for programming information.

OUTPUT SIGNALS (continued)

Name	Adr.	Units	Min.	Max.	Step	Default
BIT #3	100	None	0	65535	1	0
		Bit#3 Output Control -- controls what internal bit is sent to bit output #3 (B03) on the Application Card. See Bit #1 for programming information.				

DIAGNOSTICS

DIAG1	122	None	0	65535	1	0
		Diagnostic Variable #1 -- selects 1 of 2 variables which can be displayed on the serial link or Programmer using "display diagnostics". See DAC8 output control for programming information.				
DIAG2	123	None	0	65535	1	0
		Diagnostic Variable #2 -- selects second variable which can be displayed on the serial link or Programmer using "display diagnostics". See DAC8 output control for programming information.				
SCHRL	127	None	1	26	1	18
		Left Scaled Unit Character -- determines which character is displayed as the left character of the scaled speed status message. 1 represents "A"; 2, "B"; 26, "Z".				
SCHRR	128	None	1	26	1	13
		Right Scaled Unit Character -- see ADR127-SCHRL for programming information.				
IRATD	129	None	0	3276	1	100
		Displayed Arm Current Scaling -- controls the factor by which per unit current is multiplied for display on the Programmer.				
VRATD	130	None	0	2621	1	100
		Displayed Arms Volts Scaling -- controls the factor by which per unit armature voltage is multiplied for display on the Programmer.				
FRATD	131	None	0	43690	1	1000
		Displayed Field Current Scaling -- controls the factor by which per unit field current is multiplied for display on the Programmer.				
SRATD	132	None	0	6200	1	100
		Displayed Speed Scaling -- controls the factor by which per unit speed is multiplied for display on the Programmer.				
TRATD	133	None	0	3276	1	100
		Displayed Torque Scaling -- controls the factor by which per unit torque is multiplied for display on the Programmer.				
HRATD	134	None	0	3276	1	100
		Displayed HP Scaling -- controls the factor by which per unit horsepower is multiplied for display on the Programmer.				

TABLE 3. RAM VARIABLE SCALING FOR PROGRAMMER AND DAC OUTPUTS

RAM Adr.	Variable Name	Value At 100% Signal Dec.	Value At 100% Signal Hex[1]	Volts On DAC[2] At 100% Signal
053	**SLOPO** - Speed Loop Output	100	0064	2.0
055	**CFB** - Current Feedback	100	0064	2.0
061	**ILOPO** - Current Loop Output	220	00DC	4.3
063	**NETSR** - Net Speed Reference	6553	1999	8.0
065	**TR** - Timed Reference (if used)	6553	1999	8.0
067	**DSR** - Draw Speed Reference	6553	1999	8.0
069	**JR** - Jog Reference	6553	1999	8.0
071	**SFB** - Speed Feedback	6553	1999	8.0
073	**SLEAD** - Speed Feedback Lead	6553	1999	8.0
103	**VFB** - Voltage Feedback	1638	0666	8.0
105	**CEMF** - Counter Electromotive Force	1638	0666	8.0
463	**FC** - Field Current (rated current, not scaled)	2047	07FF	10.0
477	**MSR** - Manual Speed Reference	6553	1999	8.0
486	**ASR** - Automatic Speed Reference	6553	1999	8.0
501	**TOCT** - Timed Overcurrent Timer (trips at 1024)	1024	0400	0.3

NOTES: *(1) Negative numbers are represented in Two's complement.*
(2) D/A converters DAC8 and DAC12 have an op-amp with a gain of two.

9. TROUBLESHOOTING

This section contains information on detecting and correcting problems that may be encountered when working on a DC-300 controller. The DC-300 controller contains a built-in fault detection system that indicates when a fault occurs and the nature of the fault. Tables 4 and 5 are intended to aid in the correction of problems both indicated by the drive fault system and those not indicated by the fault display. Following Tables 5, 5A, and 5B is a "Stability" Troubleshooting paragraph and Table 6, "Stability Troubleshooting".

Table 4, "BugBuster™ Troubleshooting Chart" is for aid in troubleshooting problems that have no associated fault or error indication or cannot be remedied through the standard fault system. Table 4 includes a "Quick Reference Guide", an outline of the possible problem descriptions included in the chart.

Tables 5, 5A, and 5B, "Specific Fault Troubleshooting", LAN Fault Troubleshooting, and **"Specific Error Troubleshooting"** define drive fault and error codes. Fault codes are displayed on the Diagnostic LEDs and Programmer module, and broadcast on the optional Serial Link. Error codes are only related to Serial Link errors, and therefore, only broadcast on the Serial Link.

FAULT/ERROR CODES

The DC-300 controller displays faults with a series of Diagnostic LEDs (interpreted as a binary number) or with a fault message (fault number and name) on the optional Programmer.

NOTE: *In the normal condition the drive's LEDs will flash from right to left as follows:*

***One** at a time in the standby mode.*
***Two** at a time in the RUN or JOG modes.*

Tables 5, 5A, and 5B list predefined fault codes in order of the assigned fault number. To determine the fault number from the Diagnostic LEDs, add the number value of the LEDs printed on the card above or below the LEDs (also add to this total the value of 64 if the LEDs are blinking). See EXAMPLE.

TYPES OF FAULTS

The faults listed in Tables 5, 5A, and 5B may be divided into four categories:

L - The fault is **latched**, i.e., once it occurs it will not clear itself even if the cause of the fault no longer exists. The controller must be reset to clear it.

N - The fault is **non-latched**, i.e., it will clear itself if the cause of the fault no longer exists.

T - The fault **trips the drive**, i.e., it causes the controller to remove power from the motor. The controller must be reset. **All trip faults are latched.**

A - The fault is **annunciated only**, i.e., if the drive is running it will not cause the drive to shut down. However, an annunciated fault will prevent the drive from starting if it is not running. **Annunciated** faults may or may not be **latched.**

There are several ways to reset the controller.

-- A **software reset** may be executed by serial command from a device connected to the serial link.

-- A **hardware reset** may be executed by toggling the reset switch on the Control Card, by momentarily connecting **6PL-1** to **6PL-4** (**3TB69** to **3TB66**), or by momentarily interrupting incoming power to the drive.

Most **latched** faults may be adequately cleared by a software reset, however, some faults should be cleared by a hardware reset. These are indicated in the tables by the letter "**H**". If a **Software Reset** does not seem to restore the drive to normal operation, a **Hardware Reset** should be attempted.

EXAMPLE: FAULT CODE IDENTIFICATION

LED Indication	Binary Number						Decimal Number	Fault Number
	32	16	8	4	2	1		
Steady display	0	1	1	1	0	0	16+8+4 = 28	FL28
Blinking display	0	1	0	1	0	0	16+4+64 = 84	FL84

NOTE: *0 = LED OFF 1 = LED ON 1 = BLINKING LED*

BUGBUSTER™ TROUBLESHOOTING CHART QUICK REFERENCE GUIDE

PROBLEM DESCRIPTION

DRIVE FAULTS, LED INDICATION

1. All six diagnostic LEDs off.
2. All six diagnostic LEDs on.
3. Fault messages displayed (repetitive).
4. Fault messages displayed (non-repetitive, random).

PROGRAMMER PROBLEMS

5. Programmer display off.
6. Error messages displayed.
7. Programmer ineffective - no errors displayed.
8. Parameter changes ineffective.
9. Self-tune or other diagnostics do not perform.
10. SCR test failure.

FUSE BLOWING, SCR PROBLEMS, MOV FAILURES, CARD FAILURES

11. Fuse blowing, AC or DC power fuses.
12. SCR test blowing AC or DC power fuses.
13. Field fuses blowing.
14. Power supply fuses blowing.
15. MOV failures.
16. Repetitive card failures.
17. Random SCR misfirings.
18. Line voltage problems.

OPERATIONAL PROBLEMS

19. 24 Volt logic not operating properly.
20. Drive/Motor does not run.
21. Field current or voltage wrong value.
22. Drive runs in wrong direction.
23. Drive does not reverse.
24. Drive speed is incorrect.
25. Spindle orient not positioning.
26. Serial link does not function properly.
27. Erratic operation.
28. Stability problems.
29. Signal level detector chatter.
30. Too much coast time on deceleration (regen. drives).

MOTOR PROBLEMS

31. Motor sparking.
32. Motor runs hot.
33. Motor audible noise.
34. Motor vibration.

NOISE, WIRING, POWER SUPPLIES, ETC.

35. EEROM memory rewrites
36. Noise or random nuisance faults
37. Wiring procedures
38. Checking power supplies

TABLE 4: BUGBUSTER™ TROUBLESHOOTING CHART

Problem	Check/Adjust/Replace
1. All Diagnostic LEDs OFF	1. Check Programmer display for drive fault (if used). 2. Check for 3-phase AC input. 3. Check for open AC line fuses (**FU1-3**) or breaker. 4. Check for open fuses on the Power Supply Card (**CFU1-8**) 5. **2PL** disconnected, open, loose, or intermittent. 6. Power supplies loaded down or failed - refer to "Checking Power Supplies", this section. 7. If Hand-Held Programmer is available, connect and attempt to enter the **PARAMETER** mode. Refer to Error Codes (Table 5B) if any detected. A hard reset should clear this lock-up condition. 8. Replace Control Card software (**U1, U5,** and **U6**). 9. Replace Control Card.
2. All Diagnostic LEDs ON	1. Drive faulted - check fault code list. 2. **U5** and **U6** microprocessor interchanged on the Control Card. 3. **RSET-3TB69** connected to +15V. 4. Extended memory **U1** installed - check **JP24** per the Programming Software Package. 5. Replace Control Card software (**U1, U5,** and **U6**).
3. Fault Messages Displayed (Repetitive)	1. See Specific Fault Troubleshooting, Table 5, for troubleshooting procedures related to specific drive fault number.
4. Fault Messages Displayed (Non-Repetitive, Random)	1. See problem description on "Noise or Random Nuisance Faults". 2. See "Wiring Procedure".
5. Programmer Display Off	1. Check that the Programmer plug is connected properly. **Local** - 10 conductors must be in right side of plug when connected. **Hand-Held** - red stripe must appear on the left-hand side of the cable leaving the Programmer and on the left-hand side of the cable when inserted into receptacle. 2. Check power supplies - refer to "Checking Power Supplies", this section. 3. Replace Programmer Card.
6. Error Messages	1. See Error Messages, Table 5B, for listing.
7. Programmer Ineffective No Errors Displayed	1. Check that the Programmer plug is connected properly. **Local** - 10 conductors must be in right side of plug when connected. **Hand-Held** - red stripe must appear on the left-hand side of the cable leaving the Programmer and on the left-hand side of the cable when inserted into receptacle. 2. Check jumpers **JP3** and **JP4** on the Programmer Card for proper position. 3. Verify that Programmer Card jumpers **JP1** and **JP2** are in 1-2 position. 4. Check that the drive is not in **AUTO** mode (this mode inhibits Diagnostic Run, SCR test, and certain programmed **OPERATE** mode references, i.e. **RUNRF, JOGRF,** etc.). 5. Replace Programmer.

Problem	Check/Adjust/Replace
8. Parameter Changes Ineffective	1. Check **JP13** (Write Protect) on the Control Card (must be in 2-3, the Enable position, for programming). 2. Check setting in **ADR003-DGNJP** for odd number (must be an odd number to allow reprogramming). 3. If the drive uses multi-motor operation for different gear ranges, spindle orient, etc., verify that the proper motor parameters are being adjusted. Check by setting the **IOC** trip setpoint of the motor parameters being adjusted to 10. Starting the drive will cause **IOC** fault if the proper motor parameters have been adjusted. 4. Check "Programmer Ineffective - No Errors Displayed" paragraph. 5. Replace Control Card software (**U1, U5,** and **U6**).
9. Self-Tune or Other Diagnostics Do Not Perform	1. Verify that the particular function is available (refer to the Custom Software Description, software jumper descriptions). 2. Verify that the SCR test is enabled (this is required for S-Tune, I-Tune, torque proving, and the SCR test). 3. Check that **JP13** on the Control Card is in the 2-3 position (must be in position for some **DIAGNOSTIC** tests to function, see Section 11). 4. Check that the drive is in **MANUAL** mode (must be in this mode to run **DIAGNOSTIC** tests). 5. Check **POL** on **6PL-15 (3TB55)** for reverse signal. The drive must be in forward for the Self-Tune diagnostic to operate. 6. Check "Programmer Ineffective - No Errors Displayed" paragraph.
10. SCR Test Failure	1. Check fault codes if executing an SCR test results in a drive fault. 2. Check that the drive is in **MANUAL** mode (must be in this mode to run SCR test). 3. Check that the Programmer is in **DIAGNOSTIC** mode - refer to "Programmer Ineffective - No Errors Displayed", this section. 4. Check that the motor is connected properly and no wires are open. 5. Verify that the motor is not rotating during test. 6. Verify that the SCR test is enabled - refer to the Custom Software Description, software jumper descriptions. 7. Check that the MA contactor has picked up and that there are no open contacts. 8. Check the DC fuse (**FU4**). 9. Check for any external chokes, capacitors, or dynamic braking circuits in the armature circuit. It may be necessary to remove these components for the test to operate properly.
11. Fuse Blowing (AC or DC Power Fuses)	1. Compare the fuse size with the Renewal Parts List. 2. Check for failed SCRs by running the SCR test. 3. Check the setting of **ADR080-CMFLM** on regenerative drives; should be 1704 or less. 4. Check for out of tolerance or sagging AC line voltage on regenerative drives - refer to "Line Voltage Problems" and "Noise or Random Nuisance Faults" paragraphs in this section. 5. Check the field power module for a short. 6. Using a megger, check for a ground in motor field, armature circuit or blower circuit. 7. Verify that the drive is not forced into Current Limit by an overhauling load (this disables **CEMF** limit while in regeneration).

Problem	Check/Adjust/Replace
11. Fuse Blowing (AC or DC Power Fuses) (continued)	8. Adjust the field control parameters to provide a lower armature voltage at top speed - lower **ADR079-CROSS** on field weakening drives or **ADR076-FLDMX** on base speed drives. 9. Refer to "Random SCR Misfirings" paragraph in this section.
12. SCR Test Blowing AC or DC Power Fuses	1. Check for shorted SCRs with an ohmmeter from each MA or MD contactor pole to both sides of the DC bus. 2. Check for ground fault in the motor armature circuit. 3. Check the phase rotation and connection of the AC sync wires at the incoming fuse breaker and Power Supply Card. 4. Check the phase rotation and connection of the AC power wires at the incoming fuses, breaker, MA or MD contactor, and SCRs. 5. Check that the phasing of AC voltages at line fuses, power supply, and field exciter input are all exactly in phase with each other.
13. Field Fuses Blowing	1. Check for a ground in the motor field or conduit wiring. 2. Check the field power module for a short or failure. 3. Verify that all wires on the Power Supply Card are connected to the correct stab-on. 4. Verify that **FSPL** is plugged into the proper position with the correct polarity. 5. Check for shorted MOVs on the Power Supply Card. 6. Check for shorted Control Power Transformer (**CPT**) primary.
14. Power Supply Fuses Blowing	1. Verify that the fuses are correct size. 2. Check for any shorted or over-loaded power supplies (Refer to "Checking Power Supplies" paragraph in this section). 3. If using a Hand-Held Programmer, verify that it is properly connected to the receptacle. 4. If the +15 VDC supplies a digital tach, verify that a cooling fan is installed below the card assembly on the DC-300 controller and is operating properly. 5. Replace the Power Supply Card.
15. MOV Failures	1. Refer to "Line Voltage Problems" paragraph in this section. 2. Motor fault causes AC fuses to open during regeneration. Check for grounded armature, field, or wiring.
16. Repetitive Card Failures	1. Check all connections at **2TB, 3TB,** and **4TB.** 2. Verify that all ribbon cables and plugs are connected properly (i.e., not offset by one row or pin). 3. Check the stab-on connection points on the Power Supply and Power Connection Cards. 4. Verify that all voltages (incoming AC and internal DC) are correct. 5. Check for moisture. 6. Check for damage due to airborne chemical contamination.

Problem	Check/Adjust/Replace
17. Random SCR Misfirings	1. Check for inoperative SCRs. 2. Check for loose connections on the fuses, breaker, MA contactor, SCRs, or any armature connections. 3. Check for loose connections on SCR gate leads, Power Connection Card, or **5PL**. 4. Check for notching on the AC line - this could be caused by other drives on the same incoming AC line of a horsepower equal to or greater than four to six times the horsepower of the problem drive. 5. Refer to "Line Voltage Problems" paragraph in this section.
18. Line Voltage Problems	1. Verify that the line voltage is within drive specifications, -5% to +10%. 2. Verify that the AC line voltage is balanced line-to-line and line-to-ground. 3. Check for AC line notches or spikes. 4. Verify that the line frequency is within ±1 Hz of rated. 5. When AC line is supplied by a diesel driven alternator, check for changes in line frequency during drive acceleration or regeneration. 6. Check for more than one and one-half cycles of ringing following commutation notches in the AC line caused by the drive. Amplitude should be less than or equal to 200 volts. 7. Check for an ungrounded secondary on the power system transformer directly in front of the drive (if used). (The transformers are generally a delta/wye type with the neutral of the "wye" grounded through a high resistance or directly to ground.)
19. 24 Volt Logic Not Operating Properly	1. Check the ±24 volt power supplies. 2. Check logic polarity control Jumpers, **JP19** and **JP20** on the Control Card, **JP5** and **JP6** on the optional Application Card. 3. Check control logic connections at **3TB** and **4TB**. 4. 24 volt logic being switched too quickly to optional **4TB** or optional Application Card - there must be at least a 70 millisecond delay between two logic inputs changing state. 5. Check for proper polarity and common circuit if 24 volt supply is from a source other than drive. 6. Check that the drive is in the proper motor mode (0, 1, 2, or 3) for the particular motor or function being operated.
20. Drive/Motor Does Not Run	1. No connections to REF 24, ±15V, or common should be made at **4TB**. Move any such connections to **3TB**. 2. **FU1, 2, 3,** or **4** open (or optional circuit breaker open). 3. **CFU1, 2, 3, 4,** or **5** open and/or MOVs shorted on Power Supply Card. 4. Check all hardware jumpers, particularly **ADR012-OPTJP, JP19** and **JP20** on the Control Card, and **JP5** and **JP6** on the optional Application Card. 5. Check for the speed reference at **3TB** or **4TB (MSR, JOGR,** or **ASR)**. 6. Check control logic voltage and state at **3TB** and **4TB (RUN, JOG, SLOW, MODE 0** thru **MODE 7**, etc.) as well as power supply voltage levels. 7. Check that the drive is in the proper operate mode **(AUTO, MANUAL,** or **ORIENT)**. 8. Drive may be stalled in current limit.

Problem	Check/Adjust/Replace
21. Field Current or Voltage Wrong Value	1. Check that the drive is not in field economy. 2. Check for proper motor field connections. 3. Check **JP1** and **JP2** on the Power Supply Card for proper maximum field settings. 4. Check the **P1-FIELD SCALE** pot seal - factory adjusted for field current. (Do not adjust.) 5. Check **ADR076-FLDMX** and **ADR077-FLDMN** per the Custom Software Description. 6. Check optional **4TB-20** for connection to **FRA** if external field forcing is used. 7. Check for any external parallel resistors or diodes in the field circuit. 8. Check the field power module for shorted elements. 9. Check for saturated fields when not able to produce required torque (symptom is also high current, high speed versus reference). Field current is reduced, but not enough to cause (**FL04**) Field Loss trip. (May be due to an open in one of the components feeding the MFC package, a failed fuse, wire connection, or diode in the MFC package, or too high of a required field voltage.) 10. Check the field current (Test point **FC**). Replace the Power Supply Card if failed. 11. Replace the Control Card.
22. Drive Runs in the Wrong Direction	1. Check for logic signal on **3TB15 (POL)**. 2. Armature connections reversed (correct armature and tach leads). 3. Field connections reversed (correct field and tach leads). 4. Check all software jumpers (**ADR000** to **ADR014**) per the Custom Software Description. 5. Check the reference polarity. 6. Check logic polarity control jumpers, **JP19** and **JP20** on the Control Card, **JP5** and **JP6** on the optional Application Card. 7. Check the Custom Software Description to verify that the reverse function is enabled when operating from the serial link.
23. Drive Does Not Reverse	1. Verify that the drive is regenerative. 2. Check the settings of the soft jumpers (**ADR000** to **ADR014**) per the Custom Software Description. 3. Check the reference polarity. 4. Check control logic to **POL (3TB55)** if used for reversing. 5. Check logic polarity control jumpers, **JP19** and **JP20** on the Control Card, **JP5** and **JP6** on the optional Application Card. 6. Check the Custom Software Description to verify that the function is enabled when operating from the serial link.
24. Drive Speed Is Incorrect	1. Verify that the motor design is suitable for the application. 2. Check that the proper reference (**MSR, ASR, RUNRF, R1RF, R2RF, SLWRF, JOGRF**) is enabled. Refer to Custom Software Description. 3. Check **JP1** on the Control Card for proper scaling of **MSR** input. 4. Adjust **P1-REF SCALE** on the Control Card for **MSR** input. 5. Check that **ADR068-REFSF** is set correctly. 6. Check **JP1** on the optional Application Card for proper **ASR** scaling. 7. Adjust **P1-ASR SCALE** on optional Application Card.

Problem	Check/Adjust/Replace
24. Drive Speed Is Incorrect (continued)	8. Adjust **P7-SFB ZERO** on optional Encoder Process Interface Card if analog AC or DC tach is used. 9. Confirm the setting of **SW1-TACH SCALE**, on the optional Encoder Process Interface Card if an analog AC or DC tach is used. 10. Check output voltage at the drive if a pulse tach is used. May be low due to excessive cable length. 11. **RPMSF** or **REFSF** may be misadjusted, check per the Custom Software Description. 12. Verify the proper settings in **ADR053-DTCHR** and **ADR074-TCPPR** per the Custom Software Description. 13. Check for noisy tach signal. Refer to "Wiring Procedures", this section. 14. Set **ADR001-FDBJP** per the Custom Software Description. 15. Confirm that **JP15, JP16,** and **JP19, JP20** on the optional Encoder Process Interface Card are in accordance with application requirements. 16. Check the setting of **ADR041-ILMMX** if the drive is suspected of being in current limit. 17. Check the setting of **ADR080-CMFLM** if the drive is suspected of being in CEMF limit. 18. Check **ADR002-REGJP** and **ADR079-CROSS** per the Custom Software Description. 19. Check **ADR076-FLDMX** and **ADR077-FLDMN** per the Custom Software Description and by calculations in the key parameter adjustment list. 20. Check **ADR059-RATIO** per the Custom Software Description. (Should be set at "10000" for no draw.)
25. Spindle Orient Not Positioning	1. Confirm that position loop is enabled by the control logic. 2. Check that the position loop has been enabled by the hardware and software jumpers. 3. Verify that the proper motor parameters are selected by changing the IOC trip parameter for that motor to 10 and getting an **IOC** when position commanded. 4. If spindle continues to turn or orients in only one direction, adjust **P7-SFB ZERO** on the Process Interface or Encoder Process Interface Card. 5. If an encoder is used, verify that only one marker pulse per revolution is present and not noisy. 6. Confirm that the motor field is weakened when in position.
26. Serial Link Does Not Function Properly	1. Verify that **JP11, JP12, JP13,** and **JP14** on the optional Process Interface or Encoder Process Interface Card are set correctly. 2. Check the baud rate of the drive (**ADR126-BAUDR**) compared to serial link device (hard reset must be performed after **BAUDR** is changed). 3. Check serial link connections at **3TB22, 23,** and **24 (DCOM, R232,** and **3TB22**). 4. If RS422 is used, verify that the handshaking lines are properly connected. Verify that sending device is responding to the drive's **DTR** signal (low level) when the device is allowed to transmit to the drive. 5. Replace the optional Process Interface or Encoder Process Interface Card.

Problem	Check/Adjust/Replace
27. Erratic Operation	1. Check power supplies for magnitude and ripple (refer to "Checking Power Supplies" paragraph in this section). 2. Verify that **1CPL** on the MFC/Power Supply Card is inserted correctly (not backwards or offset). 3. Check connections on all ribbon cables. 4. Tighten all connections on **2TB, 3TB,** optional **4TB,** and all power connections. 5. Confirm correct hardware jumper setting per the Custom Software Description. 6. Check all parameters per the Custom Software Description. 7. Check for noisy ground connection - refer to "Wiring Procedures". 8. Examine **CFB** and **VFB** with an oscilloscope for stability (refer to "Stability Problems" paragraph in this section). 9. If a digital tach is used, check the integrity of the signal.
28. Stability Problems	1. Run Diagnostic **Test 12** or examine **CFB** with an oscilloscope to determine if any SCRs have failed. 2. Check settings of all parameters per the software test data sheet located inside the drive door pocket. 3. Check the setting of **SW1-TACH SCALE** switch on the optional Process Interface or Encoder Process Interface Card if an analog AC or DC tach is used. 4. Check Jumpers **JP17** and **JP18** on the Process Interface Card if an analog tach with tach linearity is used (should be in 1-2 position). 5. Verify that all contacts on the MA or MD contactor are closing. 6. Check and adjust for any offset on **CFB, P2-CFB ZERO, P3-CFB ADJUST** (on the Power Supply Card), **JP3** and **JP4** on the Power Supply Card, or the **P4-OFFSET** (on the Control Card). 7. Refer to the "Troubleshooting Stability Problems" paragraph in this section.
29. Signal Level Detector Chatter	1. Is **SLD** looking at a "noisy" signal (e.g., **CFB**, tach feedback, etc.)? 2. Is enough hysteresis programmed into the appropriate parameter address?
30. Too Much Coast Time on Deceleration (Regenerative Drives)	1. Lower the value in **ADR048-RSTOP.** 2. Check settings per Custom Software Description and confirm regeneration is enabled.
31. Motor Sparking	1. Check motor commutator for contamination. 2. Check the setting in **ADR024-IRTLM**, may be too high. 3. Check the field current - refer to "Field Current or Voltage Wrong Value". 4. Check the field current stability - refer to "Stability Troubleshooting" paragraph in this section. 5. Check the commutator condition. 6. Confirm that all brushes are seated properly with proper brush tension. 7. Check the brush grade. 8. Check the brush neutral adjustment. 9. Check for sagging AC line voltage - refer to "Line Voltage Problems", paragraph in this section.

Problem	Check/Adjust/Replace
31. Motor Sparking (continued)	10. If tapered current limit is required for the application, verify that **ADR081-ADR084** are set to reflect this.
32. Motor Runs Hot	1. Motor may be overloaded, check for mechanical problems. 2. Check the motor winding temperature. If the motor has Class F insulation and the winding temperature is less than 130° C, this is normal. 3. Check the rotation of the motor blowers and that filters are not dirty or clogged. 4. Check the field current with respect to the motor nameplate rating. Adjust **ADR076-FLDMX** if necessary. 5. Check the field current when motor is not running. If the field current is full nameplate rating when the motor or blower is not running, check **ADR050-FLDEC.** 6. If motor runs hot at very low speeds, check motor application guide for full-torque speed range. 7. Confirm that all SCRs are firing properly by monitoring current feedback (**CFB**) or running SCR test (Diagnostic **Test 12**).
33. Motor Audible Noise	1. Loose pole bolts - tighten per manufacturer's torque specifications. 2. SCRs misfiring - check current feedback. 3. Cross-firing SCRs - increase deadband. 4. Unstable regulator - check "Troubleshooting Stability Problems" paragraph in this section.
34. Motor Vibration	1. Coast Stop Drive - if the vibration is electrically induced the vibration will disappear. If it does not disappear, check for mechanical causes. 2. Refer to "Stability Problems" paragraph in this section. 3. Motor not mounted level. 4. Motor base vibrating. 5. Motor not mounted as designed - i.e., horizontal design motor mounted with shaft vertical.
35. EEROM Memory Rewrites	1. **JP13** on the Control Card must be in the 1-2 position during "Power Up" or hard reset.
36. Noise or Random Nuisance Faults	1. Verify that the 5-volt DC supply is not used externally. 2. Check if **ACOM, DCOM,** or **PCOM** are used externally - **PCOMX** is the only point that should be used for common connections. 3. Verify that **PCOM** passes through a ferrite core (not required if **PCOMX** is on **3TB73-75**). 4. Check for suppression on all relays, contactors, and starters. 5. Place **JP16** on the Control Card in the 2-3 position on all drives not using a frequency reference. 6. Check all power supplies - refer to "Checking Power Supplies" paragraph in this section. 7. When an encoder is used, verify that all of the encoder wires except the shield have been passed through a ferrite core on optional balun assembly. 8. Refer to "Wiring Procedures" paragraph in this section.

Problem	Check/Adjust/Replace
37. Wiring Procedures	1. Check the wire sizes on the armature, field, and AC power connections. 2. Verify that the control wires (24-volt DC, 115-volt AC) do not run in the same conduit or run parallel closer than four inches to the power wires, causing noise pick-up. 3. Verify that all of the signal wires (i.e., tach, encoder, reference, load cell, or RS-232C/422) are twisted pairs, shielded, and shields grounded at the drive. 4. Verify that the signal wires are not grounded at more than one end or other than the drive end only. 5. Verify that the signal wires do not run in the same conduit or parallel closer than four inches to the power wires. 6. Verify that there is only one system ground. 7. Verify that the ground bus is large enough to prevent voltage drops or noise pick up.
38. Checking Power Supplies	1. Check the +15V test point on the Control Card for +15VDC, $\pm$5%. Verify that there is less than or equal to 100 millivolts of ripple with an oscilloscope. 2. Check the -15V test point on the Control Card for -15VDC, $\pm$5%. Verify that there is less than or equal to 100 millivolts of ripple with an oscilloscope. 3. Check the +5V test point on the Control Card for +5VDC, $\pm$1%. Verify that there is less than or equal to 100 millivolts of ripple with an oscilloscope. 4. When the 5V power supply is used to power an encoder or digital tach, check for power supply heating if no fan is supplied.

TABLE 5. SPECIFIC FAULT TROUBLESHOOTING

Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
000001	FL01	IOC	**Instantaneous DC overcurrent (L,T)** -- Tighten armature connections. -- Armature grounded. -- Examine motor commutator for damage or evidence of flashover. -- Verify proper brush seating and tension (refer to motor manual). -- Mechanical Problems - binding shaft, unreleased brake, vibration, gearbox, or load problems. -- IOC parameter set too low. Set at least 128 units above current limit settings. -- Current feedback isolator failure - check **1PL** and **DRPL.** -- SCR misfiring - Check **VFB** or **CFB** waveforms. -- Check input at **SJ2 (4TB16** or **13PL-16).** -- **ADR011-SJAJP** set incorrectly. -- SCR failure: Try SCR test. (Refer to Control Card **JP12** description, TABLE 1, or Use of Optional Programmer Module if a programmer is available.) -- Current regulator mistuned - adjust **ILPPG, ILPIG, ILPDG, ILPDP (ADR019, 020, 021, 022).** -- Check multimotor IOCs - **ADR043, ADR153, ADR165,** and **ADR177.**
000010	FL02	OVRSP	**Overspeed (L,T)** -- **ADR073-SPDLM** set too close to **ADR068-REFSF** and **ADR071-REFLM** - set a minimum of 5% greater than **REFSF** and **REFLM.** -- Speed overshoot occurs - speed regulator response too high, reduce gain. (Refer to TABLE 6, Troubleshooting "Stability" Problems.) -- Conflicting settings of **ADR001-FDBJP, ADR002-REGJP,** and **ADR013-POSJP** for digital tach and standard Process Interface Card (531X133PRUA-G1) or Encoder Process Interface Card (531X134EPRA-G1). Refer to Custom Software Description (Encoder Process Interface Card cannot accept encoder and digital tach feedback at same time). -- Noise on digital tach input. -- Loss of field current.
000011	FL03	----	**SPARE**

KEY: L - LATCHED N - NON-LATCHED H - REQUIRES HARD RESET
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Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
000100	FL04	FLOSS	**Loss of field current (L,T)** -- **CFU1, CFU2,** or **CFU3** on Power Supply Card open. -- **1PL** disconnected or damaged. -- Field grounded. -- Field lead connections made improperly. -- Field regulator saturation; add "16" to **ADR012-OPTJP.** -- Open field circuit components - resistors, contactors or DC fuses. -- **ADR078-FLDLS** set too high - set at least 25% below **ADR077-FLDMN** and 50% below **ADR076-FLDMX.** -- FC gain jumpers set wrong - **JP1** and **JP2** on Power Supply Card. -- External field supply - set **ADR078-FLDLS** at "60000" to inhibit field loss. -- Field regulator gain too low - adjust **BFLPG, BFLIG, WFLPG, WFLIG (ADR031, 032, 033, 034).**
000101	FL05	TPOL	**Tach polarity incorrect (L,T)** -- Reversed tach connections. -- Switch **SW1-7** on Process Interface Card set improperly - open for DC tach, closed for AC tach. -- Armature or field reversing contactors - **ADR001-FDBJP** must contain a 2; check Custom Software Description to see if this feature is supplied.
000110	FL06	JMPER	**Software jumper illegally changed while running (L,A)** -- Drive must be stopped to change parameters in **ADR000** to **ADR014.**
000111	FL07	RESET	**Reset while running (L,T)** -- Drive powered up with **RUN** or **JOG** closed - check **6PL-12** and **6PL-13 (3TB58** and **57).** -- Momentary loss of incoming power while running. -- Incorrect logic polarity, check **ADR012-OPTJP.**

Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
001000	FL08	TOC	**Timed overcurrent (L,T)** -- Motor mechanically overloaded - check for change in process load, and for worn bearings, gears, or brakes. -- **CFU2** on Power Supply Card open. -- Field current is less than 85% of nameplate rated current - readjust field current parameters. -- **ADR044-TOCTH** and **ADR045-TOCGN** set too low - check against the parameter list shipped with drive. -- Current feedback isolator failure or **1PL** disconnected or damaged. -- **P3-CFB ADJ** on the Power Supply Card misadjusted - readjust per procedure in TABLE 1. -- Armature winding failed. -- Dynamic braking failure (if used) - i.e., failure to close DB circuit. -- Check **ADR206-TOCTR**
001001	FL09	LNKR	**Serial Link trip (drive told to trip via serial link) (L,T)** -- Drive received external trip signal - check for external fault. -- **12PL** or **8PL** incorrectly connected or damaged.
001010	FL10	LNKFL	**Serial Link fault (communications failure) (L,T)** -- Programmer Card jumpers incorrect - refer to TABLE 1. -- Baud rate set incorrectly - check **ADR126-BAUDR.** -- **12PL** or **8PL** incorrectly connected or damaged. -- Intermittent connection between **3TB32 (CTSA)** and **12PL-7** or **3TB41 (CTSB)** and **12PL-14.** -- **FL36** appears momentarily previous to **FL10** latching - refer to **FL36.**
001011	FL11	DFPON	**Delayed firing power on when it should be off (L,T)** -- **Q2** failed on Control Card - check test point **DFP.** -- Replace Control Card.
001100	FL12	DFPOF	**Delayed firing power off when it should be on (L,T)** -- **Q2, Q3, R33** or **U14** failed on Control Card - replace Control Card.
001101	FL13	EEXXX	**Checksum error or parameter out of range where XXX equals a 3-digit number, indicating an out-of-range parameter address or altered checksum. (L, T, A, H)** -- Spontaneous parameter alteration - check all parameters and verify EEROM is not write "ENABLED" during power up or power down, or "Hard" reset. -- Value other than zero in **ADR255-UNDFD** - replace EEROM. **NOTE:** *Drive will* **TRIP** *if EEROM error detected during powerup. Fault is* **ANNUNCIATED** *if an attempt to program an out-of-range parameter is made.*

Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
001110	FL14	ILOOP	**Current loop regulator saturated (L,T)** -- Main contactor not picking up - check pilot relay operation. -- DC fuse open (regenerative drives only). -- Loss or incorrect scaling of voltage feedback - check jumpers on the Power Connection Card and Control Card **TP VFB** or **VFBB**. -- Loss of current feedback - check **1PL-20** and **DRPL**. -- Armature not connected. -- Regeneration enabled by soft jumper on single quadrant drive. -- Brushes missing or unseated. -- Controllers using the inner voltage loop (optional) - hardware failure in voltage regulator; replace Control Card. -- Controllers using the inner current loop, **ADR019-ILLPG**, **ADR020-ILPIG**, and **ADR021-ILPDG** set too high - check against parameter list shipped with drive. -- Current feedback miscalibrated - see TABLE 1, Power Supply Card. -- Offset voltage at **CFB** with zero armature current - adjust **P2** (**CFB ZERO**) on the Power Supply Card per TABLE 1. -- Power Supply Card failed - replace. -- Control Card failed - replace. -- If **DFM**, check **DFM** parameters **ADR297** thru **ADR328**.
001111	FL15	VFBLS	**Loss of armature voltage feedback (L,T)** -- Voltage feedback wires disconnected or damaged - check **P1** and **P2** between SCR conversion module and Power Connection Card. -- DC fuse (**FU4**) open - regenerative drives only. -- **5PL** disconnected or damaged. -- **SFB** circuit misadjusted - adjust **P7-SFB ZERO** on the Process Interface Card per TABLE 1. -- **CFB** jumpers incorrect - check **JP3** and **JP4** on the Power Supply Card per TABLE 1. -- **VFB** circuit failure - check test point **VFB**. -- Run SCR Test - refer to Control Card description of **JP12** of Use of Optional Programmer Module if available.
010000	FL16	NOTRP	**Trip from N.O. on Application card (L,T)** -- External input at **4TB8** (**13PL-8**). -- Application Card absent but selected by soft jumper **ADR010-APPJP** should be zero when no Application Card is present. -- If preceded by **FL07** and Soft Reset, check logic polarity on **ADR012-OPTJP**. -- If Microapplication Card is used, check **JP33** (should be in 1-2 position) and verify that **U33** is installed correctly. -- Replace Application Card.

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Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
010001	FL17	XSTOP	**Coast stop trip (L,T)** -- Check for input at **XSTOP** on **3TB1 (6PL-16)**. -- Check **ADR004-STPJP** - a 4 causes coast stop when **XSTOP** is selected.
010010	FL18	STALL	**Stalled overcurrent (L,T)** -- Mechanical problems - failed brake, frozen bearings, application problem, etc. -- **ADR046-STLTH** and **ADR047-STLGN** set wrong - check against "as shipped" parameter list. -- Drive running less than 1% of top speed and armature current greater than **ADR046-STLTH**. -- **CFU2** on Power Supply Card open. -- Field current is less than 85% of nameplate rated current while the drive is supplying armature current. -- Current feedback isolator failure or **1PL** disconnected or damaged. -- **P3-CFB ADJ** on the Power Supply Card misadjusted - readjust per procedure in TABLE 1. -- Armature winding failed. -- Dynamic braking failure (if used) - i.e., failure to open DB circuit.
010011	FL19	OVERV	**Excessive armature voltage (L,T)** -- **ADR080-CMFLM** set too high - compare to parameter list. -- Drive unstable - see Stability procedure in Troubleshooting section of Instruction Book. -- **VFB** circuit failure - check test point **VFB**.
010100	FL20	TLOSS	**Loss to tach feedback (L,T)** -- Tach leads disconnected. -- Tach feedback circuit failure - **7PL** disconnected or damaged. -- Process Interface Card jumpers or switches incorrect. Refer to TABLE 1. -- **P7-SFB ZERO** on Process Interface Card misadjusted - set test point **SFB** for zero volts at zero speed. -- AC or digital tach connected but CEMF regulator being used - remove **JP7** from Process Interface Card. -- **ADR053-DTCHR** set too high when AC Tach being used - lower value to 10. -- **ADR002-REGJP** set incorrectly for type of Process Interface Card used - PI Card (531X133PRUA_G1) or Encoder PI Card (531X134EPRA_G1), refer to Custom Software Description. -- **JP17** on Control Card in 1-2 position - move to 2-3 position. -- Process Interface Card failed - replace.

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Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
010101	FL21	NCTRP	**Trip from N.C. interlock on Application card (L,T)** -- Loss of external input on **4TB7 (13PL-7)**. -- Application Card absent but selected by soft jumper - **ADR010-APPJP** should be zero when no Application Card present. -- If preceded by **FL07** and Soft Reset, check logic polarity **ADR012-OPTJP**.
010110	FL22	PHLOS	**AC phase loss (L,T)** -- **JP14** on Control Card in wrong position - refer to TABLE 1. -- Fuse open in AC line - check **FU1-3**. -- **1PL** disconnected or damaged. -- Low or unbalanced AC line voltage - line-to-line or line-to-neutral more than 5% low. -- Loose AC power connections. -- **CFU1** open. -- Verify that **ADR187-FUSLS** contains a zero - change if necessary. -- $\pm$24-volt or $\pm$15-volt power supplies failed - check **CFU6** and **CFU7** on Power Supply Card. -- $\pm$15-volt power supplies outside limits ($\pm$5%) - replace Power Supply Card. -- If fuse loss detection (**ADR187**) enabled, setting may be too sensitive or threshold too high.
010111	FL23	OVERF	**Field overcurrent (L,T)** -- Field regulator unstable - refer to Stability procedure in Troubleshooting section. -- Field forcing at **FRA (4TB20, 13PL-20)** - check input scaling, see TABLE 1. -- **ADR076-FLDMX** or **ADR049-FTOCG** incorrect - check parameter list. -- Field SCR shorted - replace Motor Field Module. -- Voltage regulated field - check for miswired or grounded field.
011000	FL24	U3 FL	**Faulty RAM on U3 8155 (L,T)** -- Replace Control Card.
011001	FL25	COMTR	**SCR commutation failure (L,T)** -- Low AC line voltage (less than 90% rated). -- **ADR080-CMFLM** set too high - should not exceed 1.13 x AC RMS in-line voltage. -- **1CPL** disconnected. -- **1CPL** reversed - phase 1 CT should be wired to bottom of **1CPL** at Power Supply Card. -- **P3-HP SEL** and **JP8-11** gains set too high on Control Card. -- **1PL** disconnected or damaged. -- Gate plug(s) incorrectly connected. -- SCR(s) failed - try SCR self test. -- Disable nuisance trips by setting **STPJP** bit 6.

Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
011010	FL26	ACIOC	**Circulating AC line fault (regenerative controllers only) (L,T)** -- Low or noisy AC line. -- Incorrect setting **P3-HP SEL** and **JP8-11** on Control Card. -- Noise on SCR gate leads. -- **1PL** disconnected or damaged. -- SCR(s) failed - try SCR self test. -- Disable nuisance trips by setting **STPJP** bit 9.
011011	FL27	LYSNC	**Loss of AC line signal (L,T)** -- Fuse open in AC line - check **FU1-3**. -- **1PL** disconnected or damaged. -- Low or unbalanced AC line voltage - line-to-line or line-to-neutral more than 5% low. -- Loose AC power connections. -- **CFU1** open. -- 15-volt power supplies outside limits (±5%) - replace Power Supply Card. -- AC line frequency below 45Hz or above 63Hz (70Hz if called for by **ADR001-FDBJP**). -- Check 50/60Hz square wave at test point **SY** on Control Card - replace Control Card if no signal.
011100	FL28	RMODE	**Illegal selection of RUN mode (N,A)** -- Do not command **DIAGNOSTIC RUN** or **STATIC** while running or vice versa. -- Attempted to run **Test 15, S-TUNE** when feature not supplied. -- Also occurs if **ADR197-AIXJP** bit 4 is set without **ADR014-SPCJP** bit 0 and **ADR001-FDBJP** bit 5 set.
011101	FL29	CNLON	**Control on input opened (L,T)** -- Open circuit between **3TB3** and **3TB4**. -- **6PL** incorrectly connected or damaged. -- Noise between **3TB3** and **3TB4**.
011110	FL30	NOAPP	**Application Card failure (L,T)** -- Application Card absent but selected by soft jumper - **ADR010-APPJP** should be zero when no Application Card present. -- **3PL** incorrectly connected or damaged. -- If Microapplication Card is used (531X139APMA-M1or M3), the Control Card must be a 531X300CCHA-M2 or M3. -- If Microapplication Card is used, check **JP33** (should be in 1 2 position) and verify that **U33** is installed correctly. -- Replace Application Card.

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Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
011111	FL31	NOSEQ	**Phase sequence indeterminate (L,T)** -- **CFU1** on Power Supply Card open. -- Excessively noisy or notched AC line. -- Line unbalance, phase to ground by more than 15%. -- 15-volt power supplies outside limits (±5%) - replace Power Supply Card. -- **1PL** incorrectly installed or damaged.
100000	FL32	TOCAN	**Timed overcurrent annunciation (N,A)** -- Motor mechanically overloaded - check for change in process load, and for worn bearings, gears, or brakes. -- **CFU2** on Power Supply Card open. -- Field current is less than 85% of nameplate rated current. -- **ADR044-TOCTH** and **ADR045-TOCGN** set too low - check against the parameter list shipped with drive. -- Current feedback isolator failure or **1PL** disconnected or damaged. -- **P3-CFB ADJ** on the Power Supply Card misadjusted - readjust per procedure in TABLE 1. -- Armature winding failed. -- Dynamic braking failure (if used) - i.e., failure to close DB circuit.
100001	FL33	PHSEQ	**Phase sequence reversed (N,A)** -- Reverse any two AC input lines, recheck rotation of any three phase motors. -- Excessively noisy or notched AC line. -- Line unbalance, phase to ground by more than 15%. -- 15-volt power supplies outside limits (±5%), noisy or excessive ripple, replace Power Supply Card. -- **1PL** incorrectly installed or damaged. -- **ADR012-OPTJP** not properly set for phase sequence insensitivity option.
100010	FL34	PLLSY	**AC line PLL out of sync (N,A)** -- Excessively noisy or notched AC line. -- **CFU1** on Power Supply Card open. -- Line unbalance, phase to ground by more than 15%. -- 15-volt power supplies outside limits (±5%) - replace Power Supply Card. -- On variable frequency AC lines - **ADR029-PLLGN** too low.

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Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
100011	FL35	O SPD	**Motor rotating when RUN commanded (N,A)** -- **RUN** command initiated during motor rotation. -- **ADR048-RSTOP** set too low. -- Test point **SFB** contains an offset at zero speed - adjust **P7-SFB ZERO** on the Process Interface Card. -- **ADR004-STPJP** not set properly for "flying restarts".
100100	FL36	LNKER	**Uncipherable messages detected on serial link by drive or other device (N,A)** -- Excessive noise on **12PL-7** thru **14**. -- **12PL** incorrectly connected or damaged. -- 5V DC power supply noisy. -- Baud rate incorrectly set - **ADR126-BAUDR** normally set at 4800. -- Check Process Interface Card jumpers **JP1, JP11, JP12, JP13** and **JP14**. -- Serial link message protocol incorrect, message too long, checksum byte incorrect, or handshaking incorrect - refer to "Serial Link User's Guide". -- CRT used on **COMPL** is disconnected or has noise. -- Replace Process Interface Card. -- Replace Control Card.
100101	FL37	IHUNG	**Armature current not zero when fully retarded (N,A)** -- **JP3** and **JP4** on Power Supply Card in 2-3 position when running - should be in 1-2 position. -- Offset voltage at test point **CFB** when zero armature current flowing - adjust **P2-CFB ZERO** on the Power Supply Card. -- **DRPL** disconnected or damaged. -- **1PL** disconnected or damaged. -- **ADR080-CMFLM** set too high (only if fault occurs during regeneration). -- Current isolator failed - replace Power Supply Card.
100110	FL38	ELOCK	**Attempt to change software adjustments when soft locked (N,A)** -- **ADR003-DGNJP** must be odd to program EEROM - add one to value in **DGNJP**. -- CRT used on **COMPL** is disconnected or has noise.
100111	FL39	EFAIL	**Software adjustment not stored in EEROM (N,A)** -- **JP13** (or **SW1**) on Control Card in hard lock position. -- CRT used on **COMPL** is disconnected or has noise. -- EEROM failed - replace.

Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
101000	FL40	MXSPD	**Speed reference greater than REFLM speed adjustment (N,A)** -- Manual Reference jumper **JP1** on Control Card improperly set - see TABLE 1. -- **P1-REF SCALE** on the Control Card misadjusted - see TABLE 1. -- **ADR071-REFLM** set below **ADR070-RPMSF.** -- **P1-ASR SCALE**, on the Application Card misadjusted (if used) - see TABLE 1. -- **P4-PF GAIN** or **P5-PF OFF**, misadjusted on Process Interface Card - see TABLE 1. -- Top speed reference for parallel input (if used) is set higher than **ADR068-REFSF** and **ADR070-RPMSF.** -- Preset references (**RUNRF, R1RF, R2RF, SLWRF**) set too high (if used).
101001	FL41	KEYOF	**Keypad function disable (N,A)** -- Programmer key depressed for a function that is not enabled. -- CRT used on **COMPL** is disconnected or has noise. -- In MANUAL Mode check **SERJP.** -- In AUTO Mode also check **REFJP.**
101010	FL42	RLOGC	**Run logic error (N,A)** -- **JOG, RUN,** or **DIAGNOSTIC RUN** called for with **XSTOP** open. -- Keypad **STOP** called for during an external **RUN** cycle. -- Incorrect logic polarity - check **ADR012-OPTJP, JP19** and **JP20** on the Control Card, and **JP5** and **JP6** on the Application Card. -- CRT used on **COMPL** is disconnected or has noise.
101011	FL43	U11FL	**A-D converter failure (L,T)** -- Check for noise on **ACOM**, move wiring to **PCOMX.** -- Control Card failed - replace.
101100	FL44	U4LAT	**U4 to U5 communication error (L,T)** -- Check PROM revision compatibility. -- Check for noise on power supplies or analog inputs. -- Replace Control Card.
101101	FL45	SLOOP	**Speed tracking fault (L,T)** -- Drive entered current limit or **ADVLM** limit when tracking enabled. -- **ADR036-SERLM** set too low - should be between 1and 7. -- If using inner voltage loop, check software revision (**Diagnostic Test 10**) - if earlier than 33, replace software. -- Speed regulator unstable - see "Stability Troubleshooting" section.

Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
101110	FL46	IOFST	**Current feedback offset fault (L,T)** -- Current feedback signal at testpoint **CFB** offset by 35%. -- **P2-CFB ZERO** on MFC/Power Supply Card misadjusted.
101111	FL47	OADJ>	**A-D converter offset too high (N,A)** -- **P4-OFFSET** on the Control Card misadjusted - turn CW. -- Check for noise on **ACOM.**
110000	FL48	OADJ<	**A-D converter offset too high (N,A)** -- **P4-OFFSET** on the Control Card misadjusted - turn CCW. -- Check for noise on **ACOM.**
110001	FL49	NO U4	**U4 to U6 communications error (L,T)** -- Check for PROM revision compatibility. -- Check for noise on power supplies or analog inputs. -- Replace Control Card.
110010	FL50	U4BAD	**U4 RAM, timer or ALU is faulty (L,T)** -- Verify settings of **ADR011, 014, 056, 089, 187, 209,** and **210** per Custom Software Description. -- Incorrect software installed for application - check revision (**Diagnostic Test 10**); if 1B or earlier and values are programmed in the above settings, refer to GE DRIVE SYSTEMS Product Service. -- If **RUP** is REV46, **U4** must be 8032 processor, not 8031. -- **U4** failed - replace Control card.
110011	FL51	U1ROM	**U1 ROM is faulty (L,T)** -- Replace **U1** on the Control Card. -- Replace Control Card.
110100	FL52	NCLSD	**Annunciate N.C. interlock on Application card (N,A)** -- Check external input on **4TB7 (13PL-7).** -- Application Card absent but selected by soft jumper - **ADR010-APPJP** should be zero if no Application Card is present.
110101	FL53	SCLER	**SCALE ERROR (L,T)** -- Occurs if position feedback is not enabled from the Process Interface Card and the product of **ADR070-RPMSF** and **ADR074-TCPPR** is greater than 2,490,367 - check setting in **ADR070** and **ADR074.** -- Product can exceed 2,490,367 with "134" Encoder Process Interface Card, but must set **REGJP** bits 14 and 15.
110110	FL54	NO U5	**U5 to U4 communication error (L,T)** -- Check for PROM revision compatibility. -- Replace Control Card or **U5.**

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Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
110111	FL55	U5BAD	**U5 RAM, timer or ALU is faulty (L,T)** -- **U5** failed - replace. -- Replace Control Card.
111000	FL56	U5ROM	**U5 ROM is failed (L,T)** -- **U5** failed - replace.
111001	FL57	NOPEN	**Annunciate N.O. interlocks on Application card (N,A)** -- External input on **4TB8 (13PL-8)**. -- Application Card absent but selected by soft jumper - **ADR010-APPJP** should be zero if no Application Card is present.
111010	FL58	U5COM	**U4 not communicating correctly with U5 (L,T)** -- Check for PROM revision compatibility. -- Digital tach pulse rate exceeds 41,000 Hertz. -- Noise on tach input. -- Replace Control Card.
111011	FL59	NO U6	**U6 does not communicate properly with U4 (L,T)** -- Check for PROM revision compatibility. -- Replace Control Card.
111100	FL60	U6BAD	**U6 RAM, timer or ALU is faulty (L,T)** -- **U6** failed - replace. -- Control Card failed - replace.
111101	FL61	U6ROM	**U6 ROM is faulty (L,T)** -- **U6** failed - replace. -- Control Card failed - replace.
111110	FL62	----	**SPARE**
111111	FL63	U6COM	**U4 not communicating correctly with U6 (L,T)** -- Check for PROM revision compatibility. -- Digital tach pulse rate exceeds 41,000 Hertz when using Process Interface Card (531X133PRUA_G1) or 410,000 when using Encoder Process Interface Card (531X134EPRA_G1). -- **ADR002-REGJP** not set for Encoder Process Interface card (531X134EPRA_G1). -- Check for noise on tach input signal. -- **U1** extended memory - move **JP24** and **JP31** to the 2-3 position. -- **U5** and **U6** (TUP) in wrong sockets. -- Replace Control Card. -- Motor fields are open.

KEY: L - LATCHED N - NON-LATCHED H - REQUIRES HARD RESET
T - TRIPS DRIVE A - ANNUNCIATED ONLY

Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
000000	FL64	TUNRQ	**Tuneup Request - Tuneup requested remotely** -- Will automatically clear if Programmer Card has this func. -- If not cleared, install correct Programmer Card. -- Do not perform tuneup requests remotely.
000001	FL65	A1FOP	**SCR A1F is not operating (L,T)** -- Low current test may be enabled (**ADR012**, 256 bit set to zero) - run high current test (set **ADR012**, 256 bit to 1). -- Gate circuit **1F** disconnected or open. -- **5PL** disconnected or damaged. -- Failed SCR - check and replace if necessary. -- Replace Power Connection Card. -- Replace Control Card.
000010	FL66	A2FOP	**SCR A2F is not operating (L,T)** -- Low current test may be enabled (**ADR012**, 256 bit set to zero) - run high current test (set **ADR012**, 256 bit to 1). -- Gate circuit **2F** disconnected or open. -- **5PL** disconnected or damaged. -- Failed SCR - check and replace if necessary. -- Replace Power Connection Card. -- Replace Control Card.
000011	FL67	A3FOP	**SCR A3F is not operating (L,T)** -- Low current test may be enabled (**ADR012**, 256 bit set to zero) - run high current test (set **ADR012**, 256 bit to 1). -- Gate circuit **3F** disconnected or open. -- **5PL** disconnected or damaged. -- Failed SCR - check and replace if necessary. -- Replace Power Connection Card. -- Replace Control Card.
000100	FL68	A4FOP	**SCR A4F is not operating (L,T)** -- Low current test may be enabled (**ADR012**, 256 bit set to zero) - run high current test (set **ADR012**, 256 bit to 1). -- Gate circuit **4F** disconnected or open. -- **5PL** disconnected or damaged. -- Failed SCR - check and replace if necessary. -- Replace Power Connection Card. -- Replace Control Card.
000101	FL69	A5FOP	**SCR A5F is not operating (L,T)** -- Low current test may be enabled (**ADR012**, 256 bit set to zero) - run high current test (set **ADR012**, 256 bit to 1). -- Gate circuit **5F** disconnected or open. -- **5PL** disconnected or damaged. -- Failed SCR - check and replace if necessary. -- Replace Power Connection Card. -- Replace Control Card.

NOTE: *0 = LED OFF 1 = LED ON 1 = BLINKING LED*

Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
000110	FL70	A6FOP	**SCR A6F is not operating (L,T)** -- Low current test may be enabled (**ADR012**, 256 bit set to zero) - run high current test (set **ADR012**, 256 bit to 1). -- Gate circuit **6F** disconnected or open. -- **5PL** disconnected or damaged. -- Failed SCR - check and replace if necessary. -- Replace Power Connection Card. -- Replace Control Card.
000111	FL71	FWDOP	**All forward SCRs are not operating (L,T)** -- Low current test may be enabled (**ADR012**, 256 bit set to zero) - run high current test (set **ADR012**, 256 bit to 1). -- **5PL** disconnected or damaged. -- Replace Power Connection Card. -- Replace Control Card.
001000	FL72	ARMOP	**Armature circuit open (L,T)** -- DC fuse open (regenerative drives only). -- Armature circuit open. -- Brushes missing or disconnected. -- **5PL** disconnected or damaged. -- Main contactor not picking up - check pilot relay operation. -- Loss of current feedback - check **1PL-20** and **DRPL**. -- Armature not connected. -- Offset voltage at **CFB** with zero armature current - adjust **P2-CFB ZERO** on the Power Supply Card. -- Power Supply Card failed - replace. -- Control Card failed - replace.
001001	FL73	A1ROP	**SCR A1R is not operating (L,T)** -- Low current test may be enabled (**ADR012**, 256 bit set to zero) - run high current test (set **ADR012**, 256 bit to 1). -- Gate circuit **1R** disconnected or open. -- **5PL** disconnected or damaged. -- Failed SCR - check and replace if necessary. -- Replace Power Connection Card. -- Replace Control Card.
001010	FL74	A2ROP	**SCR A2R is not operating (L,T)** -- Low current test may be enabled (**ADR012**, 256 bit set to zero) - run high current test (set **ADR012**, 256 bit to 1). -- Gate circuit **2R** disconnected or open. -- **5PL** disconnected or damaged. -- Failed SCR - check and replace if necessary. -- Replace Control Card.

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Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
001011	FL75	A3ROP	**SCR A3R is not operating (L,T)** -- Low current test may be enabled (**ADR012**, 256 bit set to zero) - run high current test (set **ADR012**, 256 bit to 1). -- Gate circuit **3R** disconnected or open. -- **5PL** disconnected or damaged. -- Failed SCR - check and replace if necessary. -- Replace Power Connection Card. -- Replace Control Card.
001100	FL76	A4ROP	**SCR A4R is not operating (L,T)** -- Low current test may be enabled (**ADR012**, 256 bit set to zero) - run high current test (set **ADR012**, 256 bit to 1). -- Gate circuit **4R** disconnected or open. -- **5PL** disconnected or damaged. -- Failed SCR - check and replace if necessary. -- Replace Power Connection Card. -- Replace Control Card.
001101	FL77	A5ROP	**SCR A5R is not operating (L,T)** -- Low current test may be enabled (**ADR012**, 256 bit set to zero) - run high current test (set **ADR012**, 256 bit to 1). -- Gate circuit **5R** disconnected or open. -- **5PL** disconnected or damaged. -- Failed SCR - check and replace if necessary. -- Replace Power Connection Card. -- Replace Control Card.
001110	FL78	A6ROP	**SCR A6R is not operating (L,T)** -- Low current test may be enabled (**ADR012**, 256 bit set to zero) - run high current test (set **ADR012**, 256 bit to 1). -- Gate circuit **6R** disconnected or open. -- **5PL** disconnected or damaged. -- Failed SCR - check and replace if necessary. -- Replace Power Connection Card. -- Replace Control Card.
001111	FL79	REVOP	**All reverse SCRs are not operating (L,T)** -- Low current test may be enabled (**ADR012**, 256 bit set to zero) - run high current test (set **ADR012**, 256 bit to 1). -- **5PL** disconnected or damaged. -- **ADR002-REGJP** not set correctly for non-regenerative drives. -- Replace Power Connection Card. -- Replace Control Card.

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Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
010000	FL80	SCTFL	**SCR short circuit test failed (L,T)** -- High current test may be enabled (**ADR012**, 256 bit set to 1) - run high current test (set **ADR012**, 256 bit to zero). -- If torque proving enabled, add 8192 to **ADR012-OPTJP**. -- Check for shorted SCRs - replace.
010001	FL81	A1FSH	**SCR A1F or A4R is shorted (L,T)** -- High current test may be enabled (**ADR012**, 256 bit set to 1) - run high current test (set **ADR012**, 256 bit to zero). -- If torque proving enabled, add 8192 to **ADR012-OPTJP**. -- Check for shorted SCRs - replace.
010010	FL82	A2FSH	**SCR A2F or A5R is shorted (L,T)** -- High current test may be enabled (**ADR012**, 256 bit set to 1) - run high current test (set **ADR012**, 256 bit to zero). -- If torque proving enabled, add 8192 to **ADR012-OPTJP**. -- Check for shorted SCRs - replace.
010011	FL83	A3FSH	**SCR A3F or A6R is shorted (L,T)** -- High current test may be enabled (**ADR012**, 256 bit set to 1) - run high current test (set **ADR012**, 256 bit to zero). -- If torque proving enabled, add 8192 to **ADR012-OPTJP**. -- Check for shorted SCRs - replace.
010100	FL84	A4FSH	**SCR A4F or A1R is shorted (L,T)** -- High current test may be enabled (**ADR012**, 256 bit set to 1) - run high current test (set **ADR012**, 256 bit to zero). -- If torque proving enabled, add 8192 to **ADR012-OPTJP**. -- Check for shorted SCRs - replace.
010101	FL85	A5FSH	**SCR A5F or A2R is shorted (L,T)** -- High current test may be enabled (**ADR012**, 256 bit set to 1) - run high current test (set **ADR012**, 256 bit to zero). -- If torque proving enabled, add 8192 to **ADR012-OPTJP**. -- Check for shorted SCRs - replace.
010110	FL86	A6FSH	**SCR A6F or A3R is shorted (L,T)** -- High current test may be enabled (**ADR012**, 256 bit set to 1) - run high current test (set **ADR012**, 256 bit to zero). -- If torque proving enabled, add 8192 to **ADR012-OPTJP**. -- Check for shorted SCRs - replace.

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Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
010111	FL87	NOUAP	**Microapplication card failure (L,T)** -- Microapplication card absent but selected by soft jumper - **ADR014-SPCJP** should be set to disable Microapplication Card interface (refer to Custom Software Description). -- **3PL** incorrectly connected or damaged. -- **U30** or **U33** on Microapplication Card failed - replace. -- Replace Microapplication Card.
011000	FL88	UAPWD	**MUP watchdog misoperated (L,T)** -- Application Card (G2) installed when Microapplication Card (G1) is called for - check setting in **ADR014-SPCJP** per Custom Software Description. -- **U30** on Microapplication Card failed or jumpers set incorrectly - replace card if necessary. -- Refer to BugBuster, "Noise or Random Nuisance Faults" item.
011001	FL89	MUPTO	**RUP to MUP communication failure (L,T)** -- **U30** on Microapplication Card failed - replace. -- Replace Microapplication Card.
011010	FL90	NOMSG	**RUP to MUP watchdog detected no message received within time period expected (L,T)** -- Serial Link or Local Area Network transmission lines disconnected. -- **12PL** damaged or disconnected. -- External device failed to send message in time period set by **ADR005-SERJP**.
011011	FL91	----	**SPARE**
011100	FL92	----	**SPARE**
011101	FL93	----	**SPARE**
011110	FL94	----	**SPARE**
011111	FL95	WATCH	**FUP watchdog detected RUP has misoperated (L,T)** -- Same as **FL58.**
100000	FL96	PASSED	**Annunciates when SCR test has passed.**

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Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
100001	FL97	OCTFL	**SCR open circuit test failed (L,T)** -- Low current test may be enabled (**ADR012**, 256 bit set to zero) - run high current test (set **ADR012**, 256 bit to 1). -- DC fuse open (regenerative drive only). -- Armature circuit open. -- Brushes missing or disconnected. -- **5PL** disconnected or damaged. -- Main contactor not picking up - check pilot relay operation. -- Loss of current feedback - check **1PL-20** and **DRPL**. -- Offset voltage at **CFB** with zero armature current - adjust **P2-CFB ZERO** on the Power Supply Card. -- Power Connection Card failed - replace. -- Power Supply Card failed - replace. -- Control Card failed - replace.
100010	FL98	OCTMF	**Multiple open SCRs (L,T)** -- Low current test may be enabled (**ADR012**, 256 bit set to zero) - run high current test (set **ADR012**, 256 bit to 1) -- Check all gate plugs. -- **5PL** disconnected or damaged. -- Check SCRs for failure - replace. -- Replace Power Connection Card. -- Replace Control Card.
100011	FL99	ALLOP	**All SCRs tested open (L,T)** -- Low current test may be enabled (**ADR012**, 256 bit set to zero) - run high current test (set **ADR012**, 256 bit to 1). -- DC fuse open (regenerative drive only). -- Armature circuit open. -- Brushes missing or disconnected. -- **5PL** disconnected or damaged. -- Main contactor not picking up - check pilot relay operation. -- Loss of current feedback - check **1PL-20** and **DRPL**. -- Armature not connected. -- Offset voltage at **CFB** with zero armature current - adjust **P2-CFB ZERO** on the Power Supply Card. -- Power Supply Card failed - replace. -- Control Card failed - replace.

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TABLE 5A. LAN FAULT TROUBLESHOOTING

Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
100100	FLA0	MLNKR	**Serial Link (product protocol) Fault - BIU disabled (EE211 MSET1.8 set); receive buffer overflowed. (L, A)** -- Enable the **BIU.** -- Modify the application's code or specifications. **NOTE:** *Presently no application uses a serial link protocol except the LAN, and there are no plans for one in the future. Keep the BIU enabled.*
100101	FLA1	MSTCK	**Stack Space fulled due to MUP Application Software. The stack has overiflowed the specified stack space. (L, A)** -- The application can adjust the specified stack space. -- The application should use less stack for storage.
100110	FLA2	MEPRM	**EPROM Checksum is incorrect. (L, A)** -- Replace the EPROM **(U33).**
100111	FLA3	MIRAM	**The Internal Processor RAM failed. (L, A)** -- Replace the processor **(U30).**
101000	FLA4	MERAM	**The External RAM Chip (6116 or 6164) failed. (L, A)** -- Replace the RAM chip **(U49).**
101001	FLA5	MITMR	**The Internal Processor Timers failed. (L, A)** -- Replace the processor **(U49).**
101010	FLA6	MSRAM	**Unable to modify and read RAM on Port Expander Chip (8155). (L, A)** -- Replace the Port Expander Chip **(U37).**
101011	FLA7	MFMSG	**Link Fault, unable to send messages. Processor has not heard the last 128 bytes sent out on serial port. (L, A)** -- May not have **JP16, JP17,** and **JP15** placed in 1-2 position on Microapplication Card. -- May have too may drops with the termination resistors in the circuit. **(JP18** and **JP19** placed in 1-2 position, maximum drops = 5 with the termination resistors in the circuit). -- May have another drop programmed with the same drop number. -- All drops may not be programmed for the same link size **(ADR214).** -- The serial lines may be shorted. -- The driver **(U35)** or receiver **(U34)** may be defective. -- The internal timer on the processor may be defective **(U30).** -- If actively using the LAN, then **JP13** must be in 1-2 position. -- Replace any defective parts. -- Make certain the serial lines are not shorted and the common mode voltage does not exceed 7V between any two drops.

Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
110101	FLB7	LANEA	**Weighted Zero Protocol Error - Expected all drops to not use weighted zero checksums and acknowledging of group or broadcast commands, but found some did. (L, A)** -- Make all drops compatible. Weighted Zero Protocol requires REV52 or later **MUP** in drives and REV13 or later **SUP** in **BIU**. -- If adding or updating drops, can configure **MUP** REV52 to not use Weighted Zero with **ADR211**.
101100	FLA8	MFRZM	**Too many parameters programmed in the Freeze Mask. There are more than six parameters programmed in the Freeze Mask (see EE215, MSET5). (L, A)** -- Reprogram **EE215** to ask for six or less parameters.
101101	FLA9	MSMSG	**Received Serial Data insufficient in length. Required number of data bytes (for the received command) were not received from the LAN, but the message checksum was correct. (L, A)** -- Too many sets of termination resistors in the circuit (maximum = 5 with **JP18** and **JP19** in 1-2 position). -- Serial Link lines may have a momentary short. -- Driver of the transmitting device or the receiver of this LAN drop may have failed. -- Serial port or timer of processor (**U30**) may have failed. -- An invalid message may have been sent to this LAN drop. **NOTE:** *Check message that was sent to the LAN drop (displaying the fault) for correct data and length.*
101110	FLB0	MSBUF	**Not enough buffering of received bytes. The triple byte buffering scheme is insufficient for the amount of window time (between MUP and shared RAM [8155]). (L, A)** -- Contact Factory.
101111	FLB1	MWIND	**The window interrupted itself. The start of the next window is too close to the end of the previous window. (L, A)** -- Contact Factory.
110000	FLB2	MSWIN	**MUP did not have shared RAM for long enough time. The window interrupt routine took longer to complete than the drive allotted for. (L, A)** -- Contact Factory.

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Diagnostic Readout	Prog. Crd	Name	Description/Probable Causes/Remedies
110001	FLB3	MPRGR	**Trying to modify an illegal address in MUP. The user tried to modify a protected address in MUP. (L, A)** -- Do not try to modify protected addresses in **MUP**.
110010	FLB4	MAPP#	**The MUP application does not match the EE application number. The microapplication software does not match the number programmed into EE014 for an application selection. (L, A)** -- Check **EE014** for the correct value. -- Check the microapplication EPROM **(U33)** for the correct software package.

NOTES:

TABLE 5B. SPECIFIC ERROR TROUBLESHOOTING

Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
--	ER01	BUART	**Faulty UART on the 8031 U.C.** -- This is a normal response to **Test 50.** -- Hand-Held Programmer plugged directly into Control Card (can only be plugged into Diagnostic Card **18PL** or version 100 Control Card). -- Check **JP3** and **JP4** on Programmer Card. -- Programmer Card failed - replace.
--	ER02	BKPAD	**The keypad has shorted leads.** -- Replace ribbon cable. -- A key pressed during power up or reset. -- Programmer Card failed - replace.
--	ER03	BDMEM	**The random access memory is faulty.** -- Programmer Card failed - replace.
--	ER04	BPRGM	**The program memory checksum is incorrect.** -- Programmer Card failed - replace.
--	ER05	BTIMR	**The timers in 8031 U.C. are faulty.** -- Programmer Card failed - replace.
--	ER06	NADDR	**An address is not specified.** -- "**ENTER**" key pressed without specifying a parameter address.
--	ER07	2 KEYS	**Two keys were pressed together.** -- Press keys one at a time. -- Keypad failed - replace. -- Programmer Card failed - replace.
--	ER08	ILBRD	**Illegal broadcast.** -- Programmer Card failed - replace.
--	ER11	NOACK	**No acknowledgement received.** -- Device connected to the serial link not responding to the acknowledge requests from the Programmer Card. -- Wires to the serial link disconnected. -- Driver or receiver chip failed on Programmer, Process Interface Card, or external device. -- Processors locked up, use Hard Reset. -- If Hand-Held Programmer, may be failed **U6** port - replace. -- Noise on transmission wires. -- RS232C on hand-held cable.

Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
--	ER12	N-CTS	**No "clear-to-send".** -- Serial link transmission lines disconnected. -- Check **JP3** and **JP4** on Programmer Card. -- Noise on 5-volt power supply. -- Ground loops or unsuppressed relays. -- Programmer Card failed, replace.
--	ER13	ORBUF	**Receive buffer has overflowed.** -- Message sent to the drive which exceeds the maximum number of bytes (11 data bytes). -- Baud rate set incorrectly. -- Byte protocol set incorrectly on external device - refer to Serial Link User's Guide. -- Message protocol used incorrectly - refer to Serial Link User's Guide. -- Hardware configuration incorrect for serial link operation. -- External serial link device failed.
--	ER14	CKSUM	**Incorrect checksum received.** -- The serial link message checksum has been calculated improperly by external Serial Link device - refer to the Serial Link User's Guide. -- Baud rates set incorrectly. -- Byte protocol set incorrectly on external device - refer to Serial Link User's Guide. -- Message protocol used incorrectly - refer to Serial Link User's Guide. -- Hardware configuration incorrect for serial link operation. -- External serial link device failed.
--	ER15	BDMSG	**Something other than a "start of transmission" was received when "S.O.T." was expected.** -- "Start of transmission" byte of the message sent over the serial link has been determined incorrectly - refer to the Serial Link User's Guide. -- Baud rates set incorrectly. -- Byte protocol set incorrectly on external device (1 start bit, 8 data bits, and 1 stop bit). -- Message protocol used incorrectly - refer to Serial Link User's Guide. -- Hardware configuration incorrect for serial link operation. -- External serial link device failed.

Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
--	ER21	MTRNG	**The motor is rotating at start of "self-tune" tune-up program.** -- Test point **SFB** contains an offset to zero speed - adjust **P7-SFB ZERO** on the Process Interface Card. -- A/D offset misadjusted - adjust **P4-OFFSET** on the Control Card. -- Noise on tach leads. -- **7PL** damaged or disconnected. -- Failed Process Interface Card. -- Programmer Card failed - replace.
--	ER22	N-ACL	**Max current is reached without detecting motor acceleration.** -- Check switches and jumpers for tach on Process Interface Card. -- Drive and Programmer Card not communicating properly. -- Software jumpers misprogrammed.
--	ER23	TPSPD	**Motor has reached 50% of program's speed.** -- Check jumpers and switches on Process Interface Card. -- Motor and inertia too large for test.
--	ER24	MTRNG	**Motor accelerated for more than 10 seconds while finding the friction current.** -- Test point **SFB** contains an offset at zero speed - adjust **P7-SFB ZERO** on the Process Interface Card. -- A/D offset misadjusted - adjust **P4-OFFSET** on the Control Card. -- Motor being driven by external source.
--	ER25	MTRNG	**Motor was still turning 20 seconds after finding the friction current.** -- Test point **SFB** contains an offset at zero speed - adjust **P7-SFB ZERO** on the Process Interface Card. -- A/D offset misadjusted - adjust **P4-OFFSET** on the Control Card. -- Check jumpers or switches on the Process Interface Card. -- Pulse rate set incorrectly for digital tach. -- Soft jumpers set incorrectly.

Diagnostic Readout	Prog. Crd	Name	Description/Probable Causes/Remedies
--	ER26	N_IFB	**Reach 25 degrees retarded firing and no current feedback measured.** -- DC fuse open. -- Armature loop open. -- Check SCR. -- **CFB** circuit failed, replace Power Supply Card.
--	ER27	N_IFB	**Reach 37 degrees retarded firing and no current feedback measured.** -- DC fuse open. -- Armature loop open. -- Check SCR. -- **CFB** circuit failed, replace Power Supply Card.
--	ER28	PRGRM	**Parameter values calculated are out of range.** -- If using **S-TUNE** for speed loop, **SFB** miscalibrated or has offset. -- If using **I-TUNE** for current loop, **CFB** miscalibrated or has offset.
--	ER29	TORQP	**Torque proving called for by soft jumpers in EEROM** -- **I-TUNE** test will not work if torque proving is part of the software.

TROUBLESHOOTING STABILITY PROBLEMS

There are three broad categories of drive symptoms broadly classified as "Stability" problems. Technically, two of these are not actually stability problems but, due to their behavior, are included in this troubleshooting procedure. These categories are as follows:

"Stability" problems: "Hunting", oscillations in speed or current, etc. In general all problems in this category are cyclical in nature, with well-defined sinusoidal periods of oscillation. Usually these problems are electrically induced and can be tuned out with proportional and integral gain adjustments.

-- There are at least eight regulating loops in most DC-300 drives, any one which can be mistuned, causing stability problems under the right conditions. They are as follows:
  (1) Current regulator - continuous current
  (2) Current regulator - discontinuous current
  (3) Speed regulator - base speed
  (4) Speed regulator - weak field speed
  (5) CEMF regulator
  (6) CEMF limit regulator
  (7) Field current regulator - base speed
  (8) Field current regulator - weak field speed

NOTE: *Various forms of position regulators are also available, which, due to their variety are not addressed here.*

Mechanical Oscillation Problems: These are a direct result of mechanical deficiencies but under certain conditions can be aggravated or excited by electrical frequencies. Examples of this are vibration, audible motor noise, gear chatter, etc.

Erratic Operation Problems: Intermittent speed or current fluctuations, mechanical "bumping", or other disturbances non-sinusoidal or non-periodic in nature. These may be induced by misfiring SCRs, intermittent grounds, "noise", or mechanical problems.

NOTE: *A programmer is required to make parameter checks called for in Part I and adjustments as required in Part II.*

I. Before troubleshooting the drive, first verify the following:
  -- All software jumpers, **ADR000** through **ADR014** are in accordance with drive parameter list.
  -- All hardware jumpers are in accordance with the jumper selection chart.
  -- All wiring is correct and in agreement with the drive elementaries.
  -- All "gain" parameters are in agreement with the parameter list. Specifically, the following parameters* should be checked:

Speed Loop:

ADR015-BSLPG	ADR016-BSLIG
ADR017-WSLPG	ADR018-WSLIG
ADR037-SLEAD	ADR051-LSPAN

Current Loop:

ADR019-ILPPG	ADR020-ILPIG
ADR021-ILPDG	ADR022-ILPDP

CEMF Loop:

ADR025-CLPPG	ADR026-CLPIG

CEMF Limit Loop:

ADR027-CLMPG	ADR028-CLMIG

Field Current Loop:

ADR031-BFLPG	ADR032-BFLIG
ADR033-WFLPG	ADR034-WFLIG

***NOTE:** *These parameters assume Motor 0 operation. If Motor 1, Motor 2, or Motor 3 stability problems exist, other parameters will affect the speed and current loop (e.g.: Motor 1 base speed loop proportional gain is designated as ADR043-BSLP1).*

II. Use TABLE 6 to decide which type of problem actually exists, keeping the following pointers in mind:
  -- DC-300 drive current loops are almost always stable.
  -- Most stability problems, but not all, are corrected by adjusting speed loop parameters.
  -- It is always worthwhile to verify field current stability first.
  -- In general, when troubleshooting cascading or series regulating loops, the correct procedure is to stabilize the inner regulator first and then move outward; i.e., current regulator, then speed regulator, then position regulator (if furnished).
  -- If at any time the drive becomes stable as a result of a step, record any parameter changes, and do not continue with the remaining procedures.

TABLE 6. STABILITY TROUBLESHOOTING

Symptom	Procedures
A. Unstable when unloaded but stable when loaded, drive is stable but exhibits gearbox or motor noise **or** runs erratically.	1. Run **I-Tune (Test 14)** if available. 2. Check for failed SCRs - use the SCR test **(Test 12)** or monitor test point **CFB** on the Control Card with an oscilloscope. 3. Current reversal too sensitive - increase **ADR023-DEADB.** 4. "Light Load" instability - increase **ADR021-ILPDG.** If frequency of instability inceases, raise **ADR021** as high as possible without encountering erratic operation. (If **ADR021** is changed, set **ADR020-ILPIG** to the same value.) 5. Reduce as low as possible, even zero if necessary. 6. If instability still exists, return adjusted parameters to their original setting and skip to Symptom **E.**
B. Unstable above and /or below base speed and over a wide speed range (Stabilizing the Field Current Regulator).	1. Confirm Field Current Regulator stability - monitor test point **FC** on the Control Card with an oscilloscope or **VOM** while in Diagnostic Static **(Test 1).** If stable, skip to the next appropriate symptom. 2. Field current is unstable - reduce **ADR031-BFLPG** until **FC** is stable. Do not reduce below the setting in **ADR032-BFLIG.** **NOTE:** *ADR031 may already be lower than ADR032 in some applications, maintain this relationship. Lower ADR032 if necessary.* 3. Set **ADR033-WFLPG** equal to the new value in **ADR031-BFLPG.** 4. Set **ADR034-WFLIG** equal to the new value in **ADR032-BFIPG.** 5. Document any change and proceed to Symptom **C** or **D** as necessary.

Symptom	Procedures
C. Speed instability <u>below</u> base speed.	1. Run **S-Tune (Test 15)** if available. 2. Confirm that values in **ADR015-BSLPG, ADR016-BSLIG,** and **ADR037-SLEAD** agree with the parameter list and satisfy the following equation: BSLIG/BSLPG = .127 (SLEAD) **NOTE:** *If ADR037-SLEAD = 0, skip to step 5.* 3. If **ADR037-SLEAD** contains a value (typically "10" or less) readjust BSLPG to satisy the above equation. <u>Do not</u> change **SLEAD** unless different from the parameter list. 4. Raise/Lower **ADR016-BSLIG** and **ADR015-BSLPG** maintaining the above ratio. Raising the parameters makes the drive more responsive; lowering them makes the drive less responsive. 5. If **SLEAD** is 0 and the drive is slow and unresponsive, increase **ADR015-BSLPG** and adjust **ADR016-BSLIG** upward or downward as necessary. 6. If no improvement, return the adjusted parameters to their original settings. 7. If the drive is a CEMF regulator - lower **ADR039-IRCMP** to zero. 8. If there is some improvement but instability still exists - document changes and skip to Symptom **E.**
Symptom	**Procedures**
D. Speed stability problems <u>above</u> base speed.	1. Run **S-Tune (Test 15)** if available. 2. Confirm that the values in **ADR017-WSLPG, ADR018-WSLIG,** and **ADR037-SLEAD** agree with the parameter list and satisfy the following equation: WSLIG/WSLPG = .127 (SLEAD) **NOTE:** *If ADR037-SLEAD = 0, skip to step 5.* 3. If **ADR037-SLEAD** contains a value (typically "10" or less) readjust **BSLPG** to satisy the above equation. <u>Do not</u> change **SLEAD** unless different from the parameter list. 4. Raise/Lower **ADR018-WSLIG** and **ADR017-WSLPG** maintaining the above ratio. Raising the parameters makes the drive more responsive; lowering them makes the drive less responsive. 5. If **SLEAD** is 0 and the drive is slow and unresponsive, increase **ADR017-WSLPG** and adjust **ADR018-WSLIG** upward or downward as necessary. 6. If no improvement, return the adjusted parameters to their original settings. 7. Drive is in CEMF limit - to confirm, lower **ADR079-CROSS** (should be approximately 150 to 200 counts lower than **ADR080-CMFLM**). 8. If there is some improvement but instability still exists - document changes and skip to Symptom **E.**

Symptom	Procedures
E. Drive is unstable: over a small part of the speed range, **or** when loaded but stable when unloaded, **or** such that the oscillating frequency is proportional to speed.	1. Confirm that the problem is not tach related - run the drive as a CEMF regulator by using Diagnostic "**RUN**" (**Test 2**); if available (verify that **ADR003-DGNJP** does not have the 32 bit set by converting the decimal value to binary before performing this test). If symptoms disappear, correct the tach feedback problem (misaligned or loose coupling, belt slippage, wiring or excessive ripple). 2. Verify that the power supplies are clean with an oscilloscope (refer to "Checking Power Supplies" in the Troubleshooting section). 3. Run drive to top speed and push the "**RESET**" button on the Control Card. The drive will coast stop. Mechanically induced problems will still be present as the motor decelerates. If problem disappears, continue to next step. 4. Lower settings in **ADR015, ADR016, ADR017**, and **ADR018** as low as possible while maintaining the following ratios: a. $\frac{\text{FINAL ADR016}}{\text{FINAL ADR015}} = \frac{\text{ORIGINAL ADR016}}{\text{ORIGINAL ADR015}}$ b. $\frac{\text{FINAL ADR018}}{\text{FINAL ADR017}} = \frac{\text{ORIGINAL ADR018}}{\text{ORIGINAL ADR017}}$ 5. Reduce **ADR037-SLEAD** and double **ADR015** and **ADR016**. 6. Determine resonant frequency "Fr" in Hz with an oscilloscope, if possible. Convert to radians per second as follows: "Wr" = 6.28 x "Fr". 7. Adjust **ADR037-SLEAD** and **ADR051-LSPAN** as follows: Set ADR037-SLEAD = .615 x "Wr" Set ADR051-LSPAN = 5 and readjust as necessary 8. If no improvement, return the parameters to their original settings and skip to Symptom **G.**
F. Undirectional Stability Problems.	1. Try running the motor in the opposite direction to confirm that stubbing brushes or brush position is not the problem. 2. Inspect the load to make sure that nothing is wrong mechanically. 3. If a regenerative drive, make certain that all SCRs are firing in both directions by running **Diagnostic Test 12** (SCR test) or monitoring test point **CFB** on the Control Card with an oscilloscope.

Symptom	Procedures
G. Additional Procedures.	If the above procedures do not result in stable operation, additional steps may be required including: 1. Checkout of the current loop. 2. Running the **Square-Wave Reference Test (Test 25).** 3. Running the field regulator open loop to verify stability.

If such procedures are required, contact GE DRIVE SYSTEMS, Product Service Engineering, for instructions. (703-387-7595)

NOTES:

NOTES:

10. TYPICAL WAVEFORMS

The following are examples of typical waveforms that can be seen on an oscilloscope. Each figure is accompanied by the name of the signal, the test point which it is derived from (in parenthesis), and the mode of operation if applicable.

Also attached is the proper time setting for the waveform. (For information on using an oscilloscope, consult the owner's manual.)

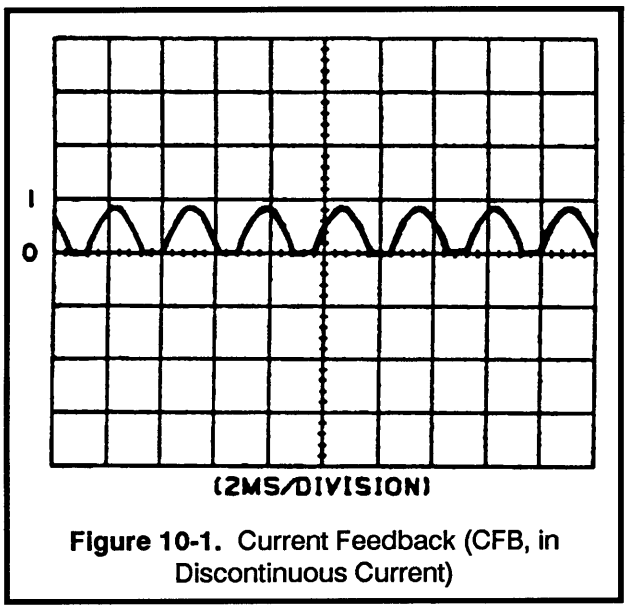


Figure 10-1. Current Feedback (CFB, in Discontinuous Current)

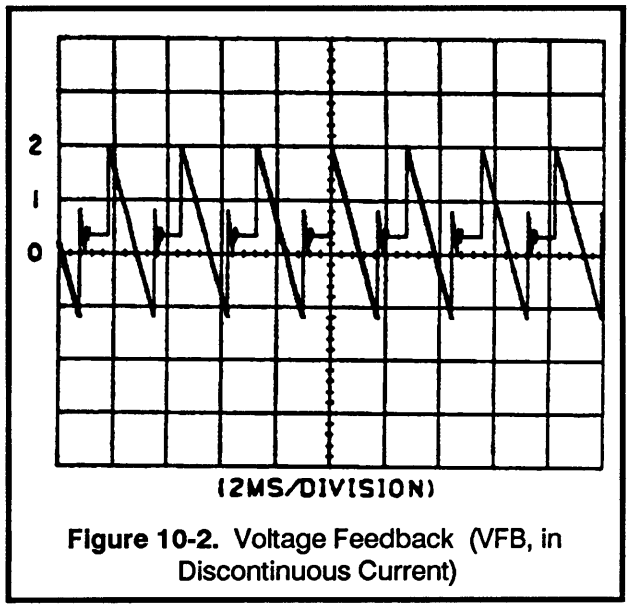


Figure 10-2. Voltage Feedback (VFB, in Discontinuous Current)

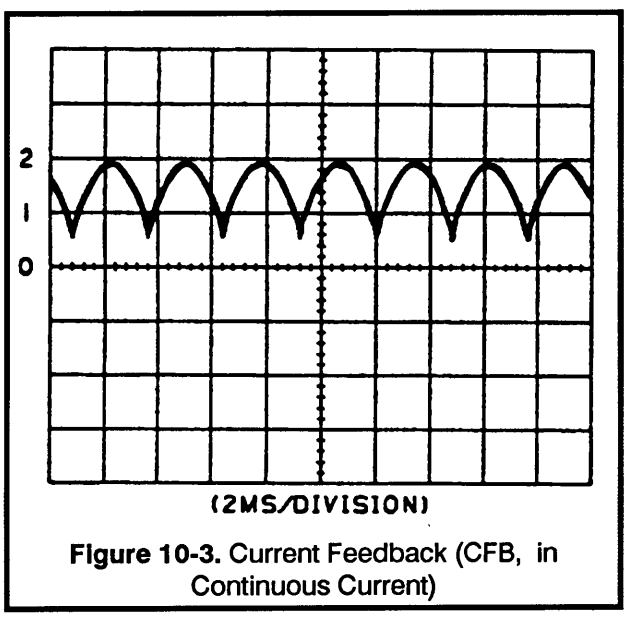


Figure 10-3. Current Feedback (CFB, in Continuous Current)

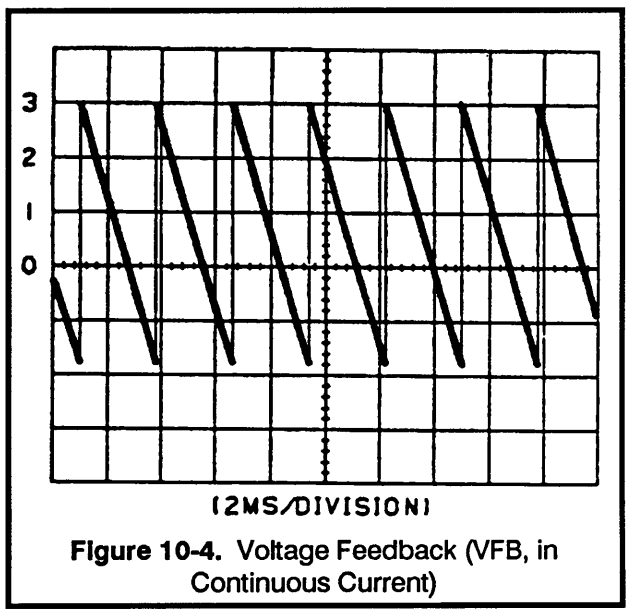


Figure 10-4. Voltage Feedback (VFB, in Continuous Current)

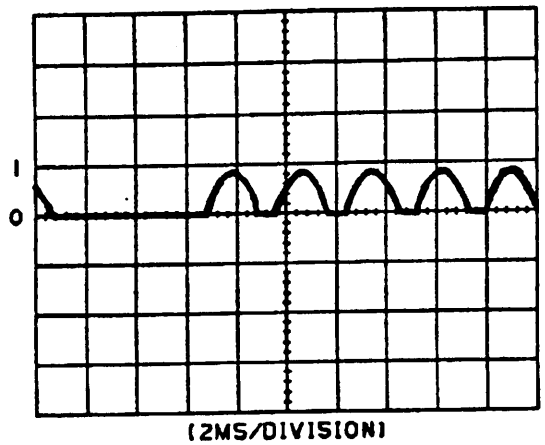


Figure 10-5. Current Feedback (CFB, with One SCR Open or Disconnected)

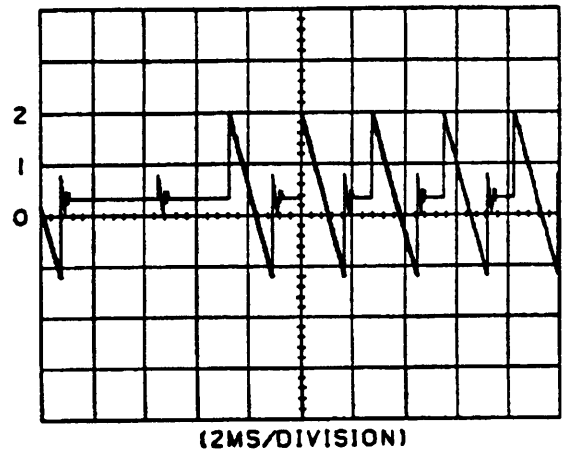


Figure 10-6. Voltage Feedback (VFB, with One SCR Open or Disconnected

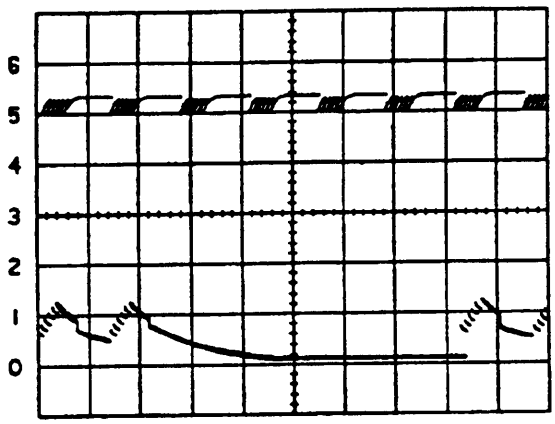


Figure 10-7. TOP - Delayed Firing Power
BOTTOM - Gate Pulses For A1 SCR
(Discontinuous Current)

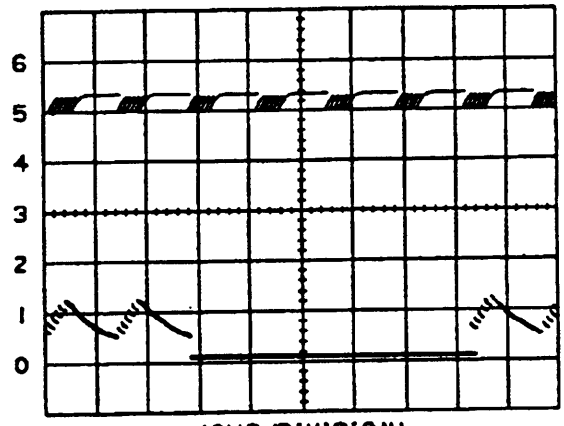


Figure 10-8. TOP - Delayed Firing Power
BOTTOM - Gate Pulses For A1 SCR
(Continuous Current)

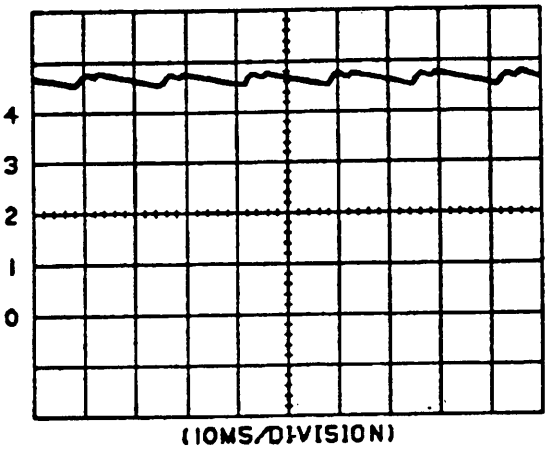


Figure 10-9. Field Current (FC)
At Full Field

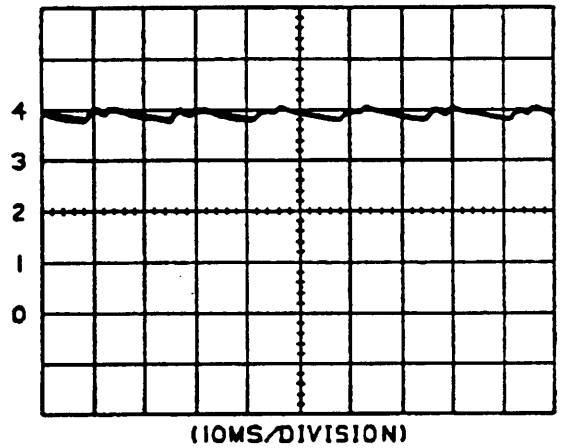


Figure 10-10. Field Current (FC)
At Weak Field

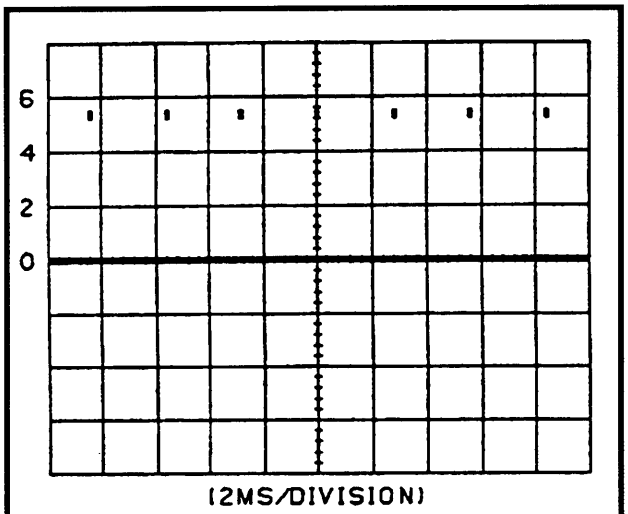


FIGURE 10-11. Initial Pulse Unit (IPU) (Indicated Initial Pulse of Each Armature SCR Firing Pulse Burst)

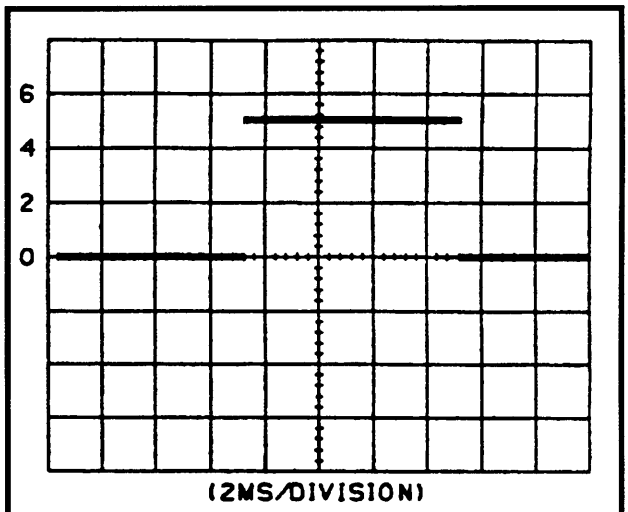


Figure 10-12. Sync Signal (SY) (Square Wave Synchronized to AC Line Frequency)

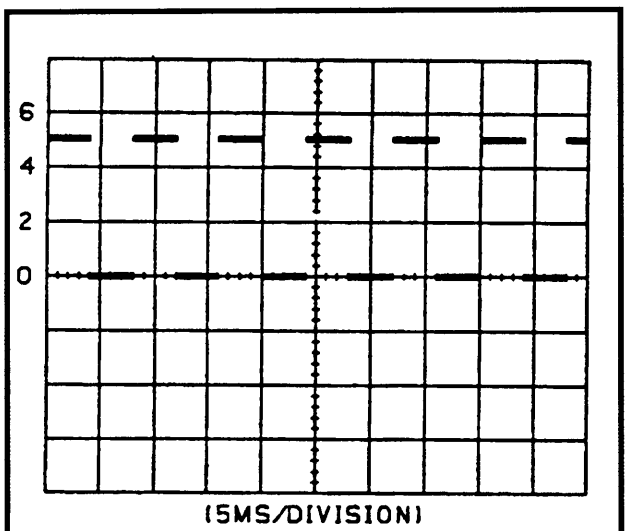


Figure 10-13. AC Tach Signal (TPN) At 450 RPM

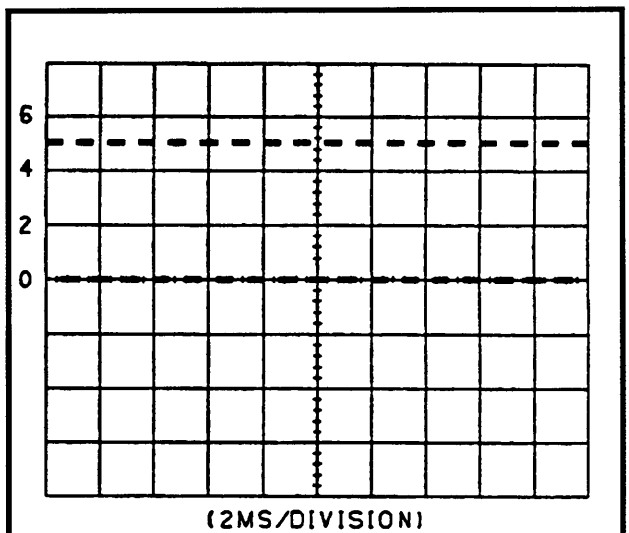


Figure 10-14. AC Tach Signal (TPN) At 3000 RPM

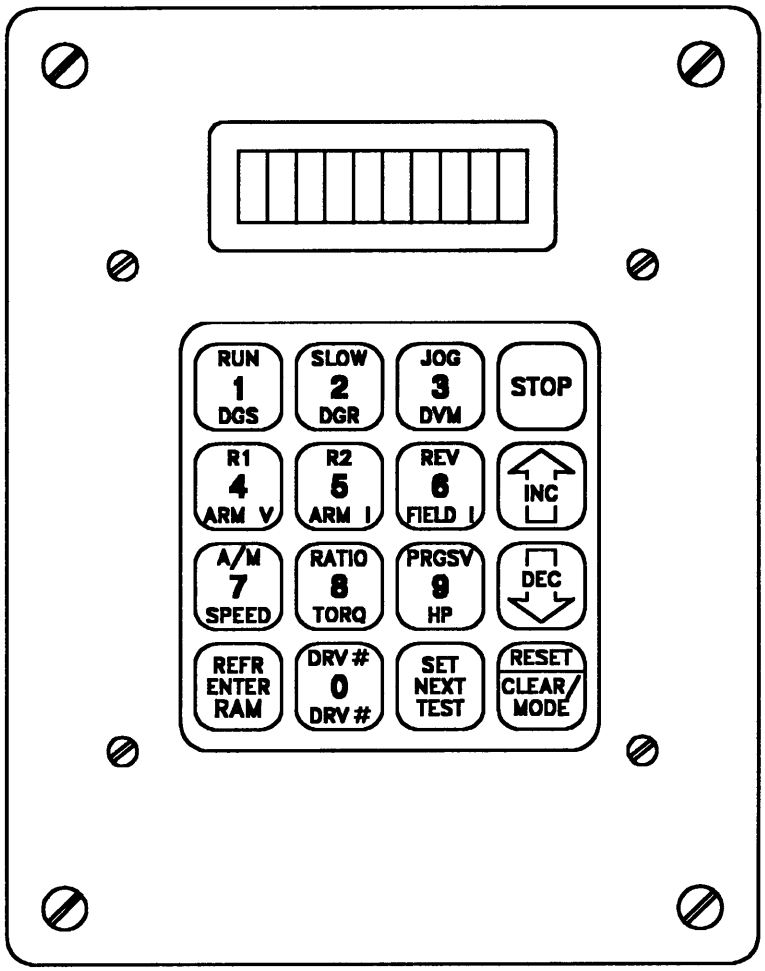


Figure 11-1. Optional Programmer

11. USE OF THE OPTIONAL PROGRAMMER MODULE

The optional Programmer assembly is available in two configurations. The **Local** configuration is mounted on the Control card and plugged into **18PL**. The **Hand-Held** configuration is contained in a box and plugged directly into the version "300" Control Card at **18PL** (when using version "100" Control card, plug into 18PL on the Diagnostic Readout Card only). The Hand-Held Programmer uses a different cable than the Local Programmer and care must be exercised when the Hand-Held Programmer is plugged in.

CAUTION: The Hand-Held Programmer cable must be oriented properly or the Control and Programmer Cards will be damaged.

Orient the Hand-Held Programmer cable as follows:

1. Hold the unit in front of the drive and extend the ribbon cable back to **18PL**.
2. Verify that the cable is not twisted, and plug into **18PL**.
   -- The colored stripe in the ribbon cable should be on the left when looking at the front of the Hand-Held Programmer and should be on the left when plugged into **18PL**.

Orient the Local Programmer cable as follows:

1. Hold the unit in front of the drive and extend the ribbon cable back to **18PL**.
2. Verify that cable is oriented with the colored wire and unused connector pins on the left side of the ribbon cable when facing the drive, and plug into **18PL**.

The optional Programmers provide an alphanumeric readout of Fault Codes (instead of the diagnostic LEDs) and operation/adjustment of the drive through three modes - **OPERATE, PARAMETER,** and **DIAGNOSTIC**. (See Figure 11-1.)

OPERATE Mode - The "Normal" Mode, allows the **RED** and **BLACK** keypad functions to be active (the **RED** keys are dominant). This is the mode the drive will enter when power is applied.

PARAMETER Mode - Allows the user to examine and change as necessary any of 254 EEROM (E^2) Parameter addresses to "customize" the drive for a given application. In this mode all **BLACK** keys are active plus **[INC], [DEC],** and **[STOP]**.

DIAGNOSTIC Mode - Allows the user to utilize any of the internal digital displays, plus run a series of diagnostic checks and self-tuning procedures. In this mode, both the **BLUE** and **BLACK** key functions are active (with **BLUE** keys dominant), plus **[INC], [DEC],** and **[STOP]**.

CHANGING BETWEEN MODES

NOTE: *Throughout this section, a number or function encased in brackets [ ] indicates a programmer key pressed. The drive will normally enter the **OPERATE** mode as soon as power is applied.*

To enter the PARAMETER Mode, perform the following:

1. Confirm that the display reads **OPERATE.**
2. Press **[SET], [DRV#], [7], [7],** and **[ENTER]**.
   -- The display will show **PARAMETER** .

NOTE: *In order to change parameter values, two conditions must be present:*

1. *The EEROM (E^2) enable device must be in the "ENABLE" position.*
   JP13: 1-2; position; E^2 write "PROTECT"
   2-3; E^2 write "ENABLE"
2. *Parameter address #3 (designated as **ADR003-DGNJP**) must contain an **odd** number (1, 3, 5, etc.). If an even number exists, add "1" to the present value and enter the new number.*

To enter the DIAGNOSTIC Mode, perform the following: (The drive must first be in the **PARAMETER** mode.)

-- If the display shows **PARAMETER**, press **[CLEAR/MODE]** and the display will immediately show **DIAGNOSTIC.**

-- If a particular parameter is being displayed, press **[CLEAR/MODE]** causing **PARAMETER** to be displayed, then press **[CLEAR/MODE]** again, and the display will show **DIAGNOSTIC.**

To return to the OPERATE Mode, perform the following:

-- If in the DIAGNOSTIC Mode, press **[CLEAR/MODE]** once and the display will return to **OPERATE.**

-- If in the PARAMETER Mode, press **[CLEAR/MODE]** either two or three times as required until the display returns to **OPERATE.**

NOTE: *The Programmer may appear to "hang up" due to programming errors. In such cases it may be necessary to initiate a "Hardware Reset" using the **RESET** switch (**SW2**) on the Control Card.*

WARNING - POTENTIAL DRIVE ACTIVATION AND INADVERTENT EQUIPMENT MOVEMENT WHEN TESTING: Throughout the remainder of this section, do not use DIAGNOSTIC RUN, SQUARE-WAVE REFERENCE TEST, or any programmer RUN functions on any crane or elevator drive or other applications where mechanical movements cannot be adequately monitored.

OPERATE MODE - (RED Keys dominant)

In the **OPERATE** Mode, any of the **RED** key functions, if activated, can be used as follows: (If key is not activated, the Programmer will display "**FL41 KEYOF**" momentarily.)

[RUN] - Run at speed set by **RUNRF.**

[SLOW] - Run at speed set by **SLWRF.**

[JOG] - Jog at speed set by **JOGRF.**

[STOP] - Controlled stop.

[R1] - Run at speed set by **R1RF.**

[R2] - Run at speed set by **R2RF.**

[REV] - Reverse **RUN** or **JOG** as required by first pressing **[REV]**, then either **[RUN]**, **[SLOW]**, **[JOG]**, **[R1]**, **[R2]**, or **[REFR].**

[A/M] - Changes drive from **AUTO** to **MANUAL**, or vice-versa, enabling use of either **ASR** or **MSR** external speed reference inputs.

[RATIO] - Displays the drive run ratio or the percentage of speed reference at which the drive will run. Ratio is used to set a draw between multiple drives monitoring the same external speed reference. Normally set at 10,000 indicating a ratio of 1:1.

[PRGSV] - For future use.

[REFR] - Run at speed reference at which drive was last run, (includes preprogrammed keypad references or changes caused by using the **[INC]** and **[DEC]** keys, **MSR**, or **ASR**).

[DRV#] - Used to examine the drive number (in most cases "00"). Also part of access code to **PARAMETER** Mode.

[SET] - Used to adjust the settings of various keypad functions; activated by pressing **[SET]**, then **[RUN]**, **[SLOW]**, **[JOG]**, **[R1]**, **[R2]**, or **[RATIO]**, then the desired setting in decimal numbers, then **[ENTER]**. Also, used for entering **PARAMETER** Mode. The EEROM must be write "ENABLED" to make these adjustments (see "Changing Between Modes" paragraph in this section).

[INC] & [DEC] - Increases or decreases the speed of any of the keypad **RUN** or **JOG** functions, or the **RATIO** function; simply enter the desired function and then press **[INC]**or **[DEC]** as desired.

[RESET] - Resets the drive after fault or error messages appear. The **RESET** key is referred to as a "Software Reset".

The normal display in the **OPERATE** Mode will be **M XX% YY%** where **M** indicates **MANUAL** Mode using the Manual System Reference, **MSR** (alternatively an **A** will indicate **AUTO** Mode using the Auto System Reference, **ASR** on the optional Application Card [or Microapplication Card]). **XX%** indicates percent of rated speed and **YY%** indicates percent of rated load.

PARAMETER MODE - (BLACK Keys)

In the **PARAMETER** Mode, any of the **BLACK** key functions plus **INC, DEC & STOP**, if activated, can be used as follows: (If key is not activated, the Programmer will display "**FL41 KEYOF**" momentarily.)

1. Enter the **PARAMETER** Mode by pressing **[SET], [DRV#], [7], [7],** and **[ENTER]**.

2. Examine the present value of any parameter by keying in its three-digit parameter address (do not press **[ENTER]**).
   - -- The name for that address will immediately be displayed along with the present contents of that address.
   - -- Press **[CLEAR/MODE]** to clear displayed parameter (display will show **PARAMETER** again).
   - -- Key in the address number for the next parameter desired.

NOTE: *Occasionally, it may become necessary to check a series of parameters. Simply call up the lowest parameter address and press* ***[NEXT]*** *as often as required until the highest parameter number is reached.*

NOTE: *After a period of inactivity, the display may return to* ***'M.XX%YY%*** *(as displayed in the* **OPERATE** *Mode). The drive is still in the* **PARAMETER** *Mode (as indicated by the apostrophe before the* ***M****) awaiting further keypad entries. To proceed, enter the next parameter address to be examined.*

CAUTION: Do not apply or remove power from the drive or perform a "HARDWARE RESET" with the E^2 enable device in the "ENABLE" position.

3. Change parameters as shown in steps of the following Example.

NOTE: *In order to change parameter values, two conditions must be present:*

1. *The EEROM (E^2) enable device must be in the "ENABLE" position.*
   JP13: 1-2; position; E^2 write "PROTECT"
   2-3; E^2 write "ENABLE"
2. *Parameter address #3 (designated as* **ADR003-DGNJP***) must contain an* ***odd*** *number (1, 3, 5, etc.). If an even number exists, add "1" to the present value and enter the new number.*

Example:

Examine and change the drive acceleration and deceleration rates. Assume the drive acceleration rate is set at 2 seconds and the deceleration rate is set at 10 seconds. Change the acceleration rate (**ADR057-ACELT**) from 2 seconds to 7.5 seconds. Reduce the drive deceleration rate (**ADR058-DCELT**) from 10 to 5.2 seconds.

1. Confirm that the E^2 enable device is in the "ENABLE" position and that **ADR003-DGNJP** contains an odd number.
2. Press **[CLEAR/MODE]** and **[0], [5], [7]**.
   - -- The display will show **ACELT.20**, corresponding to a 2-second acceleration rate.
3. Press **[INC]** until the display value increases to **75** or press **[7], [5]**.
   - -- In either case the display will now show **ACELT. 75.**
4. Press **[ENTER]** to place the new value into EEROM memory.
5. Press **[NEXT]** to command the drive to go to the next parameter, **ADR058-DCELT.**
   - -- The display will show **DCELT. 100**, corresponding to a 10-second deceleration rate.
6. Press **[DEC]** until the display shows **DCELT. 52**, or press **[5], [2]**.
7. Press **[ENTER]** to place the new value into memory.
8. Press **[CLEAR/MODE]** to return the display to **PARAMETER.**
9. Key in the appropriate three-digit parameter address number to examine other addresses.

NOTE: *This next step is* ***extremely important*** *to prevent accidental reprogramming of the drive.*

10. **Seal** the new parameter values as follows:
    a. Remove the "1" from **ADR003-DGNJP** (returning it to the "as shipped" even number).
    b. Return E^2 enable device to the "PROTECT" position (**JP13** in the 1-2 position).

4. Exit the **PARAMETER** Mode as follows:
   - -- Press **[CLEAR/MODE]** either once or twice to return to the **DIAGNOSTIC** Mode.
   - -- Press **[CLEAR/MODE]** again to return to the **OPERATE** Mode.

DIAGNOSTIC MODE

The Diagnostic Tests listed below will aid in the start-up, tune-up, and troubleshooting of the drive. To use these diagnostic features either a Local or Hand-held Programmer is required. Availability of certain tests may be dependent on hardware or software configurations provided.

WARNING - POTENTIAL DRIVE ACTIVATION AND INADVERTENT EQUIPMENT MOVEMENT WHEN TESTING: Throughout the remainder of this section, do not use DIAGNOSTIC RUN, SQUARE-WAVE REFERENCE TEST, or any programmer RUN functions on any crane or elevator drive or other applications where mechanical movements cannot be adequately monitored.

STANDARD DIAGNOSTICS TESTS:

Test 1	**[DGS]** - Diagnostic Static.
Test 2	**[DGR]** - Diagnostic Run.
Test 3	**[DVM]** - Displays Digital Voltmeter.
Test 4	**[ARM V]** - Displays armature voltage.
Test 5	**[ARM I]** - Displays armature current in amps.
Test 6	**[FIELD I]** - Displays field current in amps.
Test 7	**[RPM]** - Displays motor speed.
Test 8	**[TORQ]** - Displays motor torque in 1 lb-ft.
Test 9	**[HP]** - Displays motor horsepower.
Test 10	Displays software revision of Control card microprocessors.
Test 11	Displays diagnostic variables (selectable).
Test 12	SCR test.
Test 51	Revision date of Programmer software.
Test 52	Revision level of Programmer software.
RAM TEST	Displays single diagnostic variable (selectable).
DAC 0, 8, & 12	Digital to Analog converter outputs.
BITS 1, 2, & 3	Status bit outputs.

ADVANCED DIAGNOSTICS TESTS:

Test 14	I-Tune (Optional Current Loop Self-Tune-Up).
Test 15	S-Tune (Optional Speed Loop Self-Tune-Up).
Test 20	Electronic Wire-Check.
Test 21	Diagnostic Variable Pairs.
Test 22	Diagnostic Variables through DAC Outputs.
Test 25	Square-Wave Reference.

STANDARD DIAGNOSTIC TESTS DESCRIPTIONS:

NOTE: *For Tests 2 through 9, simply enter the* **DIAGNOSTIC** *mode and press the appropriate key. To enter all other diagnostic tests, press* ***[TEST], [X],*** *where "X" is the number of the particular test to be performed, and then press* ***[ENTER].***

WARNING - POTENTIAL DRIVE ACTIVATION AND INADVERTENT EQUIPMENT MOVEMENT WHEN TESTING: Throughout the remainder of this section, do not use DIAGNOSTIC RUN, SQUARE-WAVE REFERENCE TEST, or any programmer RUN functions on any crane or elevator drive or other applications where mechanical movements cannot be adequately monitored.

TEST 1 - **Diagnostic Static (DGS)**

Diagnostic Static (**DGS**) allows observation and adjustment of the Diagnostic Run reference (**Test 2**) without actually running the drive. The Diagnostic Run reference is determined by parameter **ADR003-DGNJP** and may be set by either **MSR**, the Programmer **INCREASE/DECREASE** keys, or the internal programmed references, **ADR092** and **ADR093 (TSTR1** and **TSTR2)**. Refer to the Custom Software Description to determine which reference is used.

-- Diagnostic Static is initiated from a stopped condition.

-- When initiated, the drive goes from field economy set by **ADR050-FLDEC** to full field current set by **ADR076-FLDMX.**

-- Diagnostic Static also fires the gate pulse transformers to the power conversion bridge without energizing the MA contactor and running the motor. This allows adjustment of the field settings and field regulator gain as well as checking the pulse transformer outputs on the Power Connection Card.

Pressing **[STOP]** discontinues Diagnostic Static.

TEST 2 - Diagnostic Run (DGR)

Diagnostic Run (**DGR**) causes the drive to run as a CEMF regulator. The Diagnostic Run reference is determined by parameter **ADR003-DGNJP** and may set by either **MSR**, the Programmer **INCREASE/DECREASE** keys, or the internal programmed references, **ADR092** and **ADR093** (**TSTR1** and **TSTR2**).

-- Refer to the Custom Software Description to determine which reference is used.

-- Pressing **[STOP]** will initiate a coast stop.

-- These features, CEMF regulation and coast stopping, will aid significantly when trying to separate mechanical problems from drive stability problems.

TEST 3 - DVM Mode (Requires Optional Process Interface or Encoder Process Interface Card; Also See Diagnostic Mode for Drive Setup)

Test 3 provides the capability of utilizing the Local or Hand-held Programmer and Process Interface Card (or Encoder Process Interface Card) as a digital voltmeter (**DVM**) for checking various analog test points (**TP**) on the drive cards. DC voltage in the +/-40 volt range can be measured.

-- Usage of **DVM Mode** requires a test probe wire (supplied in the miscellaneous parts packet shipped with the drive) routed from test point **DVM** on the Process Interface Card (or Encoder Process Interface Card) to the particular test point to be monitored.

-- The **DVM** may be calibrated by monitoring the +5 volt test point and adjusting **P3-DVM TRIM** on the Process Interface Card.

-- Pressing **[TEST]** discontinues **DVM Mode** and displays **CLR TEST**.

TESTS 4 thru 9 - Various Displays (ARM V, ARM I, FIELD I, SPEED, TORQ, HP)

Tests 4 thru 9 are initiated by setting the drive in **DIAGNOSTIC** Mode and then entering the desired test number.

-- To change to a different test simply press another number from 4 thru 9.

-- To cancel the last test press **[TEST]** once; the Programmer will display **CLR TEST**.

TEST 10 - Software Revision

Test 10 will display the revision level of the microprocessors on the Control Card. For example, **R20F20T20** says the regulator microprocessor (**RUP**) is version 2.0 software, the firing microprocessor (**FUP**) is version 2.0, and the tachometer microprocessor (**TUP**) is version 2.0. This can be verified by comparing the information with the white labels on the individual microprocessor chips located on the Control Card.

TEST 11 - Diagnostic Variable Display (also see TEST 21 and TEST 22)

Test 11 can be used to display (in hexadecimal numbers) certain drive variables for the purpose of comparison. Perform steps of the following example to display/compare variables.

Example:

To compare speed reference and speed feedback, perform the following:

1. Place the drive in **PARAMETER** Mode.
2. Determine (from the list that follows) the RAM address of the variables to be monitored.
3. Enter the first variable RAM address number into parameter **ADR122-DIAG1.**
4. Enter the second variable RAM address into parameter **ADR123-DIAG2.**
5. Place the drive in **DIAGNOSTIC** Mode and select **Test 11.**
   -- The programmer will display **DGXXXX. YYYY**, where **XXXX** is the hexadecimal value of the RAM address entered in **ADR122**, and **YYYY** is the hexadecimal value of the RAM address entered in **ADR123.**

NOTE: *The value displayed is in 2's complement. Values from 0000 to 7FFF are positive magnitude (0 to +32767). Values from FFFF to 8000 are negative magnitude (FFFF = -1 and 8000 = -32768).*

RAM ADDRESSES FOR "ANALOG" SIGNALS

RAM ADR	DESCRIPTION	RAM ADR	DESCRIPTION
053	Speed Loop Output	103	Voltage Feedback
055	Current Feedback	105	CEMF
061	Current Loop Output	463	Field Current
065	Timed Speed Reference	494	ACOM
069	Jog Reference	501	TOC Timer
071	Speed Feedback		

NOTE: *These addresses will contain a variable number, representing an "analog" signal. Refer to "RAM Variable Scaling", TABLE 4 in the Adjustment Section, for the relative hexadecimal values of the displayed variables.*

RAM ADDRESSES FOR "LOGIC" SIGNALS

RAM ADR	DESCRIPTION	RAM ADR	DESCRIPTION
137	Fault	148	Reference Enable
140	Current Limit Flag	152	Precondition
141	CEMF Limit Flag	175	Start/Stop Control
147	Ready To Start Flag		

NOTE: *This list contains addresses of signals which will be either "high" (display* **"8080"***) or "low" (display* **"0000"***). If the signal is high, then the statement is true. For example, if "137" is in* **ADR122***, Test 11 is initiated, and a fault condition exists, then the display will show* **DG8080.YYYY***, (YYYY value will depend on the number in parameter* **ADR123***). If the drive is not faulted then the display will be* **DG0000.YYYY***.*

WARNING - ELECTRICAL SHOCK HAZARD: The following test will cause armature current to circulate in the motor armature loop.

TEST 12 - SCR Test

Test 12 will cause the MA contactor to pick-up and fire each SCR once. During this test, current feedback is monitored to determine if any SCRs are open or shorted.

-- If problems exist, a fault code will be displayed to help isolate the nature and area of the problem. For example, "**FL73 A1ROP**" means that SCR **A1R** did not conduct current. SCR **A1R** can be identified from the Information Sheets on the inside of the drive door.

-- Refer to the Troubleshooting Section for a complete list of fault codes.

TEST 51 - Programmer Card Software Revision Date

Test 51 provides a check of the software date of the Programmer microprocessor (**PUP**).

TEST 52 - Programmer Card Software Revision Level

Test 52 provides a check of the software level of the Programmer microprocessor (**PUP**).

RAM TEST - Displays RAM Location Value

RAM test allows a "one-shot" look at a RAM address (refer to **TEST 11** list of for variables). The value displayed will be in decimal numbers. To display a RAM address, perform the following:

1. Press **[RAM]** and the Programmer will display *****RAM*****.
2. Press the keys corresponding to the desired RAM variable and the drive will display **XXX YYYY**, where **XXX** is the RAM address and **YYYY** is the RAM value.
3. Press **[CLEAR/MODE]** to clear the display.

WARNING - DRIVE MISOPERATION HAZARD: These outputs may already be utilized by the drive. Refer to the Drive Elementaries and Custom Software Description.

DAC0, DAC8, and DAC12 (DAC8 and DAC12 Require an Optional Application or Microapplication Card)

Three selectable analog outputs, **DAC0**, **DAC8**, and **DAC12**, are available for test purposes and are programmed similar to **Test 11**. To program **DAC0**, perform the following:

1. Enter the PARAMETER Mode and call up **ADR090-DAC0**.
2. Enter the signal to be monitored from TABLE 4, "RAM Variable Scaling". (Putting that RAM Address in **ADR090** will cause that signal to appear at the **DAC0** test point.)

   NOTE: ***DAC8*** *and* ***DAC12*** *are monitored in a similar manner, except the signal RAM address is loaded into* **ADR096-DAC8** *or* **ADR097-DAC12** *and the output appears at the* ***DAC8*** *or* ***DAC12*** *terminal on* ***4TB.***

WARNING - DRIVE MISOPERATION HAZARD: These outputs may already be utilized by the drive. Refer to the Drive Elementaries and Custom Software Description before using this feature.

STATUS BITS 1, 2, and 3 (Requires Optional Application or Microapplication Card)

ADR098-BIT#1, **ADR099-BIT#2**, and **ADR100-BIT#3** are programmed in the same manner as **DAC8** and **DAC12**. They are different in that they are logic outputs rather than analog outputs.

-- Signals listed in "RAM Addresses for Logic Signals" chart can be used to select addresses to enter in **ADR098**, **ADR099**, and **ADR100**.

-- Variables listed in "RAM Addresses for Analog Signals" chart may also be used in the bit outputs to indicate the polarity of the signal (positive or negative).

-- Outputs will appear on **4TB** as "**B01**", "**B02**", and "**B03**" respectively.

ADVANCED DIAGNOSTIC TEST DESCRIPTIONS:

WARNING - POTENTIAL DRIVE ACTIVATION AND INADVERTENT EQUIPMENT MOVEMENT WHEN TESTING: Throughout the remainder of this section, do not use DIAGNOSTIC RUN, SQUARE-WAVE REFERENCE TEST, or any programmer RUN functions on any crane or elevator drive or other applications where mechanical movements cannot be adequately monitored.

WARNING - DRIVE MISOPERATION HAZARD: Signal level detectors and bit outputs may control external devices. Refer to the Drive Elementaries and Custom Software Description before using this feature.

TEST 14 - I-Tune

This test performs a self-tune-up of the DC-300 current regulator. Before starting the test, EEROM programming must be enabled and the external control logic must be set forward if **POL** input on **3TB** is used. Also, the drive current scaling, **ADR129-IRATD**, and the DC output voltage scaling, **ADR130-VRATD**, must be set properly. If not, this test will not function properly.

I-Tune is initiated from the **DIAGNOSTIC** Mode as follows:

1. Key in **[TEST]**, **[1]**, **[4]**, and **[ENTER]**. (The test can be aborted at any time by pressing **[CLEAR/MODE]**.)
   -- The programmer will display **"I TUNE-UP"** momentarily, then **"MTR/GX"**, where **X** is the number of the motor group (0 thru 3) selected by the external control wiring on multimotor applications.
2. If the motor group desired for adjustment is displayed, press **[ENTER]**.
   or
   If the incorrect motor group is displayed, abort the test and activate the proper control logic to select the desired motor group (refer to the drive elementaries and Custom Software Description for an explanation of how to activate multimotor parameters, if used).
   -- Do not enter a number or the test will abort.
3. Press **[ENTER]** after the proper motor group has been displayed.
   -- The programmer will then display **"RESP.300"**, which is the desired tune-up response in radians/second.
   -- The current loop response can be set from 0 to 300, 300 being the fastest or most responsive setting for the current loop.

WARNING - ELECTRICAL SHOCK HAZARD: The following step will cause armature current to circulate through the motor.

4. Key in the desired response and press **[ENTER]** to begin the auto tune-up test.
   -- The drive will step the current regulator a number of times and then display **"ILPIG.00XXX"**, the current loop integral gain parameter for the motor group being tuned (**XXX** is the value that the test adjusted the integral gain to).
   -- Pressing **[ENTER]** will program the new current loop values into the appropriate parameters.
   -- Pressing **[CLEAR/MODE]** will abort the test.

Performing I-Tune can change the following parameters:

Group	ADR	Mnemonic	Description
MTR/G#0	019	ILPPG	Current Loop Proportional Gain
	020	ILPIG	Current Loop Integral Gain
	021	ILPDG	Current Loop Integral Discontinuous Gain
MTR/G#1	147	ILPP1	Motor #1 Current Loop Proportional Gain
	148	ILPI1	Motor #1 Current Loop Integral Gain
	149	ILPD1	Motor #1 Current Loop Integral Discontinuous Gain
MTR/G#2	159	ILPP2	Motor #2 Current Loop Proportional Gain
	160	ILPI2	Motor #2 Current Loop Integral Gain
	161	ILPD2	Motor #2 Current Loop Integral Discontinuous Gain
MTR/G#3	171	ILPP3	Motor #3 Current Loop Proportional Gain
	172	ILPI3	Motor #3 Current Loop Integral Gain
	173	ILPD3	Motor #3 Current Loop Integral Discontinuous Gain

WARNING - POTENTIAL DRIVE ACTIVATION AND INADVERTENT EQUIPMENT MOVEMENT WHEN TESTING: Throughout the remainder of this section, do not use DIAGNOSTIC RUN, SELF-TUNE (I-TUNE or S-TUNE), SQUARE-WAVE REFERENCE TEST or any programmer RUN functions on any crane or elevator drive.

TEST 15 - S-TUNE

This test performs a self tune-up of the DC-300 drive speed regulator. Before starting the test, EEROM programming must be enabled and the external control logic must be set forward if the **POL** input on **3TB** is used.

S-Tune is initiated from the **DIAGNOSTIC** Mode as follows:

1. Key in **[TEST]**, **[1]**, **[5]**, and **[ENTER]**. (The test can be aborted at any time by pressing
   -- The programmer will display **"DR TUNE-UP"** momentarily, then **"B.SPD*****"**.
2. Key in the base speed of the motor and press **[ENTER]**.
   -- The programmer will then display **"T.SPD*****"**.
3. Key in the top speed of the motor and press **[ENTER]**.
   -- If the base and top speeds are the same, the two entries will be equivalent. The programmer will then display **"MTR/GX"**, where **X** is the number of the motor group (0 thru 3) selected by the external control wiring on multimotor appplications.
4. If the motor group desired for adjustment is displayed, press **[ENTER]**.
   or
   If the incorrect motor group is displayed, abort the test and activate the proper control logic to select the desired motor group (refer to the drive elementaries and Custom Software Description for an explanation of how to activate multimotor parameters, if used).
   -- Do not enter a number or the test will abort.
5. Press **[ENTER]** after the proper motor group has been displayed.
   -- The programmer will then display **"RESP.XX"**. This is the speed regulator response in radians/second and may be adjusted depending on desired response. Larger numbers will make the drive more responsive and smaller numbers will slow the drive response.
   -- The drive will attempt to tune as close as possible to the requested response but may be limited by size and type of motor being used.

WARNING - MOTOR ROTATION HAZARD: The next step will cause the motor shaft to rotate.

6. Press **[ENTER]** to begin self-tuning for the response requested.
   -- The drive will step the motor a number of times.
   -- Upon completion of the tune-up, the Programmer will indicate the response attained during the tune-up procedure.
   -- There are two possibilities available at this point:
      (1) The EEROM enable device may be set to "PROTECT" and **[ENTER]** pressed. This will load the tuned parameters into RAM and the drive can be run to check performance.
      (2) The EEROM enable device may be set to "ENABLE" and **[ENTER]** pressed. This will load the tuned parameters into the EEROM memory, retaining the settings for all future operation.
   -- In either case, the drive will display **DIAGNOSTIC** when programming is completed.

NOTE: *If the parameters have been changed in RAM only (EEROM write protected) and drive performance is satisfactory, perform S-Tune again and program the parameters into the EEROM (EEROM Write Enabled) for permanent storage.*

Performing S-Tune can change the following parameters:

Group	ADR	Mnemonic	Description
MTR/G#0	015	BSLPG	Base Speed Loop Proportional Gain
	016	BSLIG	Base Speed Loop Integral Gain
	017	WSLPG	Weak Field Speed Loop Proportional Gain
	018	WSLIG	Weak Field Speed Loop Integral Gain
	025	CLPPG	CEMF Loop Proportional Gain
	026	CLPIG	CEMF Loop Integral Gain
	027	CLMPG	CEMF Limit Proportional Gain
	028	CLMIG	CEMF Limit Integral Gain
	037	SLEAD	Speed Feedback Lead
MTR/G#1	143	BSLP1	Motor #1 Base Speed Loop Proportional Gain
	144	BSLI1	Motor #1 Base Speed Loop Integral Gain
	145	WSLP1	Motor #1 Weak Field Speed Loop Proportional Gain
	146	WSLI1	Motor #1 Weak Field Speed Loop Integral Gain
	150	SLEA1	Motor #1 Speed Feedback Lead
MTR/G#2	155	BSLP2	Motor #2 Base Speed Loop Proportional Gain
	156	BSLI2	Motor #2 Base Speed Loop Integral Gain
	157	WSLP2	Motor #2 Weak Field Speed Loop Proportional Gain
	158	WSLI2	Motor #2 Weak Field Speed Loop Integral Gain
	162	SLEA2	Motor #2 Speed Feedback Lead
MTR/G#3	167	BSLP3	Motor #3 Base Speed Loop Proportional Gain
	168	BSLI3	Motor #3 Base Speed Loop Integral Gain
	169	WSLP3	Motor #3 Weak Field Speed Loop Proportional Gain
	170	WSLI3	Motor #3 Weak Field Speed Loop Integral Gain
	174	SLEA3	Motor #3 Speed Feedback Lead

TEST 20 - Electronic Wire Check

Test 20 allows for six different levels of wire checking and output forcing tests as follows:

- **20-1:** Signal Level Detector (**SLD**) forcing.
- **20-2:** Status Bit monitoring.
- **20-3:** Fault Code testing.
- **20-4:** Logic input monitoring (Application Card).
- **20-5:** Logic input monitoring (Control Card).
- **20-6:** Microapplication Card input monitor ing (future).

Proceed as follows to perform all Electronic Wire Check tests:

1. Confirm that programmer is in **DIAGNOSTIC** Mode and that the EEROM is write "ENABLED" (see TABLE 1, Control card, JP13). (Change EEROM to write "PROTECT" after test is completed).
2. Key in **[TEST]**, **[2]**, **[0]**, and **[ENTER]**.
   - This causes **Test 20** to be active and the display to show **1.'.00111111**. The left-most digit indicates the test level, in this case, the **1.'.** indicates level 1 or Signal Level Detector (**SLD**) forcing.
3. Press the **[INC]** or **[DEC]** key to change the type of test until the desired test number is displayed in the left-most digit.
   - For example, if **Test 20** is called for and the initial display is shown above, press **[INC]** once to run the next level (20-2), Status Bit Monitoring. The programmer display will be similar to **2.'.11111100**.
   - Each test is described in the following paragraphs.

NOTE: ***Test 20-1*** *temporarily changes the value of Parameter* **ADR009-SLDJP**. *If the AC control power is removed while keys 0 thru 5 are depressed,* **ADR009-SLDJP** *may be changed and will not contain the correct value when AC power is reapplied. The value of* ***SLDJP*** *should be recorded prior to initiating this test.*

Test 20-1 - Signal Level Detector (SLD) Forcing (Requires Application or Microapplication Card)

The display will show **'1.'.00111111** when **Test 20-1** is initiated.

- -- **SLD0** is energized if the right most digit is "1." and dropped out if it is a "0".
- -- The status of all six SLDs (**SLD0** through **SLD5**) can be seen by scanning from right to left. For example, the second digit from the right is **SLD1**, the third digit from the right is **SLD2**, etc.
- -- To change the state of **SLD0**, depress **[0]** key.

NOTE: *SLDs cannot be toggled while the drive is running or the drive will fault. While the* ***[0]*** *key is depressed,* ***SLD0*** *will be maintained in the opposite state it was originally. When the* ***[0]*** *key is released,* ***SLD0*** *returns to its original state. Keys 1-5 operate in the same manner for* ***SLD1*** *thru* ***SLD5.***

Test 20-2 - Status Bit Outputs (Requires Application or Microapplication Card)

Status bits **B01, B02,** and **B03 (ADR098, ADR099,** and **ADR100** respectively) are accessed as described in **Test 20.**

- -- When **Test 20-2** is initiated, **B01** will appear as the right most character in the display, **B02** will be shown on the left of **B01**, and **B03** is shown on the left of **B02.**
- -- If **Bit#1** is "high" the right most character will be "1"; if it is "low", this character will be "0".

Test 20-3 - Fault Code Outputs (Requires Application or Microapplication Card)

Test 20-3 is used to test remote fault code lines and their effects on external equipment.

1. Access **Test 20-3** as described in **Test 20.**
2. Enter the desired fault code by keying in **[0].**
   - -- The programmer will display **FAULT.**
3. Key in the number of the Fault Code desired and press **[ENTER].**
   - -- Note that Fault 63 will cause **FC5 (4TB-45)** to be high.

NOTE: ***Test 20-3*** *does not cause the drive to fault unless a number greater than 127 is entered. This test is a way of forcing the fault code output lines on the Application Card (or Microapplication Card) to check interfaces with external equipment. The Diagnostic LEDs will also display the appropriate Fault Code.*

Test 20-4 - Logic Input Monitoring (Requires Application or Microapplication Card)

The display will show **'4.'.XXXXXX** when **Test 20-4** is initiated. The Xs will be either **1** or **0.**

- -- A **1** will be displayed if a particular input is tied to **REF24** and the drive is programmed for negative logic (**JP19** and **JP20** on the Control Card are in 1-2 and 3-4 position respectively, and **JP5** and **JP6** on the Application Card [or Microapplication Card] are in the 1-2 and 3-4 position respectively).
- -- From right to left the digits are **MODE 0** thru **MODE 7.**

Test 20-5 - Logic Input Monitoring (Control Card)

The **Test 20-5** display is similar to the **Test 20-4** display except that the left four digits do not apply. Starting at the first digit from the right, the status of four logic inputs will be displayed as **"5.XXXXDCBA"** where:

- **A =** Status of **RUN** command inverted ("0" indicates **RUN** is initiated).
- **B =** Status of **JOG** command inverted ("0" indicates **JOG** is initiated).
- **C =** Status of **POL** command inverted ("0" indicates **POL** is initiated).
- **D =** Status of **XSTOP** command ("1" indicates **XSTOP** initiated).

Test 20-6 Microapplication Card Input Monitoring (Microapplication Card required)

Test 20-6 is reserved for future use.

TEST 21 - Diagnostic Variable Pairs

NOTE: ***Test 21*** *will cause* ***ADR122-DIAG1*** *and* ***ADR123-DIAG2*** *to change.*

Test 21 performs in a similar manner to **Test 11** except that several pairs of RAM addresses are already pre-programmed in memory. **Test 21** is entered from the **DIAGNOSTIC** Mode as follows:

1. Key in **[TEST], [2], [1],** and **[ENTER].**
   - -- The programmer will display **TST21.**

2. Key in **[0], [1], [2], [3], or [1], [0]** (corresponding to **Tests 21-0, 21-1, 21-2, 21-3, or 21-10** respectively), and **[ENTER]**, to display the pairs of RAM addresses shown in the chart below.
   - -- As in **Test 11**, the display will show **DGXXXX.XXXX**. The left and right four characters are as selected from the listing.
   - -- **Test 21-3** will display the variables programmed into **ADR096-DAC8** and **ADR097-DAC12**.
   - -- When **[1], [0]** is entered (calling for **Test 21-10**), the display will prompt with **ADDR1**. Proceed as follows for this test:
     a. Key in the first RAM address you wish to monitor and press **[ENTER]** (typical RAM addresses are listed under **Test 11**). The display will then prompt with **ADDR2**.
     b. Key in the second RAM address and press **[ENTER]**. The display will now show **DGXXXX.XXXX**, where the left four digits are the value of the signal address stored in **ADDR1**, and the right four digits are the value of the signal address stored in **ADDR2**.

TEST 22 - Diagnostic Variables Through DAC Outputs (Requires Application or Microapplication Card)

Test 22 performs in a similar manner to **Test 21** with the addition that the various signal pairs are automatically sent to the **DAC8** and **DAC12** terminals on **4TB**. **Test 22** is entered from the **DIAGNOSTIC** Mode as follows:

1. Key in **[TEST], [2], [2]**, and **[ENTER]**.
   - -- The programmer will display **TST22**.
2. Key in **[0], [1], [2], [3], or [1], [0]** (corresponding to **Tests 22-0, 22-1, 22-2, 22-3, or 22-10** respectively), and **[ENTER]**, to display the pairs of RAM addresses shown in the chart below.
   - -- As in **Test 21**, the display will show **DGXXXX.XXXX**. The left and right four characters are as selected from the listing.
   - -- Calling for **Test 22-3** causes +5V to appear at the **DAC8** and **DAC12** terminals in order to quickly check the integrity of **DAC8** and **DAC12**.
   - -- When **[1], [0]** is entered (calling for **Test 22-10**), the display will prompt with **ADDR1**. Proceed as follows for this test:
     a. Key in the first RAM address you wish to monitor and press **[ENTER]** (typical RAM addresses are listed under **Test 11**). The display will then prompt with **ADDR2**.
     b. Key in the second RAM address and press **[ENTER]**. The display will now show **DGXXXX.XXXX**, where the left four digits are the value of the signal address stored in **ADDR1**, and the right four digits are the value of the signal address stored in **ADDR2**.

NOTE: ***Test 22*** *temporarily changes the value of* **ADR096-DAC8, ADR097-DAC12, ADR122-DIAG1,** *and* **ADR123-DIAG2.** *If AC control power is removed while conducting* ***Test 22,*** *then* ***DAC8, DAC12,*** *and* ***DIAG2*** *will not contain the original value when power is reapplied. The value of* ***DAC8, DAC12, DIAG1,*** *and* ***DIAG2*** *should be recorded prior to conducting this test.*

WARNING - POTENTIAL DRIVE ACTIVATION AND INADVERTENT EQUIPMENT MOVEMENT WHEN TESTING: Throughout the remainder of this section, do not use DIAGNOSTIC RUN, SQUARE-WAVE REFERENCE TEST, or any programmer RUN functions on any crane or elevator drive or other applications where mechanical movements cannot be adequately monitored.

RAM VARIABLE PAIRS FOR TEST 21 AND TEST 22

Test Number	Left 4 Digits	(RAM Adr)	Right 4 Digits	(RAM Adr)
0	Speed Loop Output	(053)	Current Loop Output	(061)
1	Speed Reference	(065)	Speed Feedback	(071)
2	Position Error	(471)	In Position Flag	(154)
3	DAC8	(ADR096)	DAC12	(ADR097)
10	Diagnostic Var. #1	(ADR122)	Diagnostic Var. #2	(ADR123)

TEST 25 - Square Wave Test Reference

Test 25 causes the drive to run and follow an internal drive square wave reference set up by the values in **ADR092-TSTR1**, **ADR093-TSTR2**, and **ADR094-TSTT1**. **TSTR1** and **TSTR2** are run reference levels and **TSTT1** is the time duration of each half cycle. (See Example)

- Run references are 0 to 127 with 100 equal to 100% speed with positive reference. Negative reference is initiated with values from 128 to 255 with the magnitude equal to 256 minus the setting (for example, 156 corresponds to -100%).
- The time period is set with **TSTT1** and will have a 50% duty cycle for powers of 2 (1, 2, 4, 8, 16, etc.). The cycle time will be 262* **TSTT1** milliseconds. That is, with **TSTT1** set to 8, the test will cycle between **TSTR1** and **TSTR2** with approximately 1 second for each reference.
- Pressing **[STOP]** will end the test.

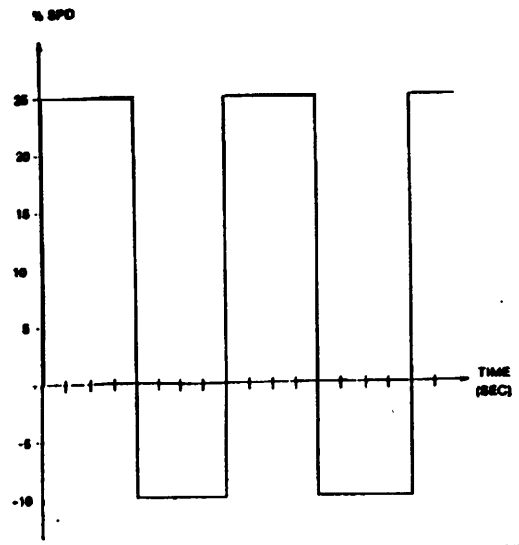
NOTE: ***Test 25*** *temporarily changes the value of* **ADR003-DGNJP**. *If AC control power is removed while conducting this test, then* ***DGNJP*** *will not contain the correct value when AC power is reapplied. The value of* ***DGNJP*** *should be recorded prior to conducting this test.*

Example:

Program the square wave reference to run 25% of top speed forward for approximately 4 seconds and 10% reverse for approximately 4 seconds.

ADR092-TSTR1= 25 (25% speed)
ADR093-TSTR2= 246 (-10% speed)
ADR094-TSTT1= 4000 ms/262 = 16

(The reference will look like the sketch below.)



DIAGNOSTIC MODE FOR DRIVE SETUP

Step 1 of the following procedure utilizes Diagnostic **Test 21-10**. All other steps utilize the **DVM** Mode. These procedures allow accurate tuning of all analog pots as listed below.

Control Card	OFFSET	P4
Control Card	REF SCALE	P1
Process Interface Card*	SFB ZERO	P7
Process Interface Card*	PF OFF	P5
Process Interface Card*	PF GAIN	P4
Application Card**	ASR SCALE	P1
Application Card**	FRA SCALE	P2

**Or Encoder Process Interface Card*

***Or Microapplication Card*

1. Calibrate **P4-OFFSET** on the Control Card and **P7-SFB ZERO** on the Process Interface Card (or Encoder Process Interface Card) as follows:
   a. Place the drive in **DIAGNOSTIC** Mode.
   b. Enable EEROM and key in **[TEST]**, **[2]**, **[1]**, **[ENTER]**, then **[1]**, **[0]** and **[ENTER]**.
      -- The drive will display **ADDR1**.
   c. Key in **[4]**, **[9]**, **[4]** and **[ENTER]**.
      -- The drive will display **ADDR2**.
   d. Key in **[0]**, **[7]**, **[1]** and **[ENTER]**.
      -- The drive will display **DG.XXXX.XXXX**.
   e. Use the left four characters and **P4-OFFSET** on the Control Card to adjust the drive offset to zero.
   f. Use the right four characters and **P7-SFB ZERO** on the Process Interface Card (or Encoder Process Interface Card) to adjust speed feedback for zero.

2. Place the drive in the **DVM** Mode by pressing **[DVM]**.

3. Calibrate **P1-REF SCALE** on the Control Card as follows:
   a. Connect the **DVM** test probe from **TP DVM** on Process Interface Card (or Encoder Process Interface Card) to **TP SRS**, located on the Control Card.
   b. Set the input at **MSR** on **3TB65** to its maximum value and adjust **P1-REF SCALE** on the Control Card for 4.0 VDC at **TP SRS** (also located on the Control Card).

NOTE: *If 4.0 VDC cannot be obtained at* ***SRS****, then check* ***JP1*** *on the Control Card for proper position. If MSRs maximum voltage is 5-8 VDC, then position 2-3 is correct.*

4. Calibrate **P1-TACH SCALE** on the Process Interface Card (or Encoder Process Interface Card) as follows:
   a. Calculate the tachometer voltage at top speed from the tachometer nameplate and motor top speed.
   b. Refer to TABLE 1 for the Process Interface Card (or Encoder Process Interface Card) **SW1** setting and determine which switch positions should be closed for desired drive top speed

      NOTE: ***SW1****, position 7 should be open for DC tachs and closed for AC tachs.*
   c. Start the drive and run at half speed (2.0 VDC at **TP SRS** on Control Card if running in **MANUAL** Mode, or 2.0 volts at **TP ASR** on Application Card [or Microapplication Card] if running in **AUTO** Mode).
      -- Monitor motor speed with either a direct reading mechanical tach or with a voltmeter between **1TB-1** and **1TB-3** on the Process Interface Card (or Encoder Process Interface Card).
   d. Adjust **P1-TACH SCALE** on the Process Interface Card (or Encoder Process Interface Card) for 50% of motor speed.
   e. Run the drive at top speed and recheck the motor speed.
   f. Readjust **P1-TACH SCALE** as necessary for the correct top speed.
      -- **TP SFB** on the Process Interface Card (or Encoder Process Interface Card) should now read 4.0 VDC and be equal to **TP SRS** voltage (**TP ASR** if running in **AUTO** Mode).

This completes the calibration of the speed feedback circuit and setup of the drive's analog pots.

5. Check the setup of the Process Follower option (if used) as follows:
   a. Verify that **JP2** and **JP3** on the Process Interface Card (or Encoder Process Interface Card) are set as required according to the following chart:

JP2	JP3	Follower Mode
1-2	1-2	5-65 MA
1-2	2-3	5-50 MA
2-3	1-2	4-20 MA
2-3	2-3	1-5 MA or 0-10V

   b. Set the programmer in **DIAGNOSTIC** Mode and turn on the **DVM** (press **[DVM]**).
   c. Connect the test probe from **TP DVM** on the Process Interface Card (or Encoder Process Interface Card) to **TP PFRF** (also on the card).
   d. Adjust **P5-PF OFF** on the Process Interface Card (or Encoder Process Interface Card) for zero volts on the **DVM**.
   e. Raise the process follower input to its maximum value and adjust **P4-PFGAIN** on the Process Interface Card (or Encoder Process Interface Card) for 7.5 VDC on the **DVM**.

NOTE: *It is not necessary to run the drive to calibrate the process follower, only to vary its input as described above.*

6. Adjust the **FRA** input to the Application Card (**If used**) for the proper amount of field forcing as follows (refer to the elementary drawings for your specific drive):
   a. Verify that the proper voltage is applied to the **FRA** terminal on **4TB**.
   b. Adjust **P2-FRA SCALE** on the Application Card for the correct field current with field forcing applied.
      -- Use **Test 6** in **DIAGNOSTIC** mode to monitor field current while adjusting **P2-FRA SCALE**.

12. PARTS REPLACEMENT

REPLACEMENT OF SCR MODULES

WARNING - SHOCK AND BURN HAZARD:

Always disconnect power to the controller before performing any work on controller components. Failure to do so may cause serious injury to personnel and damage to the controller or the controlled machinery.

1. Lower the card carriers to expose the rear of the unit.
2. Disconnect the wires from the damaged module and remove the bus bar and mounting screws.
3. Clean the mounting surface of old lubricant and dirt.
4. Apply a thin film of General Electric GL322 Versilube® Plus lubricant (Part No. 104X214AB059) to both contacting surfaces and install the module. Tighten screws to the torque specifications below.

CAUTION: Treat all cards with static sensitive handling techniques. Use "grounding" strap when changing cards and store cards in the anti-static bags they are shipped in.

CARD REPLACEMENT PROCEDURE

1. Carefully disconnect all cables.
2. Release all hold-down tabs and remove card.
3. Set all jumpers on the replacement card exactly as the jumpers on the failed card.
4. Install the new card verifying that all tabs snap tightly into position.
5. Reconnect all cables verifying that they are properly seated at both ends.
6. Set all pots in the same physical position as the pots on the failed card and fine tune as required per instructions in TABLE 1 of the Adjustments section.
7. If the Control Card is changed, remove the EEROM from the failed card and install in the new card.
   - -- If the EEROM from the failed card is suspected to be damaged, it will be necessary to reprogram a new EEROM to the custom parameter list.
   - -- Use a proper EEROM removal tool and approved grounding techniques.

SCR MODULE PART NUMBER	HEATSINK MOUNTING SCREWS	*BUS BAR MOUNTING SCREWS
104X125DC020 104X125DC021	Max. 40 inch pounds Max. 53 inch pounds	Max. 106 inch pounds Max. 133 inch pounds
104X125DC022 104X125DC023	Min. 28 inch pounds Max. 30 inch pounds	Hand Tighten
104X125DC026 104X125DC027	Min. 40 inch pounds Max. 49 inch pounds	Min. 22 inch pounds Max. 31 inch pounds
** Support the bus bar while tightening screws to avoid excess torque on connected modules.*		

NOTES:

13. SPARE AND RENEWAL PARTS

Renewal parts information for this equipemnt is included in the **CUSTOM INSTRUCTION BOOK** provided for the drive. The renewal parts list is generated specifically for the equipment furnished on each order and provides information on parts relative to the particular rating and options incorporated into the drive at the time of shipment. If the renewal parts information is missing, a replacement list may be obtained by contacting your local **GE Sales or Service Office** or **GE DRIVE SYSTEMS Authorized Distributor** with the drive **model** and **requisition numbers**

It is recommended that the following parts lists be reviewed and a set of spare parts be maintained to minimize down time should repairs be required. Renewal parts may be ordered by contacting your local **GE Sales or Service Office** or an **Authorized GE Parts Distributor in your area.**

PART NUMBER IDENTIFICATION

The parts listed are divided into four categories, each with its own part number structure:

Order-Specific Assemblies
Common Assemblies
Component Parts
Circuit Cards

Order-Specific Assemblies are generally major assemblies consisting of a group of common assemblies that make up the specific controller provided. Other items which are obtained specifically for the order may also be provided with a similar part number structure.

Example:

7VXYZ999CD01

7V	--	assembly provided by GE DRIVE SYSTEMS
XYZ999	--	identification number (original order)
CD	--	assembly type (**CD** = Core Drive)
01	--	unit identification number

Common Assemblies are subassemblies, consisting of a group of component parts, which are in common use in **GE DRIVE SYSTEMS** products and are not specific to a particular order. The part number consists of the number "**36**" followed by an alpha character and may be up to 14 characters in total length.

Example:

36C77345AAG01 is the part number for an External Motor Field Control Module.

Component Parts are the basic components used in the assemblies and generally represent the lowest discrete level. Component part numbers consist of a combination of alphanumeric characters, defining the class of component and the specific item.

Example:

104X109AD038

104X	--	component part
109	--	component type (fuse)
AD038	--	specific item (500 V, 3.2A FNQ)

Printed Circuit Cards are identified by a catalog number giving the product class, card function and revision levels. It should be noted that some cards contain software which is designated by the suffix "**Mn**" where "**n**" is the software revision number. These cards are also available less software with suffix "**Gn**" designation. The factory may substitute later versions of cards based on availability and design enhancements.

Example:

531X300CCHABM5

531X	--	card for GE DRIVE SYSTEMS microprocessor based controller
300CCH	--	Control Card for DC-300 Controller
ABM5	--	revision level, including software

As an aid in expediting part identification, a partial listing of the parts, functions and associated root part numbers is shown in the following charts.

STANDARD PRINTED CIRCUIT CARDS*	CATALOG NUMBER (Root)
Control Card (with microprocessors)	531X300CCH_ _M_
Power Connection Card (4 Quadrant)	531X121PCR_ _G_
Power Connection Card (1 Quadrant)	531X122PCN_ _G_
Power Connection Card	531X123PCHA_G1/G2
Power Connection Card	531X308PCSA_G1/G2
Power Supply Card	531X111PSH_ _G2 (with external field control)

OPTIONAL PRINTED CIRCUIT CARDS*	CATALOG NUMBER (Root)
Application Card	531X139APMA_G2
Microapplication Card	531X139APMA_M_
Process Interface Card	531X133PRUA_G1
Encoder Process Interface Card	531X134EPRA_G1
3TB Terminal Board Card	531X170TBS_ _G_
4TB Terminal Board Card	531X171TMAA_G2
Programmer Card	531X135PRGA_M_
"Local" Programmer Assembly	36B605548A_G01
"Hand-Held" Programmer Assembly	6VHHP10A1
LTB	531X189LTB
RTB	531X191RTB

**Several versions of some cards are available. Check old part to verify catalog number before ordering renewal part.*

INTERCONNECTION CABLES

Connector	Catalog
1PL	36A358208AAG02
2PL	104X154CB111
3PL	36A358208AAG04
4PL-3	36A358208AAG07
5PL	36A358208AAG01
6PL	36A358208AAG08
7PL	36A358208AAG05
8PL-10	36A358208ABG01
9PL	36A358208CXG01
10PL-3	36A358208AAG09
11PL	36A358208AAG11

Connector	Catalog
12PL	36A358208AAG10
13PL	36A358208AAG12
14PL	36A358208EYG01
16PL (6 Foot)	36A358208AAG18
16PL (8 Foot)	36A358208AAG19
16PL (10 Foot)	36A358208ARG01
16PL (15 Foot)	36A358208ARG02
18PL	36A358208ADG01

NOTE: *Connectors used may vary with options provided.*

SUBASSEMBLIES AND COMPONENTS

Subassembly/Component	Catalog Number
External Motor Field Control Module (remote mounted)	36C774345AAG01
External Motor Field Control Module (mounted on "C" frame)	36C774345ABG01
Motor Field Power Module	104X125DC012*
Motor Field Power Module	104X125DC062*
K6, K7, K8, Relays**	104X166AA077
Thermoswitch (optional)	104X152AC067
Control Power Transformer (230 VAC, 1-30 HP)	36B605538AAG01
Control Power Transformer (230 VAC, 40-100 HP)	36B605538ACG01
Control Power Transformer (460 VAC, 3-75 HP)	36B605538AAG02
Control Power Transformer (460 VAC, 100-250 HP)	36B605538ACG02
Fan (small, 4-1/2")	104X215CA003
Fan (large, 7")	104X215CA004
Blower Motor (230 VAC)	36B605593AAG03
Blower Motor (460 VAC)	36B605593AAG04
Current Transformer Assembly (75 HP, 460 VAC)	36B05537AAG01
Current Transformer Assembly (150 HP, 460 VAC)	36B05537ABG01
Current Transformer Assembly (200/250 HP, 460 VAC)	36B05633AAG01
Fuses (Control Power)	
3 Amps (CFU6)	104X109AD004
1.5 Amps (CFU7)	104X109AD088
0.5 Amps (CFU8)	104X109AD009
3.2 Amps (CFU4 & 5)	104X109AD038
15 Amps (CFU1, 2 & 3)	104X109AD076

* ***104X125DC012*** *was supplied on equipment shipped* ***thru April, 1991.***
104X125DC062 *is supplied on equipment shipped* ***after April, 1991.***
The 104X125DC062 is a functional replacement only. It is larger than the 104X125DC012 and physically cannot be used as a direct substitute. Whichever unit was originally supplied with the equipment should be used as the replacement.

** *Once relay contacts are used on high current or voltage signals, they cannot be used for low level signals.*

DC-300 SPARE AND RENEWAL PARTS - "A" FRAMES* (HP Related, 230 VAC)

Part Description	Cat. # 104X-	Horsepower 2	5	7.5	10
AC Power Fuse	109AG-	003	003	006	006
DC Power Fuse**	109AJ-	001	005	005	007
Arm SCR Modules***	125DC-	031	031	032	032
MA Contactor	105CB-	003	003	003	003
MA Auxiliary	105CB-	007	007	007	007

DC-300 SPARE AND RENEWAL PARTS - "A" FRAMES* (HP Related, 460 VAC)

Part Description	Cat. # 104X-	Horsepower 3	10	20	25
AC Power Fuse	109AG-	003	003	006	006
DC Power Fuse**	109AJ-	001	003	007	009
Arm SCR Modules***	125DC-	031	031	032	032
MA Contactor	105CB-	003	003	003	003
MA Auxiliary	105CB-	007	007	007	007

* *"A" and "B" Frames no longer available as of 12/93.*

** *Regenerative drives only.*

*** *Some drives may use a different SCR module. The difference may be distinguished by color and size. The SCR Modules listed in the table are ivory in color and measure 2-1/4" X 1-3/4" X 1-3/4". Older drives may have black modules that measure 4" X 2-1/2" X 1-3/4". For these modules, substitute 022 for 031 and 023 for 032 in above charts.*

DC-300 SPARE AND RENEWAL PARTS - "B*" AND "C" FRAMES (HP Related, 230 VAC)

Part Description	Cat. # 104X-	Horsepower 5	7.5	10	15	20	25	30	40
AC Power Fuse	109AG-	007	007	007	007	011	011	011	021
DC Power Fuse**	109AG-	005	005	007	007	009	015	015	021
Arm SCR Modules									
Fused Drives	125DC-	026	026	026	026	026	027	027	020
Fuseless Drives***		027	027	027	020	020	020	020	021
Circuit Breaker	202AA-	015	015	015	015	009	009	009	----
MA Contactor	105CB-	003	003	003	003	005	006	006	013
MA Contactor	105CA-	----	----	----	----	----	----	----	----
MA Aux. Interlock	105CB-	007	007	007	007	007	008	008	011
MA Aux. Interlock	105CA-	----	----	----	----	----	----	----	----
MA Aux. Interlock	118BA-	----	----	----	----	----	----	----	011

Part Description	Cat. # 104X-	Horsepower 50	60	75	100	125
AC Power Fuse	109AG-	021	021	021	027	027
DC Power Fuse**	109AG-	021	021	021	025	025
Arm SCR Modules						
Fused Drives	125DC-	020	020	020	021	048
Fuseless Drives***		021	021	021	021	----
Circuit Breaker	202AA-	----	----	----	----	----
MA Contactor	105CB-	013	013	013	----	016
MA Contactor	105CA-	----	----	----	069	----
MA Aux. Interlock	105CB-	011	011	011	----	----
MA Aux. Interlock	105CA-	----	----	----	025	----
MA Aux. Interlock	118BA-	011	011	011	----	----

* *"A" and "B" Frames no longer available as of 12/93.*

** *Regenerative drives only.*

*** *Fuseless Drives have circuit breakers in place of the three-phase AC line fuses.*

DC-300 SPARE AND RENEWAL PARTS - "B*" AND "C" FRAMES (HP Related, 460 VAC)

Part Description	Cat. # 104X-	Horsepower 5	7.5	10	15	20	25	30	40
AC Power Fuse	109AG-	007	007	007	007	007	007	007	007
DC Power Fuse**	109AJ-	003	003	003	007	007	009	009	013
Arm SCR Modules									
Fused Drives	125DC-	026	026	026	026	026	026	026	026
Fuseless Drives***		026	026	026	027	027	027	020	020
Circuit Breaker	202AA-	015	015	015	015	015	015	015	015
MA Contactor	105CB-	003	003	003	003	003	003	003	005
MA Contactor	105CA-	----	----	----	----	----	----	----	----
MA Aux. Interlock	105CB-	007	007	007	007	007	007	007	007
MA Aux. Interlock	105CA-	----	----	----	----	----	----	----	----
MA Aux. Interlock	118BA-	----	----	----	----	----	----	----	----

Part Description	Cat. # 104X-	Horsepower 50	60	75	100	125	150	200	250	300
AC Power Fuse	109AG-	011	011	015	021	021	021	027	027	027
DC Power Fuse**	109AJ-	013	014	015	015	017	017	019	021	025
Arm SCR Modules										
Fused Drives	125DC-	027	027	027	020	020	020	021	021	048
Fuseless Drives***		020	020	020	020	021	021	021	021	----
Circuit Breaker	202AA-	009	009	009						
MA Contactor	105CB-	005	006	012	013	013	013	----	----	016
MA Contactor	105CA-	----	----	----	----	----	----	069	069	----
MA Aux. Interlock	105CB-	007	008	008	----	----	----	----	----	----
MA Aux. Interlock	105CA-	----	----	----	----	----	----	025	025	----
MA Aux. Interlock	118BA-	----	----	----	011	011	011	----	----	----

* *"A" and "B" Frames no longer available as of 12/93.*

** *Regenerative drives only.*

*** *Fuseless Drives have circuit breakers in place of the three-phase AC line fuses.*

DC-300 SPARE AND RENEWAL PARTS - "G" FRAMES (HP Related, 230 VAC)

Part Description	Cat. #	Horsepower 5	7.5	10	15	20	25	30
AC Power Fuse	323A2432	P50	P50	P50	P50	P51	P52	P52
DC Power Fuse*	323A2432	P50	P50	P50	P50	P51	P52	P52
Arm SCR Modules								
Fused Drives	104X125DC	026	026	026	026	026	027	027
Fuseless Drives**		027	027	027	020	020	020	020
Circuit Breaker	104X202AA	015	015	015	015	009	009	009
MA Contactor	104X105CB	003	003	003	003	005	006	006
MA Contactor	104X105CA	----	----	----	----	----	----	----
MA Aux. Interlock	104X105CB	007	007	007	007	007	008	008
MA Aux. Interlock	104X105CA	----	----	----	----	----	----	----
MA Aux. Interlock	104X118BA	----	----	----	----	----	----	----

* *Regenerative drives only.*

** *Fuseless Drives have circuit breakers in place of the three-phase AC line fuses.*

DC-300 SPARE AND RENEWAL PARTS - "G" FRAMES (HP Related, 460 VAC)

Part Description	Cat. #	Horsepower 5	7.5	10	15	20	25	30	40
AC Power Fuse	323A2432	P50	P50	P50	P50	P50	P50	P50	P51
DC Power Fuse*	323A2432	P70	P70	P70	P70	P70	P70	P70	P71
Arm SCR Modules									
Fused Drives	104X125DC	026	026	026	026	026	026	026	026
Fuseless Drives**		026	026	026	027	027	027	020	020
Circuit Breaker	104X202AA	015	015	015	015	015	015	015	015
MA Contactor	104X105CB	003	003	003	003	003	003	003	005
MA Contactor	104X105CA	----	----	----	----	----	----	----	----
MA Aux. Interlock	104X105CB	007	007	007	007	007	007	007	007
MA Aux. Interlock	104X105CA	----	----	----	----	----	----	----	----
MA Aux. Interlock	104X118BA	----	----	----	----	----	----	----	----

Part Description	Cat. #	Horsepower 50	60	75
AC Power Fuse	323A2432	P52	P52	P52
DC Power Fuse*	323A2432	P71	P72	P72
Arm SCR Modules				
Fused Drives	104X125DC	027	027	027
Fuseless Drives**		020	020	020
Circuit Breaker	104X202AA	009	009	009
MA Contactor	104X105CB	005	006	012
MA Contactor	104X105CA	----	----	----
MA Aux. Interlock	104X105CB	007	008	008
MA Aux. Interlock	104X105CA	----	----	----
MA Aux. Interlock	104X118BA	----	----	----

* *Regenerative drives only.*

** *Fuseless Drives have circuit breakers in place of the three-phase AC line fuses.*

14. WARRANTY PARTS AND SERVICE

1. Consult the **GE DRIVE SYSTEMS Terms and Conditions** for details of the product warranty, including warranty period and parts or service coverage.

2. Warranty replacement parts or service assistance may be obtained direct from the factory during regular working hours **(8 a.m. - 5 p.m.; M-F)** by calling:

GE DRIVE SYSTEMS
PRODUCT SERVICE ENGINEERING
SALEM, VA
(703) 387-7595

3. So that we may serve you more expediently, please have the following information ready whenever contacting the factory:

DRIVE SERIAL # and MODEL #
PART # and DESCRIPTION

4. **GE DRIVE SYSTEMS** may request that failed parts replaced under warranty be returned to the factory. A **Return Authorization** tag or number is to be obtained and included with any parts being returned. When returning parts, pack them carefully to prevent damage in transit and ship to*:

GE DRIVE SYSTEMS
1501 ROANOKE BLVD
SALEM , VA 24153
ATTENTION: G27

**It is not necessary to return failed parts by "Overnight" or "Priority" shipment.*

NOTES:

NOTES:

15. RECOMMENDED POWER STUD WIRING, TERMINALS & CRIMPING TOOLS

RECOMMENDED POWER STUD WIRING AND CRIMPING TOOLS

ITEM	AMP CRIMPING TOOL CAT. NO.	TYPE	FOR WIRE SIZE	TERMINAL TYPE
1	49592	Hand	14--12 AWG	Solistrand*
2	49935	Hand	12--10 AWG	Solistrand
3	69062	Hand-hydraulic with self-contained dies	8--2 AWG	Solistrand

ITEM	AMP ELECTRO-HYDRAULIC POWER UNIT 115VAC (CAT. NO.) Crimp Tool	Head	Die	FOR WIRE SIZE	TERMINAL TYPE
4	69120-1	69065	46323-2	1--10 AWG	Solistrand
5	69120-1	69065	46323-2	2/0 AWG	Solistrand
6	69120-1	69065	46324-2	3/0 AWG	Solistrand
7	69120-1	69065	46325-2	4/0 AWG	Solistrand
8	69120-1	69065	46326-2	250 MCM	AMPower
9	69120-1	69065	46327-2	300 MCM	AMPower

**Trademark of AMP Incorporated.*

RECOMMENDED POWER STUD WIRING AND TERMINALS

Drive	230 VAC, 240 VDC Stud Dia. (c)		Power Wire (a, b)			Terminal - AMP Solistrand GE Cat.#	AMP	Hole	Crimping Tool
HP	AC	DC	Qty.	AWG	MCM	104X161AA	Cat.#	Dia.	See tool list
1-3	.25		1	12	---	009	33458	.25	Item 2
		.25	1	12	---	009	33458	.25	Item 2
5	.25		1	10	---	009	33458	.25	Item 2
		.25	1	10	---	009	33458	.25	Item 2
7.5	.25		1	8	---	012	33461	.25	Item 3
		.25	1	8	---	012	33461	.25	Item 3
10	.25		1	6	---	015	33465	.25	Item 3
		.25	1	6	---	015	33465	.25	Item 3
15	.25		1	4	---	059	31811	.25	Item 3
		.25	1	4	---	059	31811	.25	Item 3
20	.25		1	2	---	057	320383	.25	Item 3
		.25	1	2	---	057	320383	.25	Item 3
25	.25		1	1/0	---	067	36915	.25	Item 4
		.25	1	1/0	---	067	36915	.25	Item 4
30	.25		1	1/0	---	067	36915	.25	Item 3
		.25	1	1/0	---	067	36915	.25	Item 3
40	.38		1	2/0	---	023	36923	.38	Item 5
		.38	1	3/0	---	025	36927	.38	Item 6
50	.38		1	3/0	---	025	36927	.38	Item 6
		.38	1	4/0	---	068	321878	.38	Item 7
60	.38		1	4/0	---	068	321878	.38	Item 7
		.38	2	4/0	---	021	36917	.38	Item 4
75	.38		2	1/0	---	021	36917	.38	Item 4
		.38	2	2/0	---	023	36923	.38	Item 5
100	.38		2	3/0	---	025	36927	.38	Item 6
		.38	2	4/0	---	068	321878	.38	Item 7

NOTES:

(a) *Wire size from NEC Table 310-16. Copper wire rated 60 degrees C for 100 amps and less, 75 degrees C for over 100 amps, in 30 degrees C ambient and 1.25 times drive rated amps. These are minimum wire sizes. Consult and conform to local and national codes.*

(b) *Quantity of wires and terminals in parallel per stud.*

(c) *AC stud is not used for wiring on drives that have circuit breakers.*

RECOMMENDED POWER STUD WIRING AND TERMINALS

Drive	460 VAC, 500 VDC Stud Dia. (c)		Power Wire (a, b)			Terminal - AMP Solistrand GE Cat.#	AMP	Hole	Crimping Tool
HP	AC	DC	Qty.	AWG	MCM	104X161AA	Cat.#	Dia.	See tool list
1-3	.25		1	12	---	009	33458	.25	Item 2
		.25	1	12	---	009	33458	.25	Item 2
5	.25		1	12	---	009	33458	.25	Item 2
		.25	1	12	---	009	33458	.25	Item 2
7.5	.25		1	12	---	009	33458	.25	Item 2
		.25	1	12	---	009	33458	.25	Item 2
10	.25		1	10	---	009	33458	.25	Item 2
		.25	1	10	---	009	33458	.25	Item 2
15	.25		1	8	---	012	33461	.25	Item 3
		.25	1	8	---	012	33461	.25	Item 3
20	.25		1	6	---	015	33465	.25	Item 3
		.25	1	6	---	015	33465	.25	Item 3
25	.25		1	6	---	015	33465	.25	Item 3
		.25	1	6	---	015	33465	.25	Item 3
30	.25		1	4	---	059	31811	.25	Item 3
		.25	1	4	---	059	31811	.25	Item 3
40	.25		1	2	---	057	320383	.25	Item 3
		.25	1	2	---	057	320383	.25	Item 3
50	.25		1	1	---	067	36915	.25	Item 4
		.25	1	1	---	067	36915	.25	Item 4
60	.25		1	1	---	067	36915	.25	Item 4
		.25	1	1	---	067	36915	.25	Item 4
75	.25		1	1/0	---	067	36915	.25	Item 4
		.25	1	2/0	---	075	36921	.25	Item 5
100	.38		1	3/0	---	025	36927	.38	Item 6
		.38	1	4/0	---	068	321878	.38	Item 7

NOTES:

(a) *Wire size from NEC Table 310-16. Copper wire rated 60 degrees C for 100 amps and less, 75 degrees C for over 100 amps, in 30 degrees C mabient and 1.25 times drive rated amps. These are minimum wire sizes. Consult and conform to local and national codes.*

(b) *Quantity of wires and terminals in parallel per stud.*

(c) *AC stud is not used for wiring on drives that have circuit breakers.*

RECOMMENDED POWER STUD WIRING AND TERMINALS (cont.)

	460 VAC, 500 VDC					Terminal - AMP Solistrand			Crimping Tool
Drive	Stud Dia. (c)		Power Wire (a, b)			GE Cat.#	AMP	Hole	
HP	AC	DC	Qty.	AWG	MCM	104X161AA	Cat.#	Dia.	See tool list
125	.38		1	4/0	---	068	321878	.38	Item 7
		.38	2	1/0	---	021	36917	.38	Item 4
150	.38		2	1/0	---	021	36917	.38	Item 4
		.38	2	2/0	---	023	36923	.38	Item 5
200	.38		2	3/0	---	025	36927	.38	Item 6
		.38	2	4/0	---	068	321878	.38	Item 7
250	.38		2	4/0	---	068	321878	.38	Item 7
		.38	2	---	300		325803	.38	Item 9

NOTES:

(a) *Wire size from NEC Table 310-16. Copper wire rated 60 degrees C for 100 amps and less, 75 degrees C for over 100 amps, in 30 degrees C mabient and 1.25 times drive rated amps. These are minimum wire sizes. Consult and conform to local and national codes.*

(b) *Quantity of wires and terminals in parallel per stud.*

(c) *AC stud is not used for wiring on drives that have circuit breakers.*

16. GLOSSARY OF TERMS

Base Speed	--	Maximum speed of a motor at rated armature volts and field amps.
DFP	--	Delayed Firing Power to armature SCRs.
EEROM	--	Electrically Erasable Read-Only Memory, also called EE or E^2 (E squared). Contains the drive Parameters.
EE Enable Device	--	Device used to "Enable" or "Protect" EEROM programming. A hardware jumper (**JP13**) located on the Control Card.
Field Weakening	--	Lowering of field current at rated armature volts to allow higher speeds. Actual armature voltage where field weakening begins is called the crossover point (adjusted by Parameter **ADR079-CROSS**)
FUP	--	Firing Microprocessor, **U5** on the Control Card.
"Hard" Reset	--	Hardware Reset, resets the drive cards. Used to clear some faults and allow certain parameter changes. Activated by depressing **SW2** on the Control Card or connecting **3TB69** to **3TB66**.
K1	--	"**RUN**" Relay located on Power Supply Card.
K2	--	**MA** Auxiliary Relay, pilot relay for **MA** contactor. **K2** is located on the Power Supply Card.
K3	--	Fault Relay located on the Power Supply Card.
LED	--	Light Emitting Diode.
MA	--	Main AC Contactor (see General Description).
MFC	--	Motor Field Control (see General Description).
MOV	--	Metal Oxide Varistor.
Parameters	--	Adjustable software settings used to program and tune the drive regulators.
PI	--	Process Interface Card.
PL	--	Plug (may be plug or receptacle).
PSC	--	Power Supply Card.
PUP	--	Programmer Microprocessor, **U1** (and **U3** PROM) on Programmer Card.
RAM	--	Random Access Memory
Regenerative	--	Ability to return power from the armature to the AC line. Creates braking effect on motor. (Regeneration, regen., etc.)
RS232C	--	Serial Link Communication Standard.
RS422	--	Serial Link Communication Standard.
RUP	--	Regulator Microprocessor, **U4** (and **U1** PROM) on Control Card.
SCR	--	Silicon Controlled Rectifier (Thyristor).
SLD	--	Signal Level Detector, **SLD0** thru **SLD5**, (**RX1** or **K1** thru **RX5** or **K5**) on optional Application Card and **SLD5** on **3TB**.
"Soft" Reset	--	Software Reset initiated by pressing the **RESET** key on the optional Programmer.
TUP	--	Tachometer Microprocessor, **U6** on the Control Card.

NOTES:



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